

**STUDIES ON INTEGRATED MANAGEMENT
OF *Rhizoctonia bataticola* (Taub.) Butler
CAUSING DRY ROOT ROT OF CHICKPEA**

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B.Sc. (Ag.)

**MASTER OF SCIENCE IN AGRICULTURE
(PLANT PATHOLOGY)**



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OF *Rhizoctonia bataticola* (Taub.) Butler
CAUSING DRY ROOT ROT OF CHICKPEA**

BY
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B.Sc. (Ag.)

**THESIS SUBMITTED TO THE
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IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE OF**

**MASTER OF SCIENCE IN AGRICULTURE
(PLANT PATHOLOGY)**

CHAIRPERSON: Dr. N.P. ESWARA REDDY



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2012

DECLARATION

I, **G. AMRUTHA VEENA**, hereby declare that the thesis entitled **“STUDIES ON INTEGRATED MANAGEMENT OF *Rhizoctonia bataticola* (Taub.) Butler CAUSING DRY ROOT ROT OF CHICKPEA”** submitted to Acharya N. G. Ranga Agricultural University, Hyderabad for the award of degree of **MASTER OF SCIENCE IN AGRICULTURE** is the result of original research work done by me. I also declare that no material contained in this thesis has been published earlier in any manner.

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Ms. G. AMRUTHA VEENA has satisfactorily prosecuted the course of research and that the thesis entitled “**STUDIES ON INTEGRATED MANAGEMENT OF *Rhizoctonia bataticola* (Taub.) Butler CAUSING DRY ROOT ROT OF CHICKPEA**” submitted is the result of original research work and is of sufficiently high standard to warrant its presentation to the examination. I also certify that neither the thesis nor its part thereof has been previously submitted by her for a degree of any university.

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CERTIFICATE

This is to certify that the thesis entitled “**STUDIES ON INTEGRATED MANAGEMENT OF *Rhizoctonia bataticola* (Taub.) Butler CAUSING DRY ROOT ROT OF CHICKPEA**” submitted in partial fulfillment of the requirements for the award of degree of **MASTER OF SCIENCE IN AGRICULTURE** to the Acharya N.G. Ranga Agricultural University, Hyderabad, is a record of the bonafide research work carried out by **Ms. G. AMRUTHA VEENA** under our guidance and supervision. The subject of the thesis has been approved by the Student’s Advisory Committee.

No part of the thesis has been submitted by the student for the award of any other degree or diploma. The published part has been fully acknowledged. All assistance and help received during the course of the investigations have been duly acknowledged by the author of the thesis.

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LIST OF ABBREVIATIONS AND SYMBOLS

%	-	Per cent
@	-	At the rate of
°C	-	Degree Celsius
µg	-	Micro gram
µl	-	Microlitre
µm	-	Micrometre
a.i	-	Active ingredient
CD	-	Critical difference
cfu	-	Colony forming units
cm	-	Centimetre
DAI	-	Days after inoculation
EC	-	Emulsifiable concentrate
<i>et al.</i>	-	and others people
Fig.	-	Figure
g	-	gram (s)
h	-	Hour(s)
ha	-	Hectare
<i>i.e.,</i>	-	That is
IDM	-	Integrated disease management
kg	-	Kilogram
l	-	Liquid
mg	-	Milligram
mha	-	Million Hectares
ml	-	Millilitre
mm	-	Millimetre
mM	-	Millimolar
N	-	Normality
ng	-	Nanogram
No.	-	Number
PDA	-	Potato dextrose agar
PDI	-	Per cent disease incidence
pH	-	Power of hydrogen ion concentration
ppm	-	parts per million
psi	-	Pounds per square inch
RAPD	-	Random Amplified Polymorphic DNA
RBD	-	Randomized Block Design
SEm	-	Standard error of mean
Sp. or Spp.	-	Species (singular or plural form)
<i>viz.</i>	-	Namely
WP	-	Wettable powder

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ABSTRACT

Chickpea (*Cicer arietinum* L.) is an important pulse crop grown in tropical, subtropical and temperate regions of the world. Chickpea is affected by *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot an important disease with yield losses ranging from 50 to 71 per cent.

A detailed study was carried on isolation of the pathogen and establishment of pathogenicity of *Rhizoctonia bataticola*. *In vitro* antagonism of fungal and bacterial antagonists isolated from rhizosphere and root habitats against *Rhizoctonia bataticola* and their compatibility with different fungicides was studied. *In vitro* evaluation of fungicides against pathogen, integrated disease management of *Rhizoctonia bataticola* and molecular characterization of potential biocontrol agents by using RAPD and 16S rDNA analysis was also carried.

The Pathogen was isolated from infected plant showing typical dry root rot symptoms viz., withering and drying of the plants, presence of dark tap root showing signs of rotting and devoid of its lateral and finer roots, purified by single hyphal tip method and identified as *Rhizoctonia bataticola*.

In vitro efficacy of four fungicides viz., copper oxychloride, captan, hexaconazole, tebuconazole and one antifungal antibiotic validamycin was

evaluated against *Rhizoctonia bataticola* using poisoned food technique at different concentrations. Among the four fungicides and one antibiotic tested against the pathogen copper oxychloride, captan, hexaconazole and tebuconazole showed 100 per cent inhibition of mycelial growth, whereas the antibiotic validamycin showed 91.66 per cent inhibition.

Among ten fungal antagonistic isolates, *Trichoderma* isolate-7 (CT7) was found to be superior with highest per cent (83.33) inhibition of the growth of *R. bataticola*. Among 20 rhizosphere bacterial isolates CRB-13 showed highest inhibition (86.66 %). Regarding 20 root endophytes, the isolate CREB-13 showed maximum inhibition (95.55%) followed by CREB-6 (94.44 %) and CREB-16 (93.33 %).

In vitro compatibility of potential fungal and bacterial antagonists was tested against four fungicides and one antifungal antibiotic at different concentrations using poisoned food technique and spectrophotometry respectively. *Trichoderma* isolate-7 (CT7) was more compatible with validamycin (72.22 %), followed by copper oxychloride (61.12 %). Among CREB-6, CREB-13, CREB-16 and CRB-13 bacterial isolates, CREB-16 was found to be most compatible antagonistic bacteria with validamycin (84.14 %) followed by copper oxychloride (78.28 %) and least compatibility was recorded in case of tebuconazole (25.52 %) in spectrophotometry.

The *in vitro* mode of action of *Trichoderma* isolate-7 (CT7) against *Rhizoctonia bataticola* showed hyphal interaction between the antagonist and the pathogen.

The *in vitro* experimental results revealed that volatile metabolites produced by antagonistic bacteria inhibited the mycelial growth of *Rhizoctonia bataticola* to an extent of 77.77 per cent (CREB-16) over control. In the case of non-volatile test, culture filtrate of antagonistic bacteria did not show any inhibition on the growth of *Rhizoctonia bataticola* over control *in vitro*.

Trichoderma isolate-7 (CT7) showed highest compatibility with the bacterial isolate CREB-16 (72.22 %) in compatibility tests between antagonistic fungus and bacteria.

In blotter technique, 100 per cent seed germination was recorded in all the treatments under the present investigation.

The efficacy of potential fungal antagonist *Trichoderma* isolate-7 (CT7), bacterial antagonist CREB-16, fungicide copper oxychloride and organic amendments like FYM, vermicompost and neem cake were tested in pot culture against dry root of chickpea. The results revealed that treatment T₈ i.e seed treatment with fungicide (Copper oxychloride) + soil application of potential fungal (*Trichoderma* isolate-7) and bacterial biocontrol agent

(CREB-16) was found to be superior as it recorded the highest germination percentage, highest initial and final population of chickpea, least PDI, maximum plant height, root length and maximum shoot and root dry weights.

The population dynamics of both antagonists and pathogen were estimated at two different intervals initially at 7 DAS and then on 45 DAS in pot culture experiment. With the increase in antagonists population, the population dynamics of pathogen were significantly reduced from 7 DAS to 45 DAS over control in all the treatments with the maximum reduction in T_8 treatment.

The RAPD banding profiles with random primers viz., OPA-12, OPA-18, OPD-4 and OPD-5 reflected the genetic diversity among the antagonistic bacterial isolates with formation of three main clusters. Amplified 16S rDNA with universal primers 63F and 1387R produced approximately 1300 bp fragments as expected. The amplified product was sent for sequencing to Eurofins Genomics Private Limited, Bangalore.

Chapter ~ I

Introduction

Chapter I

INTRODUCTION

Chickpea (*Cicer arietinum* L.) is an important food legume crop, which has world wide acceptance as a major source of protein and essential amino acids for human as well as for animal consumption.

It is an integral part of the cropping system of the farmers all over the country, because this crop fits well in the crop rotation and mixed cropping. It has multipurpose use and ability to grow under the condition of low fertility and varying conditions of soil and climate.

Chickpea is the third important legume of the world grown in about 50 countries around the world covering an area of 11 m ha with an average global productivity of 881 kg ha⁻¹ (Anonymous, 2010).

India is the leading producer of chickpea contributing to about 70 per cent of the world's chickpea production. Among the major pulse crops, chickpea contributes nearly 42 to 47 per cent of total pulse production. Nearly 90 per cent of the countries area and 91 per cent of the production is from six states viz., Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Karnataka and Andhra Pradesh.

In India, it is grown over an area of 8.74 m ha with an annual production of 7.35 million tonnes and productivity of 841 kg ha⁻¹ (Anonymous, 2009-10). In Andhra Pradesh, it is grown in an area of 5.84 lakh ha with an annual production and productivity of 7.19 lakh tonnes and 1233 kg ha⁻¹ respectively (Annual Report, 2010-2011).

Chickpea suffers from several seed borne fungal diseases viz., black root rot, dry root rot, wet root rot, seed rotting, stem rot, crown rot, foot rot, *Sclerotinia* wilt and gray mould (Nene *et al.*, 1996). Among these root rot

diseases, dry root rot (*Rhizoctonia bataticola* (Taub.) Butler) cause considerable yield losses in chickpea which may be as high as 50 to 71 per cent (Ahmed and Mohammad, 1986).

Rhizoctonia bataticola is always considered as the most destructive pathogen of chickpea in semi-arid region (Ashraf *et al.*, 2005). The incidence of dry root rot is higher in rainfed crop (Kotasthane *et al.*, 1979) when there is moisture stress *i.e.* moisture content less than or equal to 60 per cent and temperature 33 to 35°C (Taya *et al.*, 1988) during the early stages of the crop growth period.

Dry root rot generally appears around flowering and podding time in the form of scattered dried plants, but seedlings are also infected. The symptoms include drooping of petioles and leaflets which are confined to top of the plants. The tap root is dark, shows signs of rotting and is devoid of most of its lateral and finer roots. The most important diagnostic symptom is shredding of bark which comes out in the form of flakes (Haware, 1990).

Presently there are no resistant varieties to dry root rot disease in chickpea. Hence, this work has been initiated to search for an alternative approach other than use of fungicides for the management of the disease.

Biological control using microbial antagonists is considered as good alternative of management of root diseases in many crops (Dileep *et al.*, 1999; Mukhopadhyay *et al.*, 1992; Sheela *et al.*, 1998; Khot *et al.*, 1996; Gholve and Kurundkar, 2002). It can be successfully utilized especially within the frame work of integrated disease management system. The disease suppressing and beneficial effects of rhizosphere colonizing bacteria have been well studied (Ryder *et al.*, 1994; Vidhyasekaran, 1998; Rangeshwaran and Prasad, 2000). The work on isolation and identification of endophytic bacteria in chickpea was recently started (Rangeshwaran *et al.*, 2008).

Not much attention has been paid on management of this economically important disease using native fungal antagonists, hence it is worthwhile to isolate mycoflora having antagonistic activity from new habitat of root.

Integrated disease management is gaining importance which involves blending of compatible systems of control measures for effective management of disease from profitability to food and environmental safety (Jacksen and Backman, 1993).

Organic amendments are important not only for the reduction of quantity of chemical fertilizers, but also act as excellent sources of nutrition which favour native antagonist to proliferate and suppress the soil borne diseases (Mukhopadhyay, 1994).

Attempts were also made to resolve the influence of organic amendments on disease control efficacy of bioagents and population dynamics of pathogen and antagonists.

Recently, molecular techniques gained importance in characterization and diagnosis of microbial population. Hence, molecular characterization of the potential biocontrol agents using Random Amplified Polymorphic DNA (RAPD) and 16S rDNA which helps in identification of antagonists are of immense use. These molecular techniques will also help in developing SCAR markers for the diagnosis of potential biocontrol agents in future.

Taking into consideration the above facts, the present investigation has been formulated to develop an integrated disease management package against dry root rot in chickpea with the following objectives:

Objectives of Investigation:

- 1) Isolation and identification of the dry root rot pathogen from infected chickpea plants.
- 2) To isolate fungicidal compatible potential bio-control agents from rhizosphere and root endophytes under *in vitro* conditions.
- 3) To study the effect of different organic amendments individually and in combination with potential fungicidal tolerant bio-control agent on the incidence of *Rhizoctonia bataticola* under green house conditions.
- 4) Molecular characterization of potential biocontrol agents by RAPD and 16S rDNA analysis.

Chapter ~ II

Review of Literature

Chapter II

REVIEW OF LITERATURE

A brief review of literature pertaining to the aspects under investigation is presented in this chapter.

Wherever, the literature on particular aspects of *Rhizoctonia bataticola* on chickpea was scanty, it was amply supplemented with and supported by relevant literature from other pathogen and other crops.

2.1 THE PATHOGEN

The pathogen occurs in the sclerotial form with the taxonomic nomenclature as *Rhizoctonia bataticola* (Taub.) Butler. The pycnidial stage of this pathogen is *M. phaseolina* (Tassi.) Goid. Mc Rae (1929) reported the occurrence of *Macrophomina phaseolina* on wilted plants of mung bean and urdbean and established the genetic similarity between *M. phaseolina* and *R. bataticola*.

2.1.1 Morphology

Philip *et al.* (1969) noticed the differences in morphology of *Macrophomina phaseolina* isolates from various parts of host plants. They include the pycnidial development *in vitro* on mung bean roots infected with *Macrophomina*, as black, globose or depressed and 150 to 200 µm in diameter. Pycnidiospores were oval or elliptical, hyaline, non-septate, thin walled with 10-24 x 6-10 µm in size. The pycnidiospores develop into *R. bataticola* in culture at room temperature.

Occurrence of numerous small black sclerotia of this pathogen under *in vitro* conditions was also reported by Jain *et al.*, (1973). According to him, sclerotial morphology of *Macrophomina phaseolina* of urdbean was found to be different when isolated from different plant parts. The soil and seed isolate

produced small sclerotia which were more pathogenic than leaf isolate which produced larger sclerotia.

Vishwadar and Sarbhoy (1993) studied the variation in isolates of *R. bataticola* isolated from soybean plants collected from 11 locations at IARI, New Delhi. About 44 isolates made from different parts of the plants and it was observed that the isolates differed with reference to colour of the colony, mycelial growth, shape and size of sclerotia.

Survey reports in India showed that there is a lot of variability on cultural character, growth rate and morphology of sclerotia, pycnidia, pycnidiospore and disease severity of *Rhizoctonia bataticola*. (Singh and Dubey, 2005 and Chauhan *et al.*, 2002).

Kanchan and Biswas (2009) studied the morphological variability of *Rhizoctonia bataticola* in pigeon pea and reported that the nature of mycelium varied from fluffy dark brown to partially fluffy mycelium with smooth margin.

2.2 DISTRIBUTION AND ECONOMIC IMPORTANCE OF *R. bataticola*

Rhizoctonia bataticola is a soil borne pathogen with a wide host range (Dhingra and Sinclair, 1978) and causes charcoal rot (seedling phase), root rot and stem blight in more than 500 species of plants (Sinclair, 1982).

In severe epidemics, crop losses due to the wilt complex may go as high as 60 to 70 per cent. Damage is up to 61 per cent at seedling stage and 43 per cent at flowering stage (Haware and Nene, 1980).

Ahmed and Mohammad (1986) reported a yield loss to the tune of 50 to 71 per cent due to *R. bataticola* in chickpea growing areas of Bihar.

Due to severity of the root rot disease (13%) in cluster bean yield losses upto 32% have been reported in arid zones of Rajasthan (Lodha *et al.*, 1986).

Kataria and Grover (1977) and Tyagi *et al.* (1988) reported a yield loss of 10.8 per cent and 24.1 per cent due to *R. bataticola* in mungbean from the states of Haryana and Rajasthan respectively.

Plant losses upto 77 per cent have been reported due to *Rhizoctonia bataticola* in soybean (Muthuswamy and Mariappan, 1991).

Dry root rot and charcoal rot caused by *Macrophomina phaseolina* are found to be serious diseases in chickpea (Lodha, 1993).

Usha Rani *et al.* (2009) reported that root rot in groundnut caused by *Macrophomina phaseolina* causes serious economic losses ranging from 20 to 30 per cent under dry warm conditions.

2.3 SYMPTOMATOLOGY

Nene *et al.* (1978) observed continuous black discoloration of pith and xylem vessels of roots and basal shoots as one of the main characteristic symptoms of dry root rot of chickpea.

Baldev *et al.* (1988) reported symptoms of dry root rot of chickpea which consisted of straw colored leaves and stems. Tap root of infected plants was observed to be dry and devoid of lateral and finer roots. Roots were turned dark and showed signs of rotting. The dead root was observed brittle towards the tip and showed shredding of bark.

Haware (1990) reported that dry root rot of chickpea appears around flowering and podding time in the form of scattered dried plants but seedlings are also infected. The symptoms induced were drooping of petioles and leaflets which were confined to top of the plants. Shredding of bark in form of flakes was observed.

Rangaswamy (1996) reported field symptoms which include yellowing of plants with drooping of leaves. Due to decay of roots, plants can be easily uprooted. Root portion appears brownish from outside. The stem and root below the region shows rotting with frequently pinkish white mycelial growth. Dried plants scattered through out the field are indicative of root rot incidence.

Singh (1999) reported that dry root rot of chickpea occurred from flowering to podding stage. Infected plants were suddenly dried in the field. The tap roots turned dark brittle and devoid of lateral roots.

Singh and Agarwal (2002) observed the withering and drying of chickpea plants in the field due to the infestation of *Rhizoctonia bataticola*.

2.4 EFFECT OF ENVIRONMENT ON DISEASE DEVELOPMENT

The root rot due to *Rhizoctonia bataticola* was reported to be high under low moisture regime (Ramaswamy *et al.*, 1976).

Decreased levels of sand and increased levels of clay lead to decreased number of infected plants by *Rhizoctonia bataticola*. Further in frequently irrigated field of chickpea dry root rot incidence was observed to be low (Taya *et al.*, 1988).

Samiyappan (1988) reported that urdbean root rot caused by *M. phaseolina* was low in clay loam when compared to sandy loam soils and the per cent disease incidence was 35 and 100, respectively.

Vanita *et al.* (2009) studied the effect of nutritional and physiological factors on growth and sclerotial formation of *Rhizoctonia bataticola* isolates and reported that out of six media PDA, Czapek's dox, Richard's, Oat meal and peptone agar media tested, PDA supported the maximum growth and sclerotial formation of *Rhizoctonia bataticola*. All the six isolates of *Rhizoctonia bataticola* gave maximum mycelial growth at 35°C and

minimum at 20°C. Sclerotia formation was also better at 35°C and nil or rare at 20°C and 40°C, pH 7.0 was suitable for the growth of pathogen.

2.5 ISOLATION OF THE PATHOGEN

Rhizoctonia bataticola can be isolated from different plant parts viz., diseased seeds and seedlings (Pant and Mukhopadhyay, 2001; Bagri *et al.*, 2004; Vinod Kumar *et al.*, 2007), roots (Sajeena *et al.*, 2004), stem (Pan and Bhagat, 2008) and leaves (Sarkar and Pradhan, 1999).

2.5.1 Maintenance of the pathogen

Potato dextrose agar (PDA) was found to be the best supporting medium for *R. bataticola* (Vinod Kumar *et al.*, 2007; Konde *et al.*, 2008).

2.6 PATHOGENICITY TESTS

Prameela Devi and Singh (1998) reported that 10 per cent inoculum density causing 71 per cent seedling mortality in greengram and blackgram.

Mohammadi and Banihashemi (2005) isolated 154 *Fusarium* spp. from root, crown and stem of infected chickpea plants. Among which, *F. oxysporum* (Fo) and *F. solani* (Fs) were highly pathogenic on chickpea causing 45.64 and 39.60 per cent disease incidence respectively. These pathogenicity studies revealed that 53 isolates of *F. solani* as causal agent of dry root rot and 38 isolates of *F. oxysporum* as causal agent of wilt in chickpea.

Katariya *et al.* (2007) assessed nine isolates of *R. bataticola*, the incitant of dry root rot of chickpea and revealed that all nine isolates were found pathogenic on C-235 chickpea in both sterilized and unsterilized soil. The root rot incidence varied from 26 to 100 per cent in sterilized soil and 20 to 80 per cent in an unsterilized soil.

Jaiman and Jain (2008) conducted pathogenicity test for *M. phaseolina* in cluster bean seeds and observed that seed inoculation with *M. phaseolina*

resulted in pre (33%) and post emergence mortality with less of seed germination (59%) and vigour index (856) in comparison to control (93%).

Kanchan and Biswas (2009) studied pathogenic variability of *Rhizoctonia bataticola* in pigeon pea and reported that the potency of infection of eight isolates of *Rhizoctonia bataticola* varied on four different cultivars of pigeon pea viz. Prabhat, Bahar, T-7 and T-21. The isolates, T-4 and T-5 showed maximum disease severity with 48.3 per cent each on Bahar, T-1 isolate on T-21 and Prabhat with 54.4 and 53.3 per cent respectively. The maximum disease severity was recorded when the plant had attained the age of 47 days followed by 57 and 67 days.

2.7 IN VITRO EVALUATION OF FUNGICIDES AGAINST PATHOGEN

Gautam and Narain (1996) reported fungitoxicity of carbendazim, carboxin and captan to *Macrophomina phaseolina*.

The effect of six fungicides on growth of *M. phaseolina* causing root rot disease in greengram was studied under *in vitro*. Carbendazim (0.1%) completely inhibited the growth of the pathogen. The fungicides copper oxychloride (0.25%), thiram (0.2%), thiophanate methyl (0.1%), captan (0.2%) and methoxy ethyl chloride (0.05%) also suppressed the pathogen by 95.89, 91.11, 89.67, 78.56 and 77.44 per cent respectively (Ebenezar and Wesely, 2000).

Pant and Mukhopadhyay (2001) evaluated the fungicides against *Macrophomina phaseolina*, the causal organism of soybean dry root rot and revealed that vitavax was found highly inhibitory to *Macrophomina phaseolina*.

Chattopadhyay and Kalpana (2002) reported that carbendazim could cause 100 per cent inhibition of mycelial growth of *Macrophomina phaseolina* even at 25 ppm concentration.

Konde *et al.* (2008) evaluated the fungicides against *R. bataticola* causing root rot of soybean and revealed that combination of carbendazim + thiram (0.1 + 0.2 % concentration), penconazole (0.1%) and thiophanate-M (0.1%) were significantly effective in completely (100%) inhibiting the radial growth of *R. bataticola*.

Khan and Gangopadhyay (2008) tested the efficacy of fungicides like carbendazim, carboxin, captan, chlorothalonil on mycelial growth of *Macrophomina phaseolina*. The results indicated that carbendazim and captan were highly inhibitory to the pathogen.

Paul *et al.* (2008) evaluated the fungicides against the root rot of pea caused by *F. solani*, *Rhizoctonia solani* and *S. sclerotiorum* and results revealed that carbendazim was most effective fungicide against all the pathogens.

2.8 POTENTIAL BIOCONTROL AGENTS

2.8.1 Isolation of antagonistic mycoflora from rhizosphere soil and root endophytes

Jayalakshmi *et al.* (2003) isolated *Trichoderma harzianum* from rhizosphere of healthy pigeonpea plants.

Ramesh and Korikanthimath (2006) isolated biocontrol agents like *Trichoderma viride* and *T. harzianum* from rhizosphere of various crops and tested their efficacy against *Macrophomina phaseolina* causing root rot of groundnut.

2.8.2 Isolation of antagonistic bacteria from rhizosphere soil and root endophytes

Singh and Mehrotra (1980) isolated the *Bacillus* and actinomycetes from chickpea rhizosphere that were found to be antagonistic to *R. bataticola* under *in vitro* conditions.

Nautiyal (1997) isolated 478 chickpea rhizosphere competent bacteria for suppression of chickpea pathogenic fungi *Fusarium oxysporum* f. sp. *ciceri*, *Rhizoctonia bataticola* and *Pythium* sp. and found 386 strains that effectively colonize chickpea roots.

Parmer and Dadarwal (1997) isolated *Pseudomonas* and *Bacillus* sp. from rhizosphere and rhizoplane of healthy chickpea plants.

Seed bacterization with fluorescent pseudomonads isolated from rhizosphere of crop plants reduced the number of wilted chickpea plants in wilt sick soil (Kumar, 1998).

Downing and Thomson (2000) isolated an endophytic strain of *Pseudomonas fluorescens* from apple plantlets and introduced to control the phytopathogenic fungus *Rhizoctonia solani* on bean (*Phaseolus vulgaris*) seedlings.

Rangeshwaran and Prasad (2000) isolated three hundred antagonistic bacteria against root rot and wilt from chickpea rhizosphere.

Hallmann (2001) reported different genera of endophytic bacteria which include *Bacillus*, *Pseudomonas*, *Burkholderia*, *Erwinia*, *Clavibacter*, *Xanthomonas*, *Phyllobacterium*, *Enterobacter*, *Flavobacterium*, *Agrobacterium*, *Corynebacterium*, *Curtobacter*, *Pantoea*, *Micrococcus* and *Lactobacillus* that have been isolated from various crops.

Rangeshwaran *et al.* (2008) isolated five endophytic (*Erwinia herbicola*, *Erwinia agglomerans*, *Bacillus megaterium*, *Bacillus circulans*, *Bacillus* sp.) and two rhizospheric bacteria (*Bacillus subtilis*, *Pseudomonas fluorescens*) from healthy chickpea plants that were identified based on morphological, physiological and biochemical tests.

2.8.3 *In vitro* evaluation of the efficacy of fungal antagonists against *R. bataticola*

Parakhia and Vaishnav (1986) reported that the growth of *R. bataticola*, causal organism of dry root rot of chickpea was inhibited by *T. harzianum* in dual culture and the antagonist over grew the pathogen.

Dubey and Dwivedi (1988) reported the antagonistic potential of *T. viride* against *M. phaseolina*.

Selvarajan and Jeyarajan (1996) screened three *Trichoderma* spp. *in vitro* against chickpea root rot causing organisms viz., *Fusarium solani* and *Macrophomina phaseolina*. They observed that *T. viride*, *T. harzianum* and *T. hamatum* formed inhibition zones against both the chickpea root rot pathogens. The antagonist reduced sclerotial size and germination of *Macrophomina phaseolina*.

Patel and Anahosur (2001) observed that *T. harzianum* formed inhibition zone against *Fusarium* sp. and *Macrophomina phaseolina*.

Bandyopadhyay *et al.* (2003) reported that a strain of *Trichoderma* checked the growth of *Rhizoctonia bataticola* by 51.1%.

Ananthan *et al.* (2003) studied the antagonistic effect of four *Trichoderma* sp. in dual culture technique. All the *Trichoderma* spp. significantly inhibited the growth of *M. phaseolina* to an extent of 16.11 per cent as compared to control.

Suriachandraselvan (2004) tested the efficacy of *Trichoderma* sp. against *Macrophomina phaseolina* the causal organism of sunflower charcoal rot and reported that *T. viride* and *T. harzianum* significantly inhibited the mycelial growth of pathogen (73-74%).

Strong antagonism by *Trichoderma* spp. against a range of soil borne pathogens has been reported by Pan *et al.* (2001) and Bose *et al.* (2005).

Lokesha and Benagi (2007) observed the maximum per cent inhibition of *M. phaseolina* which causes root rot disease in pigeon pea in dual culture by *Trichoderma viride* followed by *T. harzianum*.

Konde *et al.* (2008) tested the five antagonist's of soybean and observed that *Trichoderma viride* (96.39%) was superior in inhibiting the growth of *R. bataticola*.

Kaushal (2008) reported that *T. harzianum* was effective in inhibiting the mycelial growth of *R. bataticola* the causal organism of chickpea dry root rot.

Pan and Bhagat (2008) isolated ten isolates of *Trichoderma* and evaluated their efficacy against *M. phaseolina* incitant of stem rot of jute. The results revealed that *Trichoderma* isolate-6 was most effective in parasitizing growth of the pathogen.

Pan (2009) reported the antagonistic potential of *Trichoderma* isolates through production of volatile and non-volatile substances against *Macrophomina phaseolina*.

Choudhary *et al.* (2010) tested four bioagents viz, *Trichoderma viride*, *Trichoderma harzianum*, *Aspergillus versicolor* and *Bacillus firmus* for the control of dry root rot in mungbean. In dual cultures, *Trichoderma viride*, *Trichoderma harzianum* and *Aspergillus versicolor* were effective in inhibiting the growth of *Macrophomina phaseolina* to an extent of 61 to 65 per cent.

Gaur and Sharma (2010) tested nine local and exotic *Trichoderma* isolates against dry root rot pathogen (*Rhizoctonia bataticola*) of cotton under dual culture technique. Among them Coimbatore isolate of *Trichoderma*

viride-1 and local isolate of *T. harzianum* (TG-1) proved most effective that inhibited 76.53 and 72.78 per cent growth of *R. bataticola* respectively.

2.8.4 *In vitro* evaluation of the efficacy of bacterial antagonists against *R. bataticola*

The growth of *M. phaseolina* was significantly inhibited by *Bacillus subtilis* under *in vitro* conditions (Jharia and Khare, 1986).

Ishrat-Izar *et al.* (1995) reported that the growth promoting bacterium *Pseudomonas aeruginosa* (strains Pa 6 and Pa 12) significantly reduced infection by *Macrophomina phaseolina* and *Rhizoctonia solani* on chickpea.

Khot *et al.* (1996) reported that five rhizobacteria isolated from rhizosphere of chickpea plants inhibited the growth of *Fusarium oxysporum* f. sp. *ciceri* and *Rhizoctonia bataticola* under *in vitro*.

Jayasree *et al.* (2000) obtained ten strains of fluorescent Pseudomonads from rhizosphere of blackgram and studied their efficacy against *Macrophomina phaseolina* under *in vitro*. Among these strains, pf₁ was found to be more effective in inhibiting the mycelial growth of *Macrophomina phaseolina* of blackgram.

Sendhilvel (2005) reported that *Pseudomonas fluorescens* isolate SVPF2 was found to be effective in inhibiting the mycelial growth of *M. phaseolina* incitant of cowpea root rot.

Paramasivan *et al.* (2007) tested various bacterial antagonists against *Macrophomina phaseolina* the causal organism of dry root rot of Coleus and revealed that *Pseudomonas fluorescens* was most effective in inhibiting mycelial growth of pathogen.

Vinod Kumar *et al.* (2007) tested the efficacy of *Pseudomonas fluorescens* isolates against *M. phaseolina* the incitant of charcoal rot of

chickpea under *in vitro* and reported that the isolate pf 4-99 was found to be effective in inhibiting the mycelial growth of pathogen.

Khan and Gangopadhyay (2008) studied the efficacy of *Pseudomonas fluorescens* strains against *R. bataticola*, causal organism of chickpea dry root rot and reported that *P. fluorescens* strains PFBC-25 and 26 were effective in reducing the growth of the pathogen.

Uppala *et al.* (2009) isolated 46 bacterial endophytes from amaranth and screened against *Rhizoctonia solani* under *in vitro* conditions using dual culture method. Among the endophytes screened, only six bacteria showed antagonistic activity against *R. solani*.

2.9 IN VITRO EVALUATION OF THE COMPATIBILITY OF POTENTIAL ANTAGONISTS WITH DIFFERENT FUNGICIDES

A biocontrol agent must be effective and compatible with latest crop production practices so that its use can be integrated into the production system. In such an approach, biocontrol agents have been used without any toxic effect on antagonists (Papavizas and Lumsden, 1980).

2.9.1 Compatibility of potential fungal antagonists with fungicides

Akbari and Parakhia (2001) revealed that thiram, mancozeb and tridemorph were non-inhibitory to *T. harzianum* II, *T. viride* II and *Gliocladium virens* at all concentrations tested. Carbendazim had great inhibitory effect on antagonists.

Girija and Umamaheswaran (2003) reported the compatibility of *T. virens* with carbendazim under *in vitro* conditions at three concentrations (100, 150 and 1000 ppm) and observed that the antagonist *T. virens* was compatible with carbendazim at 100 ppm concentration.

Gupta (2004) reported that carbendazim completely inhibited the mycelial growth of *Trichoderma harzianum* at 1, 10, 100, and 1000 ppm concentrations.

Tiwari and Singh (2004) evaluated the *in vitro* efficacy of different fungicides against *T. harzianum* @ 1500 ppm and reported that the mycelial growth of *T. harzianum* was completely inhibited by carbendazim and hexaconazole @ 1500 ppm and the inhibition with copper oxychloride and mancozeb was 90 and 41 per cent respectively.

Vijayaraghavan and Abraham (2004) tested the *in vitro* compatibility of *T. harzianum*, *T. viride* and *T. longibrachiatum* with nine fungicides and found that mancozeb was compatible with all the three antagonists at 0.2, 0.3 and 0.4 per cent.

Gupta *et al.* (2005) reported that carbendazim was incompatible with *Trichoderma viride* TV2 isolate while carboxin was compatible for integrated treatment.

Naseema Beevi *et al.* (2005) tested the *in vitro* compatibility of *T. harzianum* with mancozeb, carbendazim and copper oxychloride and found that carbendazim at 0.1 per cent completely inhibited the mycelial growth while mancozeb and copper oxychloride showed compatibility with the antagonist at 0.2 and 0.1 per cent respectively.

Roberti *et al.* (2006) evaluated *in vitro* sensitivity of *Trichoderma* spp. against five fungicides and found that all antagonists showed low sensitivity to carboxin and thiram and high sensitivity to prochloraz.

Parab *et al.* (2009) studied the sensitivity of *T. harzianum* to fungicides and the results indicated that all systemic fungicides were found completely inhibiting growth of *T. harzianum* whereas zineb, copper hydroxide, mancozeb and copper oxychloride were found to be safe for the growth of

T. harzianum. The copper and sulphur fungicides were again screened by mycelial dry weight method, among all fungicides tested, zineb was found to be safe for *T. harzianum*.

Gaur and Sharma (2010) tested the *in vitro* compatibility of *Trichoderma harzianum* (TG-1) and *Trichoderma viride*-1 against different fungicides. Metalaxyl, fosetyl-Al, mancozeb, cymoxanil 8% + Mancozeb 64% mixture and copper oxychloride fungicides were found compatible with bioagent *T. harzianum* (TG-1) and tolerance limits (ED₅₀) of these fungicides were >1000 µg/ml. Metalaxyl proved compatible with another potential bioagent *T. viride*-1 also where even >1000 µg/ml concentration was under safe tolerance limit (ED₅₀). Copper oxychloride, mancozeb, fosetyl-Al and cymoxanil 8% + mancozeb 64% mixture fungicides showed moderate to good compatibility with *T. viride*-1 by exhibiting tolerance limits (ED₅₀) of 848, 710, 578 and 448 µg/ml respectively.

2.9.2 Compatibility of potential bacterial antagonists with fungicides

Vyas (1987) reported that *B. subtilis* showed high degree of tolerance to thiram and carbendazim.

Vidhyasekharan and Muthamilan (1995) reported that thiram and carbendazim were not inhibitory to *P. fluorescens* under *in vitro* conditions.

Niu *et al.* (1999) reported that *Pseudomonas fluorescens* strain 32 was not sensitive to carbendazim.

Dean *et al.* (2006) evaluated *in vitro* compatibility of the antagonistic bacterium *Bacillus subtilis* BSF4 strain with some fungicides and the results showed that it was compatible with azoxystrobin, penconazole, triadimenol and sulphur fungicides.

Khan and Gangopadhyay (2008) tested the *in vitro* sensitivity of *Pseudomonas fluorescens* towards different fungicides like carboxin,

chlorothalonil, carbendazim and captan. The results showed that carboxin, chlorothalonil and carbendazim were least toxic to *Pseudomonas fluorescens* strain PFBC-25, while captan was most inhibitory.

2.10 IN VITRO MODE OF ACTION OF POTENTIAL ANTAGONISTS AGAINST *Rhizoctonia bataticola*

2.10.1 Mode of action of *Trichoderma* isolate-7 (CT7) against *Rhizoctonia bataticola* under in vitro

Manoranjitham *et al.* (2000) reported the antagonistic activity of *Trichoderma* spp. and *Gliocladium virens* against a number of soil borne pathogens.

Similarly, Pant and Mukhopadhyay (2001) reported that production of knob like, haustorial structure with infection peg followed by penetration of host hyphae and finally coagulation, vacuolation and lysis of host protoplasm have been the principal mechanism in the mycoparasitism of *Rhizoctonia solani* by *Trichoderma virens*.

Pandey *et al.* (2005) reported that parasitism of *Rhizoctonia* by *Trichoderma* characteristically showed coiling, tight looping followed by penetration pegs and knob like haustoria within the hyphae.

Chakravarthy and Nagamani (2007) reported that the inhibition of *Rhizoctonia solani* by *Trichoderma* spp. was due to competition for food, space, production of antibiotics and mycoparasitism.

Paul *et al.* (2008) reported that swollen hyphae, suppressed growth, excessive vacuolation, hyphal apex deformation are the hyphal abnormalities in *Rhizoctonia solani* by *Trichoderma koningii*.

Manjunatha and Naik (2010) selected two isolates of *Trichoderma* sp, viz, *Trichoderma viride* and *Trichoderma harzianum* for intensive studies on

the basis of their *in vitro* antagonistic activity towards *Rhizoctonia bataticola*, the inducer of chickpea dry root rot. On microscopic examination, these fungal antagonists were found parasitic on *Rhizoctonia bataticola* as evidenced by coiling, penetration, disintegration and overlapping on germinating sclerotia as well as hyphae of the pathogen.

2.10.2 Effect of volatile and non volatile metabolites of potential antagonistic bacteria against *Rhizoctonia bataticola*

Prasanna Reddy *et al.* (2010) isolated eight *Pseudomonas fluorescens* strains from rhizosphere of rice seedlings collected from Andhra Pradesh and TamilNadu. Crude metabolites from one particular isolate of *Pseudomonas fluorescens* (P.f 003) gave 78 per cent inhibition against *Rhizoctonia solani* compared to control.

2.11 COMPATIBILITY BETWEEN POTENTIAL FUNGAL AND BACTERIAL ANTAGONISTS

Bacteria-fungal pathogen interactions are well documented, but very limited literature is available on the interaction of antagonistic bacterial and fungal agents. An important way of improving biocontrol in the rhizosphere is to add mixtures or combinations of biocontrol agents, particularly if they exhibit different or complementary modes of action or abilities to colonize root microsites (Mathare *et al.*, 1999).

Interaction between *Trichoderma* isolates and *Pseudomonas fluorescens* was studied by inoculating the antagonistic fungi 24 h prior to *Pseudomonas fluorescens* inoculation on KMB or PDA. The highest growth inhibition (59%) of *Trichoderma* (Th3) was noted when *Pseudomonas fluorescens* was first inoculated in KMB 24 h before inoculation of *Trichoderma harzianum* in the same medium. In contrast, the lowest inhibition (9%) was recorded when *Trichoderma viride* (Tv1) was first inoculated in KMB before inoculation of *Pseudomonas fluorescens*. All other

isolates of *Trichoderma* irrespective of their species exhibited intermediate growth inhibition. Pre-inoculation of *Pseudomonas fluorescens* in KMB always had higher inhibitory effect on *Trichoderma* than on pre-inoculation of *Trichoderma* irrespective of media. When the cell free cultural filtrate of *Pseudomonas fluorescens* was used at 5% against different species and isolates of *Trichoderma*, it also caused growth inhibition of fungal antagonist from 6 to 26% respectively (Pan and Das, 2010).

2.12 IN VITRO EVALUATION OF EFFICACY OF SEED GERMINATION BY BLOTTER SEED TECHNIQUE

Rangeshwaran *et al.* (2001) reported that seed treatment with bioagents and fungicide gave 100 per cent germination of chickpea seedlings under *in vitro* conditions in the absence of pathogen.

Nautiyal (1997) reported that seed bacterization with *Pseudomonas fluorescence* NBRI 1303 increased the seed germination of chickpea seedlings by 25 per cent.

2.13 MASS MULTIPLICATION OF *Rhizoctonia bataticola*

Sterilized sorghum grains (Rajeswari *et al.*, 1999 and Durai, 2004), sand maize medium (Sajeena *et al.*, 2004 and Katariya *et al.*, 2007) and potato dextrose broth (Gaur *et al.*, 2005) were used for mass multiplication of *Rhizoctonia bataticola*.

Ram and Pandey (2010) reported that sand maize medium was used for mass multiplication of *Fusarium solani* incitant of black root rot of chickpea.

Rhizoctonia solani was mass cultured in autoclaved wheat bran + vermiculite (1:1) mixture containing 10% glucose (Rangeshwaran *et al.*, 2008).

2.14 IN VIVO EVALUATION OF EFFICACY OF FUNGICIDES AGAINST *Rhizoctonia bataticola*

Raut and Somani (1987) reported that seed treatment with captan (0.2%) and thiram (0.2%) were found to be effective against chickpea dry root rot.

The results of Taya *et al.* (1990) indicated that carbendazim alone or in combination with thiram performed well as seed treatment, pre-sowing soil drench and seed treatment along with drenching after sowing in the control of dry root rot of chickpea.

Sharma and Tripathi (2001) reported the efficacy of carbendazim (0.2%), propiconazole (0.1%), hexaconazole (0.1%) and propineb (0.25%) against *M. phaseolina*. Seed treatment and foliar sprays of propiconazole at 15 days interval resulted in maximum reduction in disease severity (30-32%) and increase in grain yield (950-1012 Kg/ha) and 1000 grain weight (35g).

Prajapati *et al.* (2003) reported that seed treatment with carbendazim gave good control efficacy against *Macrophomina* root rot of chickpea.

Bagri *et al.* (2004) isolated eight fungal species belonging to five genera from chickpea seeds by blotter and agar plate methods. *Macrophomina phaseolina* was effectively controlled by seed treatment with thiram. Amongst the fungicides tested, thiram gave the highest seed germination (89%) and minimum pre and post emergence mortality (6 and 3% respectively).

Sharma and Gupta (2004) reported that seed treatment with carbendazim at 1g kg⁻¹ against dry root rot in chickpea has given minimum disease incidence and highest grain yield.

Vijay Mohan *et al.* (2006) used carbendazim (0.2%) and Etaconazole (0.1%) as seed treatment, soil drenching and seed treatment plus soil

drenching against chickpea root rot incidence and recorded lowest disease incidences of 15.60 and 18.20 per cent and highest grain yield of 19.20 and 18.90 q/ha respectively.

Kaushal (2008) reported that seed treatment with thiram and carbendazim at 2g/kg was effective in control of chickpea dry root rot.

Khan and Gangopadhyay (2008) reported that seed treatment with carbendazim (58.6%) and captan (55.2%) provided a better disease control as compared to carboxin (51.7%) and chlorothalonil (49.8%) against root rot of chickpea.

Bag *et al.* (2009) studied the bio-efficacy of a new fungicide Lustre 37.5 SE, a combination of flusilazole (12.5%) and carbendazim (25%) evaluated along with propiconazole, hexaconazole and thiafluzamide against the sheath blight of rice under field conditions. The results revealed that lustre to be the best treatment with minimum disease incidence (38.3%) and highest yield/ha of 4.06t/ha.

Among the different treatments seed treatment with carbendazim showed its superiority in reducing root rot disease incidence (18%) and maximum yield (859 kg ha⁻¹) in cluster bean (Jaiman *et al.*, 2009).

2.15 MASS MULTIPLICATION OF POTENTIAL ANTAGONISTS

2.15.1 Mass Multiplication of Fungal Antagonists

Shamarao *et al.* (1998) tested ten different substrates i.e. oil cakes, FYM, wheat bran, poultry manure, dung, jaggery, groundnut cake, neem cake, niger cake and pongamia cake for the mass multiplication of *T. viride* and found that wheat bran was the most suitable substrate with a spore load of 215.03×10^4 cfu/g followed by FYM (193.33×10^4) and cow dung 115.0×10^4 cfu/g, respectively.

Gaur *et al.* (2005) multiplied the antagonist of pathogen causing dry root-rot of chick pea on faecal pellets of goat and sheep, gram shell, FYM, wheat bran, peat soil, sorghum grains, sugarcane bagasse and sodium alginate granules. Out of these, faecal pellets of goat and sheep, gram shell (decomposed) and FYM were considered as the best substrates for multiplication with a population of 71×10^7 cfu/g.

Rini and Sulochana (2007) evaluated different substrates like cow dung, neem cake, coir pith, sorghum grains, saw dust and rice bran. Of these substrates, they found that sorghum grains as highest multiplying substrate for both *T. harzianum* and *T. viride*.

The mass multiplication of *Trichoderma* on solid substrate promoted the synthesis of the desired enzymes which helps in its biocontrol mechanism (Singh *et al.*, 2007).

Among the substrates, vermicompost appeared to be the best nutrient source to support the antagonist for maximum multiplication (Bora and Bora, 2008).

2.15.2 Mass Multiplication of Bacterial Antagonists

Umamaheswari *et al.* (2002) multiplied *P. fluorescens* and *Bacillus subtilis* in King's B and nutrient broth medium respectively (Gogoi *et al.*, 2002).

Zaidi and Singh (2004) tested three substrates for multiplication of *Pseudomonas fluorescens* (PBAD-27). The results revealed that the population of *P. fluorescens* was more in cow dung (42.6×10^{12}) and least in press mud (1.6×10^6).

Diby Paul *et al.* (2005) multiplied *P. fluorescens* in nutrient broth at 28°C for 48 h.

Muthukumar and Bhaskaran (2007) multiplied *P. fluorescens* in king's B broth at room temperature ($28 \pm 2^{\circ}\text{C}$) for three days (Bora and Bora, 2008).

Rangeshwaran *et al.* (2008) multiplied endophytic bacteria in Tryptic Soya Broth (TSB) in a shaker at 150 rpm for 48h.

2.16 IN VIVO EVALUATION OF POTENTIAL ANTAGONISTS AGAINST *Rhizoctonia bataticola*

2.16.1 *In vivo* Evaluation of Fungal Antagonists against *Rhizoctonia bataticola*

Deb and Dutta (1991) reported that *T. harzianum* and *T. koningii* respectively gave 45 and 38 per cent control of root rot disease in soybean caused by *S. rolfii*.

Kheri and Chandra (1991) studied the efficacy of *Trichoderma viride* in controlling the pathogenic activity of *R. bataticola* causing dry root rot of chickpea. The antagonist applied as seed coating reduced mortality of chickpea from 8 to 9 per cent in unsterilized soils under greenhouse conditions and the biocontrol efficacy of the antagonist increased in sterilized soils.

Angappan (1992) found that seed and soil application of *T. hamatum* reduced root rot incidence by 90 per cent and increased yield by 115 per cent over control in chick pea.

Prasad *et al.* (2002) evaluated two antagonistic fungi *Trichoderma harzianum* (PDBCH 10) and *T. viride* (PDBCV) against wilt and root rot of chickpea under pot culture. The results revealed that soil application of *Trichoderma harzianum* showed only 4.9 and 1.2 per cent root rot incidence at 30 and 60 days respectively.

Gaur *et al.* (2005) revealed that soil application of wheat bran based *T. harzianum* (@15g/3kg of soil) and FYM (100g) mixture at the time of sowing was found to be significantly superior in reducing root rot of chickpea by 76 per cent and enhancing overall germination (35%) compared to control.

Prajapati *et al.* (2006) reported that seed treatment with *Trichoderma harzianum* reduced dry root rot incidence from 20.3 to 16.3 per cent in chickpea.

Jaiman *et al.* (2009) reported that seed treatment with *Trichoderma viride* resulted in minimum root rot disease incidence (25%) of cluster bean.

Srivastava *et al.* (2010) reported that the treatment with soil application of *Trichoderma viride* thrice, i.e. 7, 15, 30 DAS was found best in controlling jute *Macrophomina* root rot giving minimum per cent disease incidence (4.92) as compared to control (16.34) and highest mean dry fibre yield of 25.70 q ha⁻¹ against control (17.76 q ha⁻¹).

2.16.2 In Vivo Evaluation of Bacterial Antagonists against *Rhizoctonia bataticola*

Seed coating with *Bacillus subtilis* inhibited dry root rot incidence by 52 per cent (Saxena and Saxena, 1995), but according to Khot *et al.* (1996) seed inoculation with *Pseudomonas fluorescens* resulted in significant control of root rot due to *M. phaseolina* in chickpea under field conditions. Similarly Rangeshwaran *et al.* (2001) studied the effect of seed treatment with two bacterial antagonists viz., *Pseudomonas putida* and *P. fluorescens* in control of root rot of chickpea caused by *Rhizoctonia* sp. and results indicated that seed treatment with both the antagonists significantly reduced root rot incidence of chickpea when compared to control. Among the two bacterial antagonists *P. fluorescens* was found to be highly effective in reducing the incidence by 6.4 per cent compared to *P. putida*. Seed treatment with these two biocontrol agents also resulted in significant increase in yield. Khan and

Gangopadhyay (2008) also reported that seed treatment with *Pseudomonas fluorescens* isolates PFBC-25 and PFBC-26 significantly reduced the root rot incidence of chickpea.

Nautiyal (1997) reported that chickpea seed bacterization with *Pseudomonas fluorescens* NBRI1303 increased the germination of seedlings by 25 per cent and reduced the number of diseased plants to 45 per cent.

Rajendar singh *et al.* (1998) reported that *Pseudomonas fluorescens* was found effective in reducing the incidence of dry root rot of chickpea caused by *Rhizoctonia bataticola*.

Mondal and Bhattacharya (2003) studied the efficacy of bacterial isolates of *Bacillus* sp. in controlling the pathogenic activity of *Macrophomina phaseolina* causing dry root rot of chickpea and reported that the two bacterial isolates (S12 and S17) when applied as both seed and soil drench reduced the root rot incidence of chickpea under pot culture studies.

Ramteke *et al.* (2004) reported that the phosphate solubilizing ability of endophytic bacteria in sugarcane and importance of endophytic bacteria in control of wilt pathogens.

2.17 ORGANIC AMENDMENTS

Ushamalini *et al.* (1997) reported the soil application of neem cake @ 150 kg/ha and FYM @ 10t/ha reduced the charcoal rot incidence in cowpea.

The recovery of *Macrophomina phaseolina* propagules was the least in soil amended with mustard cake or wheat straw (Sharma, 2001).

Lodha *et al.* (2002) observed that the population density of *Macrophomina phaseolina* decreased in soil amended with compost.

Soil amendment with FYM led to increased disease control efficacy of fungal antagonist like *Trichoderma* spp. against *Fusarium* wilt of cumin (Gopal, 2002).

Kadam and Raut (2006) reported that the use of organic matter had adverse effect on *Fusarium oxysporum* f.sp. *ciceri* growth.

Trivedi and Rathi (2008) studied the effect of soil amendments like FYM, vermicompost, poultry manure on seedling emergence and mortality in chickpea wilt and reported that FYM was found most effective by increased seedling emergence (78.6) and reduced seedling mortality (42.0).

2.18 INTEGRATED DISEASE MANAGEMENT

Rajeswari (1997) reported that dry root rot of mungbean caused by *Macrophomina phaseolina* was effectively controlled by integration of *Trichoderma harzianum* as soil application (5 g/kg) + seed treatment (10^8 cfu/ml) with sub-lethal doses of carbendazim (0.02%) that significantly reduced the dry root rot incidence (95.3 %) over soil treatment (91.5%).

Venkateswar Rao (1998) reported that an integration of soil solarization (six weeks), *Trichoderma viride* (native) and mancozeb (0.3%) effectively controlled the stem and root rot of sesame to the tune of 82.2 per cent caused by *Macrophomina phaseolina* under field conditions.

Prasanthi *et al.* (2000) evaluated fungal and bacterial antagonists as seed and soil application against safflower root rot caused by *R. bataticola*. Both seed treatment and soil drenching with antagonists increased safflower seedling percentage survival with seed treatment more effective than soil drenching recording highest survival rate 83.33 per cent with *T. viride* and 86.66 per cent with *Pseudomonas fluorescens*.

Sindhan *et al.* (2002) reported satisfactory disease control when *Pseudomonas fluorescens* was used as seed treatment along with carbendazim against *Macrophomina phaseolina* in chickpea.

Many of the workers reported that soil application of *Trichoderma* spp and *Pseudomonas fluorescens* effectively controlled the diseases caused by soil borne plant pathogens and also a synergistic effect in the growth of plants was observed (Bharati *et al.*, 2004 and Poddar *et al.*, 2004)

Gupta (2006) studied the effect of seed treatment with fungicides, bioagents on disease incidence of chickpea. Combined seed treatment with *Trichoderma harzianum* and thiram were found to be effective with increased seedling emergence and reduced wilt incidence.

Prajapati *et al.* (2006) studied the effect of varieties and date of sowing on disease development. The results showed that late sown crop (15th November) significantly reduced incidence of dry root rot in all chickpea varieties. The chickpea variety Radhey showed significantly less disease (10.6%), compared to 11.8 and 17.1 per cent in JG-9 and Vijay respectively in 15th November sown crop.

Singh and Sinha (2007) evaluated field efficacy of foliar application of different formulations of *P. fluorescens* (pfr₁ and pfr₅) at three doses viz., 2, 4 and 8 g/l against sheath blight of rice. A higher dose (8g/l) was found highly effective in reducing disease severity (57%), incidence (31%) and increasing grain yield (32%) and 1000 grain weight (13%).

Rangeshwaran *et al.* (2008) isolated five endophytic bacteria in chickpea and tested their efficacy in plant growth promotion along with rhizospheric bacteria, among which *Bacillus megaterium* showed highest germination (100%) and vigour index (4116).

Harlapur and Wali (2009) studied the effect of seed treatment with fungicides thiram, carbendazim, and bioagents *Trichoderma harzianum* Sk₁, phosphate solubilizing bacteria (PSB) Acd-15 alone and combined seed treatment with *Trichoderma harzianum* Sk₁, phosphate solubilizing bacteria (PSB) Acd-15 against charcoal stalk rot of maize caused by *Macrophomina phaseolina*. The results revealed that combined seed treatment with *Trichoderma harzianum* Sk₁, phosphate solubilizing bacteria (PSB) Acd-15 is significantly superior over other treatments, which recorded lower disease incidence (9.64%) and highest grain yield (56.97 q/ha).

Choudhary *et al.* (2010) tested four bioagents viz, *Trichoderma viride*, *Trichoderma harzianum*, *Aspergillus versicolor* and *Bacillus firmus* for the control of dry root rot in mungbean. In pot experiments, when *Trichoderma viride* or *Trichoderma harzianum* was used singly as soil application, the disease incidence was reduced by more than 50 per cent in plants compared to control. A combination of soil application of biocontrol agents *Trichoderma viride* and *Trichoderma harzianum* reduced dry root rot by 23 per cent, while another combination of *Trichoderma harzianum* and *Aspergillus versicolor* resulted in 31 per cent incidence.

2.19 ESTIMATION OF POPULATION LEVELS OF PATHOGEN, FUNGAL AND BACTERIAL ANTAGONISTS

2.19.1 Population dynamics of potential fungal isolate *Trichoderma* isolate-7 (CT7)

Gaur *et al.* (2005) reported that the initial mean population of *Trichoderma* in talc based formulation stored at 25°C on 0 days of storage was 270×10^6 cfu/g and it declined to 20×10^6 cfu/g after 105 days of storage indicating a reduction of 93 per cent.

Paul *et al.* (2008) studied the effect of organic amendments on population dynamics of *Trichoderma harzianum* (JMA-4) and revealed that

high population 103×10^6 cfu/g soil was obtained in soil amended with *Eupatorium odoratum*.

Choudhary *et al.* (2010) reported that the mean population densities of *Trichoderma harzianum* (10.20×10^6) recorded minimum dry root rot incidence (16.67%) on mungbean when compared to control (66.67%).

2.19.2 Population dynamics of potential bacterial isolate CREB-16

Zaidi and Singh (2004) tested three substrates for multiplication of *Pseudomonas fluorescens* (PBAD-27). The results revealed that the population of *P. fluorescens* was more in cow dung (42.6×10^{12}) and least in press mud (1.6×10^6).

Khan and Gangopadhyay (2008) reported that the population of *Pseudomonas fluorescens* in FYM amended soil were increased from 17.3 to 28.7×10^{10} cfu/g soil with the increase in FYM dose from 5 g to 20 g/kg soil.

Choudhary *et al.* (2010) reported that the mean population densities of *Bacillus firmus* (22.04×10^8) recorded minimum dry root rot incidence (33.33%) on mungbean when compared to control (58.33%).

2.19.3 Effect of fungicide, biocontrol agents and organic amendments on population levels of *Rhizoctonia bataticola*

Mallaiah (2002) reported that combined seed treatment with *T. viride*, *P. fluorescens*, neem oil and thiram gave maximum reduction in population levels of *Rhizoctonia bataticola* (90.00%) in green gram crop under pot culture.

Choudhary *et al.* (2010) reported that high levels of suppression of *Macrophomina phaseolina* (1.10×10^3) in combination treatment of *Trichoderma harzianum* and *Bacillus firmus* at the mean population

densities of 10.14×10^6 and 23.10×10^8 respectively, when compared to control (1.60×10^3).

2.20 MOLECULAR CHARACTERIZATION OF POTENTIAL BIOCONTROL AGENTS

Ramesh Kumar *et al.* (2002) studied the genetic variability among the isolates of *Pseudomonas* by RAPD with random primers, and the primer pgs3 produced several bands, including a unique band with size of 800 bp.

Daffonchio *et al.* (2004) used ITS between 16S and 23S rRNA genes to discriminate group 1 of the genus *Bacillus* by Polymerase Chain Reaction.

Reddy *et al.* (2004) characterized the most efficient strains of *Pseudomonas fluorescens* by RAPD analysis.

Marta *et al.* (2005) studied RAPD and unspecific PCR fingerprints to differentiate *Pseudomonas fluorescens* EPS62e from other *Pseudomonas fluorescens* strain. Differential amplified fragment from EPS62e were sequence characterized as SCAR markers and two primer pairs were designed and selected for their specificity against EPS62e. A SCAR primer pair was evaluated and validated for the assessment of population dynamics of EPS62e on pear plant under greenhouse condition.

Misbah *et al.* (2005) identified the *Acinetobacter* of clinical isolates by amplification of 16S r DNA genome region consisting of approximately 1500 nucleotides using three primer pairs viz., 27F, 780R; 529F, 1099R; 925F and 1491R.

Megha *et al.* (2007) studied the diversity of fifteen isolates of fluorescent pseudomonads using RAPD - PCR with eight random primers viz., OPC-9, OPD-2, OPD-3, OPO-6, OPO-09, OPO-13, 15 and 16. The PCR amplicons of fluorescent pseudomonads obtained from eight random primers produced 127 polymorphic bands. The minimum bands (9) were produced by

the primer OPD-02 and maximum number of bands (25) were produced by OPO-16.

The results of RAPD analysis revealed the genetic diversity among the isolates of *Sclerotium rolfsii* causing stem rot of groundnut (Durga Prasad *et al.*, 2010).

Gayathri *et al.* (2010) studied the polymorphism among the antagonistic seed and root endophytes against *Sclerotium rolfsii*. These results revealed that the isolates were grouped into two clusters with 133 reproducible and scorable polymorphic bands. Moreover, a primer OPD-2 yielded a unique band of 2600 bp in potential antagonists GSE-3 and GSE-4 which can be used to develop SCAR marker.

Polymorphism among *Trichoderma* isolates were observed by RAPD technique. A total of 119 polymorphic bands ranging approximately as low as 75 bp to as high as 2500 bp were generated with 5 random primers among 9 *Trichoderma* isolates (Mathews *et al.*, 2010).

Chapter ~ III

Material & Methods

Chapter - III

MATERIALS AND METHODS

3.1 LOCATION OF WORK

The laboratory and pot culture experiments pertaining to the present research work were conducted during the year 2011-12 in the Department of Plant Pathology, S.V. Agricultural College, Acharya N.G. Ranga Agricultural University, Tirupati, Chittoor District, Andhra Pradesh.

3.2.1 Glassware

Glassware made of corning or borosil were used throughout the present investigation. They were Petri plates (90 mm dia), test tubes, conical flasks (100, 250 and 500 ml), beakers (100, 500 and 1000 ml), pipettes (1, 2, 5 and 10 ml) and measuring cylinders (10, 50, 100 and 500 ml). The glassware was first washed with detergent followed by thorough cleaning with tap water before placing them in cleaning solution for 24 h and finally rinsed with distilled water for 3 to 4 times and were air dried before use.

The composition of cleaning solution

Potassium dichromate	:	60 g
Concentrated sulphuric acid	:	60 ml
Distilled water	:	1000 ml

3.2.2 Chemicals

Chemicals used in the present study were of analytical reagent (AR) and guaranteed reagent (GR) grades of standard mark. The pH of the media was adjusted using either 0.1 N HCl or 0.1 N NaOH.

3.2.3 Equipments

Compound microscope (10x, 40x magnifications) was used for observing the fungi. Weighments were done on a single pan electronic balance (Afcoset, FX-200) with a sensitivity of 0.001g. Hot air oven and autoclave were used for sterilization of glassware and media respectively. Incubators were used for incubating test materials at different temperatures. The cultures were stored in refrigerator. The sclerotia and mycelium of *Rhizoctonia bataticola* were observed by using microscope with built in camera attachment (Motic Camera, USA). Other tools which were used in the present investigation for various purposes include camel brush, inoculation needle, inoculation loop, cork borer, slides, cover slips, plastic pots etc.

3.2.4 For molecular study

DNA amplifications in RAPD and 16S rDNA were carried out in Gradient PCR master cycler (Eppendorff) and DNA banding patterns were documented in gel documentation system. Horizontal gel electrophoresis for gel running and UV transilluminator for observing the bands were used. Centrifugation was carried out in refrigerated eppendorf centrifuge. DNA samples were stored at -20°C in deep freezer. Sterile pestle and mortar, liquid nitrogen, eppendorf tubes (1.5 ml), PCR tubes (0.2 ml), tips (10, 50, 200 and 1000 µl) and micro pipettes were used.

3.2.5 Culture media used

i. Potato Dextrose Agar (PDA) Medium (Ainsworth, 1961)

Peeled potato slices	: 200 g
Dextrose	: 20 g
Agar	: 20 g
Distilled water	: 1000 ml
pH	: 7.0

ii. Nutrient Agar (NA) Medium (Tuite, 1969)

Peptone	: 5 g
Beef extract	: 3 g
Agar	: 20 g
Distilled water	: 1000 ml
pH	: 7.0

iii. Potato Dextrose Broth (PDB)

Peeled potato slices	: 200 g
Dextrose	: 20 g
Distilled water	: 1000 ml
pH	: 7.0

iv. Nutrient Broth (NB)

Peptone	: 5 g
Beef extract	: 3 g
Distilled water	: 1000 ml
pH	: 7.0

v. Rose Bengal Agar (RBA) Medium (Martin, 1950)

Glucose	: 10 g
Peptone	: 2 g
K ₂ HPO ₄	: 1 g
MgSO ₄ .7H ₂ O	: 0.5 g
Rose Bengal	: 0.3 g
Agar	: 20 g
Distilled water	: 1000 ml
pH	: 6.0

vi. *Trichoderma* Selective Medium (TSM) (Elad and Chet, 1983)

Magnesium sulphate (MgSO ₄)	: 0.2 g
Di-Potassium Hydrogen Phosphate (K ₂ HPO ₄)	: 0.9 g
Potassium Chloride (KCl)	: 0.15 g
Ammonium Nitrate (NH ₄ NO ₃)	: 1.0 g
Glucose	: 3.0 g
Rose Bengal	: 0.15 g
Penta Chloro Nitro Benzene (PCNB)	: 0.2 g
Chloramphenicol	: 0.25 g
Metalaxyl	: 0.3 g
Agar	: 15.0 g
Distilled water	: 1000ml
pH	: 6.5

vii. *Rhizoctonia bataticola* Selective Medium:

Chloroneb Mercury Rose Bengal Agar Medium (CMRA) (Meyer et al., 1973)

Polished rice	: 10 g
Chloroneb	: 300 mg
Mercuric chloride (Hgcl ₂)	: 7 mg
Rose Bengal	: 90 mg
Streptomycin sulphate	: 40 mg
Agar Agar	: 20 g
Distilled water	: 1000ml
pH	: 7.0

3.2.6 Sterilization

Glassware used for the present investigation were kept in sterilization tins or wrapped in aluminum foil and were sterilized in hot air oven at 160°C for 90 minutes.

Surface of laminar air flow (LAF) was sterilized by wiping with cotton swab dipped in alcohol. Inoculation loop, cork borer and scalpel were sterilized by dipping in alcohol and heating to red hot.

The culture media and distilled water were sterilized in an autoclave at 15 p.s.i. for 20 minutes.

The soil required for pot culture experiment was sterilized in autoclave at 20 p.s.i. for 30 minutes for two consecutive days.

3.2.7 Laboratory techniques

The general laboratory techniques described by Dhingra and Sinclair (1995), Rangaswamy and Mahadevan (1999), Nene and Thapliyal (1993) and Aneja (1993) were followed for preparation of media, sterilization, isolation and maintenance of fungal cultures with slight modifications wherever necessary.

3.2.8 Source of seed

Chickpea seeds of variety JG-11 popularly cultivated in Kadapa and Kurnool districts of Andhra Pradesh were obtained from Regional Agricultural Research Station, Nandyal and used for pot culture studies.

3.3 ISOLATION OF PATHOGEN

The pathogen was isolated from dry root rot infected chickpea plants by using tissue segment method (Rangaswamy and Mahadevan, 1999).

Small pieces of tissue from the infected collar region along with some healthy tissue were cut with sterile scalpel. Then the pieces were surface sterilized with 1 per cent sodium hypochlorite for 2 minutes, followed by three washings in sterile distilled water to remove traces of sodium hypochlorite on the bits of tissue. The sterilized pieces were transferred to PDA plated Petri plates. Plates were incubated at $28 \pm 2^{\circ}\text{C}$ and observed periodically for growth of the fungus. Alternatively, the soil from infected area was used for the isolation of pathogen by serial dilution method.

The pathogen culture was purified by single hyphal tip method and maintained on PDA by periodical transfer throughout the present investigation.

3.3.1 Identification of pathogen

The pathogen was identified based on its mycelial and sclerotial characters as described by Barnett and Barry (1972).

3.3.2 Pathogenicity test

Pathogenicity test was proved by soil infestation method. The pathogen was mass multiplied on sterilized sorghum grains in 250 ml conical flasks. The flasks containing sorghum seeds were autoclaved at 15 p.s.i for 20 min. Then the flasks were inoculated with 4 discs of 5.0 mm diameter mycelial growth of three days old culture of *Rhizoctonia bataticola* grown on PDA plate. The flasks were incubated at $28 \pm 2^{\circ}\text{C}$ for seven days. Then the inoculum was mixed with sterilized soil @ 100 g kg^{-1} soil and filled in the pots (22.5 cm diameter) were filled. The seeds of chickpea were sown simultaneously with pathogen inoculation @ 10 seeds per pot and an

uninoculated control was maintained. The plants were observed for root rot symptoms. Each treatment replicated three times.

3.4 *IN VITRO* EVALUATION OF EFFICACY OF FUNGICIDES AGAINST THE PATHOGEN

In vitro efficacy of fungicides against the pathogen was evaluated by poisoned food technique (Nene and Thapliyal, 1993). The list of fungicides used in the present studies are given below:

S. No.	Common name	Trade name	Active ingredient	Concentration (%)	Source of supply
1.	Copper oxychloride	Blitox	50% WP	0.25	Rallis India Ltd., Mumbai
2.	Captan	Captaf	50% WP	0.25	Rallis India Ltd., Mumbai
3.	Hexaconazole	Contaf	5% EC	0.20	Hyderabad Chemical Supplies Ltd., Hyderabad
4.	Tebuconazole	Folicur	25.9% EC	0.10	Bayer Crop Science Ltd., Mumbai
5.	Validamycin	Sheathmar	3% L	0.10	Dhanuka Agritech Ltd., Mewat

To 50 ml of sterilized distilled water, required quantity of fungicide was added and mixed thoroughly. This solution was added to 50 ml of sterilized cool molten double strength PDA medium, mixed thoroughly and poured into Petri plates. Six mm discs of four days old culture of pathogen were inoculated at the centre of Petri plates and then incubated at $28 \pm 2^{\circ}\text{C}$. Three replications were maintained for each fungicide. Medium without fungicide was kept as control. Per cent inhibition of the growth of the fungus over the control was calculated using the formula:

$$I = \frac{C - T}{C} \times 100$$

where,

I = Per cent inhibition in growth of test pathogen

C = Radial growth (mm) in control

T = Radial growth (mm) in treatment.

3.5 ISOLATION OF POTENTIAL BIOCONTROL AGENTS

3.5.1 Isolation of native antagonistic mycoflora and bacteria from rhizosphere

Antagonistic mycoflora and bacteria were isolated by following serial dilution technique (Johnson and Curl, 1977). Composite soil sample was collected from rhizosphere of healthy plants. The soil was dried under shade and then used for serial dilution. To get 10^{-1} dilution, ten gram of this soil was dissolved in 90 ml of sterile distilled water. From this one ml of soil suspension was taken and added to nine ml of sterile distilled water to get 10^{-2} dilution. This was repeated until a final dilution of 10^{-4} for isolation of fungi and 10^{-6} for bacteria.

Antagonistic mycoflora were isolated on Rose Bengal Agar medium by using a dilution of 10^{-4} . Antagonistic bacteria were isolated on nutrient agar medium by using a dilution of 10^{-6} . One ml of final dilution of soil suspension was poured into sterilized Petri plates, and then the melted and cooled media was poured. Plates were rotated gently on the laminar air flow bench to get uniform distribution of soil suspension in the medium. Then the plates were incubated at $28 \pm 2^{\circ}\text{C}$ and observed at frequent intervals for the development of colonies.

Three days old colonies of mycoflora were picked up and purified by single spore/hyphal tip method, whereas one day old colonies of bacteria were picked up and purified by streak plate method.

3.5.2 Isolation of native antagonistic mycoflora and bacteria from root endophytes

For isolation of endophytes, five g of root was surface sterilized for 5 min with 70 per cent ethanol and homogenized in 20 ml of sterilized phosphate buffer (0.2M Na₂HPO₄ + 0.2M NaH₂PO₄) pH 7.0 using mortar and pestle. Appropriate dilutions (10⁻⁴ for fungi and 10⁻⁶ for bacteria) of these suspensions were plated on RBA/PDA and NA for the isolation of *Trichoderma* / fungi and bacteria respectively. The plates were incubated for 72 h at 28 ± 2°C (Kishore *et al.*, 2005).

Three days old colonies of mycoflora were picked and purified by single spore/hyphal tip method, whereas one day old colonies of bacteria were picked up and purified by streak plate method.

3.5.3 Identification of potential biocontrol agents

The efficacy of antagonistic mycoflora and bacteria from rhizosphere and root endophytes was determined by dual culture technique (Morton and Stroube, 1995) under *in vitro* conditions.

3.5.3.1 Dual culture technique

To test the efficacy of antagonistic fungus, twenty ml of sterilized melted PDA was plated in Petri plates (9 cm) and allowed to solidify. Mycelial discs measuring six mm diameter from three day old cultures of both fungal antagonist and the test pathogen were placed at equidistant on sterile Petri plate containing PDA medium.

To test the efficacy of antagonistic bacterium a 4 cm line was streaked at one side of the plate. On the opposite side to the antagonist, mycelial disc measuring six mm diameter from four day old culture of test pathogen was placed on sterile Petri plate containing PDA medium.

The petri plates with pathogen inoculated at one end alone, served as control. The petri plates were then incubated at $28 \pm 2^{\circ}\text{C}$. Three replications were maintained in each treatment. Growth of antagonists, pathogen and zone of inhibition were measured after recording full growth of the pathogen in control plate. Per cent inhibition of mycelial growth of test pathogen was calculated by the formula:

$$I = \frac{C - T}{C} \times 100$$

where,

I = Per cent inhibition in growth of test pathogen

C = Radial growth (mm) in control

T = Radial growth (mm) in treatment.

3.6 IDENTIFICATION OF COMPATIBILITY OF POTENTIAL BIOCONTROL AGENTS WITH FUNGICIDES *IN VITRO*

Potential biocontrol agents were tested for their compatibility with the fungicides *viz.*, copper oxychloride (0.25%), captan (0.25%), hexaconazole (0.2%), tebuconazole (0.1%) and validamycin (0.1%). The compatibility of antagonistic fungi with different fungicides were tested by poisoned food technique as described in section 3.4 (Nene and Thapliyal, 1993). The compatibility of antagonistic bacterial isolates with the fungicides were tested by spectrophotometric method (Kishore et al., 2005) under *in vitro* conditions.

3.6.1 Spectrophotometric method

Five hundred microlitres of antagonistic bacterial cultures grown in Nutrient Broth (NB) for 16 h at $28 \pm 2^{\circ}\text{C}$ and 150 rpm were added to 50 ml of NB in 250 ml flasks containing different fungicides. Inoculated flasks were incubated at $28 \pm 2^{\circ}\text{C}$ and at 150 rpm on orbital shaker. Bacterial growth was

determined in systronic spectrophotometer at 600 nm after 24 hours of incubation. Each treatment was replicated thrice.

The flasks containing nutrient broth without fungicides was kept as control (Kishore *et al.*, 2005).

3.7 MODE OF ACTION OF POTENTIAL BIOCONTROL AGENTS

3.7.1 Mode of action of *Trichoderma in vitro*

Hyphal interactions between *Rhizoctonia bataticola* and the antagonistic *Trichoderma* was observed from dual culture. Mycelial mats were lifted gently from the zone of interaction in dual culture plates with the help of a needle and kept in a drop of cotton blue with lactophenol on a microscopic slide and was observed under microscope (Pant and Mukhopadhyay, 2001).

3.7.2 Mode of action of antagonistic bacteria on *Rhizoctonia bataticola* under *in vitro*

The following tests were performed to study the mode of action of antagonistic bacteria under *in vitro* conditions.

3.7.2.1 Effect of volatile metabolites of bacterial antagonist on radial growth of *Rhizoctonia bataticola*

Modified version of sealed Petri plate technique described by Dennis and Webster (1971) was used to study the influence of volatile substances on the growth of *Rhizoctonia bataticola*.

In this case, the bacterial antagonist from 48 h old culture was streaked at the centre of the petri plate containing nutrient agar and the lid of this Petri plate was replaced by another bottom plate containing PDA medium inoculated with six mm actively growing culture disc of *Rhizoctonia bataticola*. The two dishes were sealed together with an adhesive tape. The lid of control plate, which had not been inoculated with antagonist served as a

control. Three replications were maintained and the plates were incubated at $28 \pm 2^{\circ}\text{C}$. The radial growth of the pathogen was recorded after every 24 h and compared with the growth of pathogen in control. Based on the observations recorded, percentage inhibition of the pathogen was calculated by the formula as given in section 3.5.3.1.

3.7.2.2 Effect of non-volatile metabolites of bacterial antagonist on radial growth of *Rhizoctonia bataticola*

The effect of culture filtrate of bacterial antagonists on the growth of *Rhizoctonia bataticola* was studied as per the method given by Dennis and Webster (1971).

Filtered and sterilized culture filtrate of the bacterial antagonist grown for two days on Nutrient broth was added to molten PDA and the medium was poured into Petri plates and plates were inoculated with six mm disc of *Rhizoctonia bataticola*. The PDA plates inoculated with the pathogen but not amended with culture filtrate served as control. Observations on the radial growth of test pathogen were recorded after seven days of incubation at $28 \pm 2^{\circ}\text{C}$. The radial growth of the test fungus in the treatment in comparison with that of control was recorded and the per cent inhibition was calculated by the formula given in section 3.5.3.1.

3.8 IDENTIFICATION OF COMPATIBILITY BETWEEN POTENTIAL FUNGAL AND BACTERIAL ANTAGONISTS *IN VITRO*

The compatibility between potential fungal and bacterial antagonists was determined by dual culture technique (Morton and Stroube, 1995) maintaining separate controls for bacteria and fungus as described in section 3.5.3.1 under *in vitro* conditions.

3.9 INTEGRATED MANAGEMENT OF DRY ROOT ROT OF CHIKPEA UNDER POT CULTURE

3.9.1 Pot culture studies

This experiment was conducted under greenhouse conditions. In this experiment, the potential fungicidal compatible antagonists along with effective fungicides and different organic amendments was evaluated as per the treatments given below:

Treatment No.	Treatment
T ₁	Seed treatment with potential fungal bio-control agent @ 10 g/kg
T ₂	Seed treatment with potential bacterial bio-control agent @ 10 g/kg
T ₃	Seed treatment with potential fungal and bacterial bio-control agents @ 10 g/kg
T ₄	Seed treatment with fungicide
T ₅	Soil application of potential fungal bio-control agent
T ₆	Soil application of potential bacterial bio-control agent
T ₇	Soil application of potential fungal and bacterial bio-control agents
T ₈	T ₄ +T ₅ +T ₆
T ₉	Application of neem cake @ 12.5 q/ha
T ₁₀	Application of FYM @ 5t/ha
T ₁₁	Application of vermicompost @ 5 q/ha
T ₁₂	Seed treatment with potential bacterial bio-control agent and soil application of neem cake
T ₁₃	Seed treatment with potential bacterial bio-control agent and soil application of FYM
T ₁₄	Seed treatment with potential bacterial bio-control agent and soil application of vermicompost
T ₁₅	Control

Variety: JG-11; Design: CRD; Replications: Three

3.9.2 Mass multiplication of *Rhizoctonia bataticola*

The test pathogen *Rhizoctonia bataticola* was mass multiplied on sterilized sorghum seeds for pot culture studies. For this, 100g of sorghum seeds were washed thoroughly in tap water and soaked in water overnight in 250 ml conical flask with addition of 20 ml of 4 per cent dextrose. After removing the water, the flasks were autoclaved for 20 min at 15 p.s.i and inoculated with 2-3 discs of 4 days old culture of test pathogen. After seven days the inoculum was mixed with sterilized soil in pots @100g kg⁻¹.

3.9.3 Preparation of talc based formulation of potential fungal isolate

Two to three discs of seven day old culture of potential *Trichoderma* spp. were added separately into conical flasks containing potato dextrose broth and incubated for seven days at room temperature (28 ± 2°C). One kg of talc powder (montmorillonite) was taken in a metal tray under aseptic conditions and pH was adjusted to 7.0 by adding CaCO₃ at the rate of 15 g kg⁻¹. Ten g of carboxy methyl cellulose (CMC) was added to 1 kg of talc powder, mixed well and the mixture was autoclaved for 30 min. at 121°C for 2 successive days. The mycelial mat along with broth was homogenized, mixed with talc powder @ 1:2 v/w and shade dried until it attains 20 per cent moisture content under aseptic conditions. After mixing, the clumps were broken and homogenized uniformly, then the mixture was packed in polypropylene bags and sealed.

The initial population of the fungal biocontrol agent was evaluated by serial dilution method and used in integrated disease management.

$$\text{Number of cfu/g} = \frac{\text{Number of colonies}}{\text{Amount of sample plated} \times \text{dilution}}$$

3.9.4 Preparation of talc based formulation of potential bacterial isolate

Talc based formulation of potential bacterial biocontrol agent was prepared by following the method as described by Vidhyasekaran and Muthamilan (1995).

A loopful of potential antagonistic bacteria was inoculated into Nutrient broth and incubated in a rotary shaker at 150 rpm min⁻¹ for 48 hours at room temperature (28 ± 2°C). One kg of talc powder (montmorillonite) was taken in a metal tray under aseptic conditions and pH was adjusted to 7.0 by adding CaCO₃ at the rate of 15 g kg⁻¹. Ten g of carboxymethyl cellulose (CMC) was added to 1 kg of talc powder, mixed well and the mixture was autoclaved for 30 min. at 121°C for 2 successive days. 400 ml of the bacterial suspension containing 1 × 10⁸cfu/ml was mixed with carrier CMC mixture under aseptic conditions. After drying to 35 per cent moisture content overnight under aseptic conditions, the mixture was packed in polypropylene bags and sealed.

The initial population of the bacterial biocontrol agent was evaluated by serial dilution method and used in integrated disease management.

$$\text{Number of cfu/g} = \frac{\text{Number of colonies}}{\text{Amount of sample plated} \times \text{dilution}}$$

3.9.5 Seed treatment

Chickpea seeds were treated with talc based formulation of potential fungal and bacterial biocontrol agent @ 10 g kg⁻¹ of seed and the seeds were used for sowing. For treatment with fungicide, the chickpea seeds were treated with compatible and effective fungicide @ 2.5 g kg⁻¹ of seeds and sown in the pathogen infested soil in the pots.

The effect of seed treatment with fungal and bacterial bio-control agents and fungicide on seed germination was evaluated by blotter technique under *in vitro*. Total number of seeds plated per each plate was five.

3.9.6 Soil application

The talc based formulation of potential fungal and bacterial biocontrol agents were applied @ 100 g pot⁻¹ before sowing.

3.9.7 Observations

1. Per cent germination

$$\frac{\text{Number of seeds germinated}}{\text{Total number of seeds sown}} \times 100$$

2. Initial plant population.

3. Final plant population

4. Percent disease incidence (PDI)

$$\text{PDI} = \frac{\text{Number of diseased plants}}{\text{Total number of seeds planted}} \times 100$$

5. Plant growth parameters

i. Shoot and root length

The average shoot and root length of the plants was recorded.

ii. Dry weight of shoot and root

The samples were allowed to dry under room temperature and the dry weights of shoot and root were recorded.

6. Population dynamics of both pathogen and antagonists was evaluated at two different times during the experimental period i.e. initial population at 7 DAS and final population at 45 DAS.

i) Estimation of population levels of *Rhizoctonia bataticola*

The population levels of test pathogen, *Rhizoctonia bataticola* from different treatments in pot culture experiment was enumerated following serial dilution plate count technique (Johnson and Curl, 1977) on Chloroneb Mercury Rose Bengal Agar (CMRA) medium.

ii) Estimation of population levels of *Trichoderma*

The population levels of fungal antagonist, *Trichoderma* from different treatments in pot culture experiment was enumerated following serial dilution plate count technique (Johnson and Curl, 1977) on *Trichoderma* Specific Medium (TSM).

iii) Estimation of population levels of bacterial antagonist

The population levels of bacterial antagonist, from different treatments in pot culture experiment was enumerated following serial dilution plate count technique (Johnson and Curl, 1977) on Nutrient agar medium.

3.10 MOLECULAR CHARACTERIZATION OF POTENTIAL BIOCONTROL AGENTS

The genetic variability among the isolates of bacterial biocontrol agents was studied by Random Amplified Polymorphic DNA (RAPD) Williams *et al.* (1990). The 16S rDNA sequence was used for identification of potential bacterial biocontrol agent (Julian *et al.*, 1998).

3.10.1 Bacterial cultures

The antagonistic bacterial isolates having different degree of antagonistic activity against *Rhizoctonia bataticola* were selected for molecular characterization. Bacterial isolates were grown on nutrient broth at $28 \pm 2^\circ\text{C}$ overnight for DNA extraction.

3.10.2 Buffers used for electrophoresis

1. Composition of 10 x TBE buffer

Tris base	:	54.0 g
Boric acid	:	27.5 g
EDTA	:	4.65 g
Distilled water	:	500 ml
pH	:	8.0

Preparation

Each chemical was dissolved in separate beakers using distilled water and all were mixed finally. The pH was adjusted to 8.0 by using 0.1N Hcl or 0.1 N NaOH and volume was made upto 500 ml and sterilized by autoclaving at 15 PSI for 15 minutes.

2. Composition of loading dye (10x)

Glycerol	:	5ml
10 x TBE	:	1ml
Bromophenol blue (saturated)	:	1ml
Xylene cyanol (10%)	:	1ml
Double distilled water	:	10ml

Preparation

Contents were mixed well and divided into 1ml aliquot, sterilized and stored at -20°C for further use.

3.10.3 DNA extraction

The DNA from potential bacterial antagonists was extracted by using bacterial genomic DNA isolation kit from M/s Medox, Chennai. The quantity

and quality of DNA was verified on one per cent agarose gel and by spectrophotometer.

3.10.4 Preparation of gels

Gel plates (13 x 14 cm, Genei, Bangalore) were washed thoroughly with cleaning solution followed by distilled water and dried. The two open sides of the plates were sealed with cellophane tape. Gel solution was prepared by mixing 1g of agarose (Fermentas, Bangalore) in 100 ml of 1 x TBE buffer (1.0% gel) in a conical flask and boiled in an oven until a clear solution was obtained and 4 μ l of ethidium bromide (10mg μ l⁻¹) was added. The solution was poured onto the sealed gel casting tray, with suitable comb inserted and allowed to polymerize.

3.10.5 Loading and running of gels

The inserted comb was gently removed from the gel after polymerization. The gel plate was placed in horizontal apparatus and fixed with 1 x TBE buffer. The samples were loaded in the wells with the help of micro pipettes. After loading, the electrophoretic unit was connected to power pack (Genei, Bangalore) with a regulated electric power supply of 100V. At the end of run, the gel was carefully removed and analyzed.

3.10.6 Qualitative and quantitative verification of DNA from different isolates of potential biocontrol agents

DNA samples (5 μ l) from each isolate mixed with four μ l of 1x loading dye were loaded on the wells of the one per cent agarose gel along with five μ l of DNA marker in order to verify the quality and quantity of DNA. Alternatively the quality and quantity of DNA was also verified by spectrophotometer.

3.10.7 RAPD profiles through polymerized chain reaction (PCR)

Four different random primers belonging to operon “A” and “D” series viz. OPA-12, 18 and OPD-4, 5 (Eurofins Genomics Private Limited, Bangalore) were used to detect polymorphism among the isolates under study. The experiment was repeated thrice and results were reproducible. The primer sequences used in RAPD technique are given below.

Sl. No.	Operon	Sequence
1	OPA-12	5 ¹ (ACA TCG TCG G)3 ¹
2	OPA-18	5 ¹ (AGG TGA CCG T)3 ¹
3	OPD-4	5 ¹ (TAT CGT GTA C)3 ¹
4	OPD-5	5 ¹ (ATT GTA GAT C)3 ¹

3.10.7.1 Standardization of RAPD technique

The RAPD technique has been standardized and the following conditions were used for the amplification of DNA from different isolates.

Master mix for RAPD has been prepared as given below

1.	Assay buffer (10x)	2.5µl
2.	MgCl ₂ (25 mM)	2.0 µl
3.	dNTPS (20 mM)	1.0 µl
4.	Primer (10 p mol)	1.0 µl
5.	Taq polymerase (3u/ µl)	0.4 µl
6.	DNA sample (100 ng)	2.0 µl
7.	Millipore water	16.1 µl
	Total volume	25 µl

Conditions used for RAPD amplification

Stage-I: Initial denaturation at 94°C for 4 min.

Stage-II: Denaturation at 94°C for 1 min.

Annealing at 36°C for 1 min and

Extension at 72°C for 2 min.

Number of cycles: 40

Stage-III: Final extension at 72°C for 5 min.

Amplified PCR products were subjected to one per cent agarose gel electrophoresis with 1.0 x TBE as running buffer. The banding pattern was visualized under UV trans-illuminator with ethidium bromide (10 mg ml⁻¹) staining. The DNA banding profiles were documented in the gel documentation system (Alpha Innotech, USA) and compared with 1kb DNA ladder (Genei, Bangalore).

3.10.7.2 Scoring and data analysis

Each amplified band was considered as RAPD marker and recorded for all samples. Data were entered using a matrix in which all observed bands or characters were listed. The RAPD pattern of each isolate was evaluated by assigning character state 1 to all the bands that could be reproducible and detected in the gel and '0' for the absence of band.

The data matrix thus generated was used to calculate Jaccard's similarity coefficient for each pair wise comparison. The coefficients were calculated in Silico using the following formula.

$$\text{Similarity coefficient} = \frac{a}{n}$$

where,

a = Number of matching band for each pair of comparisons

n = Total number of bands observed in two samples.

The similarity coefficients were subjected to unweighted pair-group method on arithmetic average (UPGMA) cluster analysis to group the isolate based on their over all similarities statistical package for social sciences (SPSS) was used for the cluster analysis and subsequent dendrogram preparation.

3.10.8 Amplification of 16S rDNA from antagonistic bacterial isolates

The 16S rDNA sequence has been selected for identification of potential biocontrol agent. The 16S rDNA from potential antagonistic isolate has been amplified by using 63F and 1387R primers synthesized at Eurofins Genomics, Bangalore. The primer sequences are given below (Marchesi *et al.*, 1998).

63F – 5¹ CAG GCC TAA CAC ATG CAA GTC – 3¹

1387R – 5¹ GGG CGG (AT) GT GTA CAA GGC – 3¹

As a part of this, PCR technique has been standardized and the following components were used for the amplification of 16S rDNA.

1. Assay buffer (10x)	2.5 µl
2. MgCl ₂ (25mM)	2.0 µl
3. dNTPs (10mM)	0.5 µl
4. Forward primer (63F) (10p mol)	1.0 µl
5. Reverse primer (1387R) (10p mol)	1.0 µl
6. Taq polymerase (3U/ µl)	0.4 µl
7. DNA sample (50 ng)	2.0 µl
8. Millipore water	15.6 µl
Total volume	25.0 µl

The 16S rDNA amplification was carried under the following conditions.

Stage-I: Initial denaturation at 95°C for 4 minutes.

Stage-II: Denaturation of 95°C for 1.0 min

Annealing at 57.5°C for 1.0 min and

Extension at 72°C for 1.5 min.

Number of cycles: 30

Stage-III: Final extension at 72°C for 10 min.

The amplified products were visualized in one per cent agarose gel.

3.10.9 Gel extraction and purification

The amplified 16S rDNA of the potential bacterial antagonist was eluted from the gel using the kit Qualigens, Bangalore and then purified using the kit supplied by Qiagen, Bangalore and sent for sequencing to Eurofins Genomics Private Limited, Bangalore.

3.11 STATISTICAL ANALYSIS

Wherever necessary, the data was statistically analyzed (Gomez and Gomez, 1984). Complete Randomized Design (CRD) was used for dual culture technique, poisoned food technique, spectrophotometric method and pot culture experiment and the treatments were compared at $P \leq 0.05$.

Chapter ~ IV

Results & Discussion

Chapter-IV

RESULTS AND DISCUSSION

Chickpea is the third important legume of the world grown over an area of 11 m. ha with an average global production of 881 kg/ha. India is the leading producer of chickpea contributing to the tune of 70 per cent of the world's chickpea production.

In chickpea, dry root rot disease caused by *Rhizoctonia bataticola* (Taub.) Butler, is a major problem with typical symptoms of withering and drying of the plants, presence of dark tap root showing signs of rotting and devoid of its lateral and finer roots. The most important diagnostic symptoms is shredding of bark which comes out in the form of flakes (Haware, 1990).

Rhizoctonia bataticola is a non- specialized soil borne fungal pathogen of world-wide importance and has a wide host range of 500 plant species (Sinclair, 1982). The fungus induces a variety of symptoms such as root rots, seedling blights, stem rots, wilts etc., in different host plants.

Biological control has been considered as a potential control strategy against soil borne plant pathogens. In recent years, considerable success has been achieved by introducing antagonists to soil (or) infection court (Papavizas and Lewis, 1981). Biocontrol agents also act indirectly by inducing systemic resistance in plants, by increased nutrient uptake and making them unavailable to plant pathogens and by inactivating the pathogen enzymes (Chaube *et al.*, 2001).

Organic amendments are important not only for the reduction of quantity of chemical fertilizers but also act as excellent sources of nutrition which favour native antagonist to proliferate and suppress the soil borne diseases (Mukhopadhyay, 1994).

An integrated approach by including the fungicide tolerant native antagonists isolated from rhizosphere soil and root endophytes appears to be possible solution for effective management of dry root rot.

Molecular characterization of the potential biocontrol agents using Random Amplified Polymorphic DNA (RAPD) and 16S rDNA analysis are important to distinguish different isolates at genetic level.

The present investigation was undertaken to explore the feasibility of integrating fungicidal tolerant biocontrol agents and organic amendments as a part of management of dry root rot of chickpea caused by *Rhizoctonia bataticola*. The results of the present investigation are presented below.

4.1 THE PATHOGEN

4.1.1 Isolation of pathogen

The dry root rot affected chickpea plants were identified in the field based on key symptoms like withering and drying of plants (Plate 1). When such plants were pulled out showed blackening of tap root (Plate 2a), shredding of bark coming out in the form of flakes and devoid of lateral and finer roots (Plate 2b). Such infected plants were collected from Korrapadu village, Rajupalem mandal of Y.S.R district (Andhra Pradesh) for pathogen isolation where the disease incidence was high.

The pathogen was isolated from root rot affected plants using tissue segment method on PDA and purified by single hyphal tip method.

These symptoms were similar to those described by reports of Singh and Agarwal (2002).



Plate 1. Field view showing symptoms of dry root rot of chickpea



a) Darkening of tap root



b) Shredding of bark
c) Absence of lateral roots

Plate 2: Infected chickpea plants showing typical symptoms of dry root rot

4.1.2 Identification of pathogen

The fungus produced radial hyaline colonies, which later become carbonaceous brown to black (Plate 3). Mycelium was septate and dark brown in colour. Typical right angled branching of mycelium (Plate 4) was observed. Sclerotia (Plate 5) were black, varied from spherical to irregular in shape and measured 80 to 85µm in diameter. Pycnidial production was not observed in culture plates.

The colony characters and morphological characters of mycelium and sclerotia were in agreement with the descriptions of Barnett and Barry (1972) and Sajeena *et al.* (2004). Thus, the fungus under present investigation was identified as *Rhizoctonia bataticola* (Taub.) Butler.

4.1.3 Pathogenicity test

Soil inoculation method was used to prove the pathogenicity of *R. bataticola*. About 10 seeds of chickpea were sown in each pot with a diameter of 22.5 cm (sterilized) inoculated with pathogen @ 100 g Kg⁻¹ soil. Pre-emergence rot of seedlings was observed and the survived plants showed stunted growth followed by wilting and drying of leaves and stems (Plate 6). When the infected plants were pulled out the tap root was blackened and devoid of lateral as well as finer roots.

On re-isolation, the characters of the pathogen showed similarity with the original pathogen isolated from the field (section 4.1.1), thus fulfilling Koch's postulates.

Several workers have found soil inoculation method as most suitable in establishing the disease caused by *Rhizoctonia bataticola* (Vishwadar and Sarbhoy (1993) in soybean; Prameela Devi and Singh (1998) in greengram; Katariya *et al.* (2007) in chickpea; Kanchan and Biswas (2009) in pigeonpea).



Plate 3: Pure culture of *Rhizoctonia bataticola* (Mycelial stage)

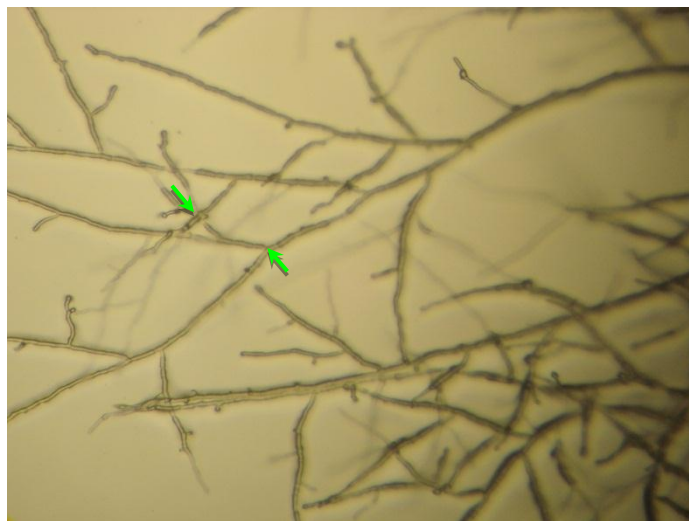


Plate 4: Photomicrograph of *Rhizoctonia bataticola* showing septate and branched hyphae (10x) (indicated by arrows)

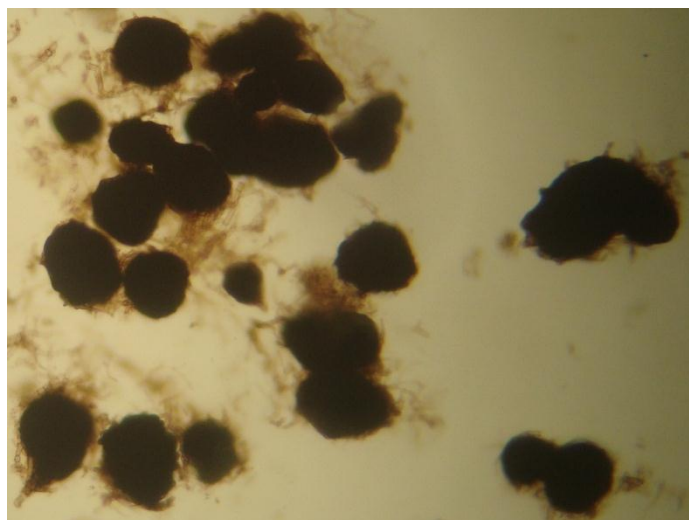


Plate 5: Photomicrograph of sclerotia of *Rhizoctonia bataticola* (10x)



Plate 6: Pathogenicity test showing dry root rot affected chickpea plants following soil inoculation



Healthy roots



Diseased roots

4.2 IN VITRO EVALUATION OF EFFICACY OF FUNGICIDES AGAINST *R. bataticola*

One of the objectives of present investigation is to include an effective fungicide in combination with potential biocontrol agent as an integrated approach to manage *Rhizoctonia bataticola* under glasshouse conditions.

Keeping this in view, an attempt was made to find out a promising fungicide against *Rhizoctonia bataticola* under *in vitro* conditions. Efficacy of two non systemic fungicides copper oxychloride, captan at different concentrations viz., 1000, 1500, 2000 and 2500 ppm, two systemic fungicides hexaconazole @ 500, 1000, 1500, and 2000 ppm whereas tebuconazole @ 250, 500, 750 and 1000 ppm and one antifungal antibiotic validamycin @ 250, 500, 750 and 1000 ppm were evaluated against *R. bataticola* by poisoned food technique as per procedure given in section 3.4. The data revealed that all the fungicides at all concentrations reduced mycelial growth of *R. bataticola* when compared to control.

It is evident from the data (Table 4.1), among all the fungicides that were tested copper oxychloride, captan, hexaconazole and tebuconazole showed 100 per cent inhibition (Plate 7, 8, 9 and 10) of mycelial growth at all the concentrations of 1000, 1500, 2000 and 2500 ppm. Whereas the fungicide validamycin (Plate 11) has shown increased inhibition from 86.66 to 95.55 per cent with increasing concentration from 250 to 1000 ppm with a mean inhibition of 91.66 per cent.

The results were in agreement with Ebenezar and Wesely (2000) who reported that the fungicides copper oxychloride (0.25%), captan (0.2%) inhibited the growth of *Macrophomina phaseolina* causing root rot disease in greengram by 95.89 and 78.56 per cent respectively. Khan and Gangopadhyay (2008) reported that carbendazim and captan @ 500 ppm were highly inhibitory to the growth of *Macrophomina phaseolina* causing

Table 4.1. *In vitro* evaluation of efficacy of different fungicides on mycelial growth of *Rhizoctonia bataticola* in poisoned food technique

S. No.	Fungicides	Concentration (ppm)	Mycelial growth of pathogen (cm)*	Per cent inhibition over control	Mean
1.	Copper oxychloride	1000	0.6	100.00 (90.00)	100 (90.00)
		1500	0.4	100.00 (90.00)	
		2000	0.3	100.00 (90.00)	
		2500	0.2	100.00 (90.00)	
2.	Captan	1000	0.0	100.00 (90.00)	100 (90.00)
		1500	0.0	100.00 (90.00)	
		2000	0.0	100.00 (90.00)	
		2500	0.0	100.00 (90.00)	
3.	Hexaconazole	500	0.0	100.00 (90.00)	100 (90.00)
		1000	0.0	100.00 (90.00)	
		1500	0.0	100.00 (90.00)	
		2000	0.0	100.00 (90.00)	
4.	Tebuconazole	250	0.0	100.00 (90.00)	100 (90.00)
		500	0.0	100.00 (90.00)	
		750	0.0	100.00 (90.00)	
		1000	0.0	100.00 (90.00)	
5.	Validamycin	250	1.2	86.66 (68.71)	91.66 (73.15)
		500	0.8	91.11 (72.68)	
		750	0.6	93.33 (75.07)	
		1000	0.4	95.55 (78.00)	
	Control	-	9.0	-	
	S. Em $\pm$			0.57	
	C.D (0.05)			1.63	

* Mean of three replications

Figures in parenthesis are angular transformed values

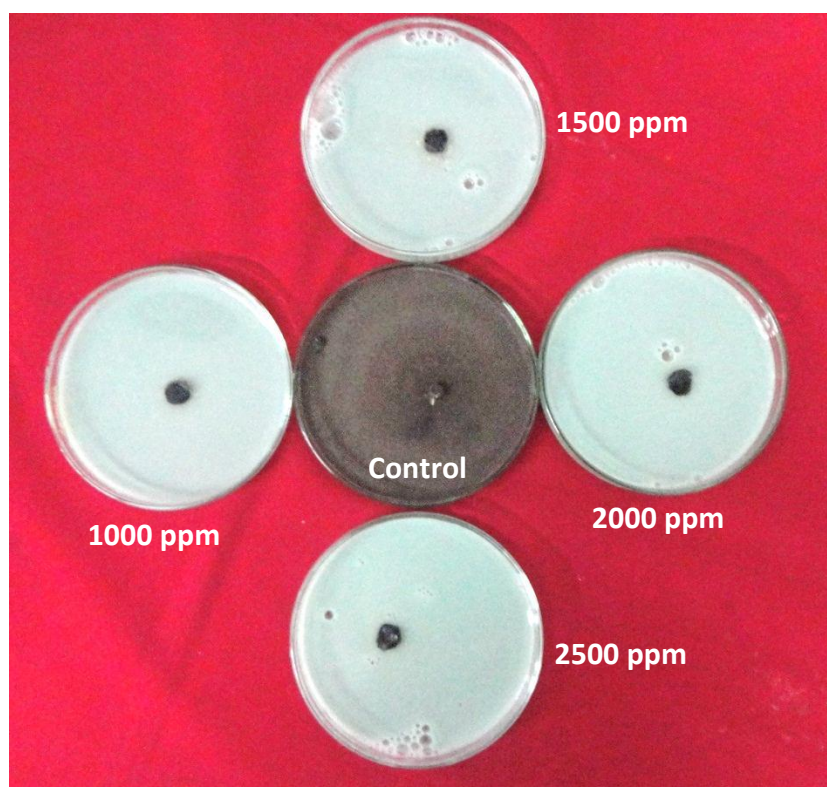


Plate 7: *In vitro* efficacy of copper oxychloride on mycelial growth of *Rhizoctonia bataticola* by poisoned food technique

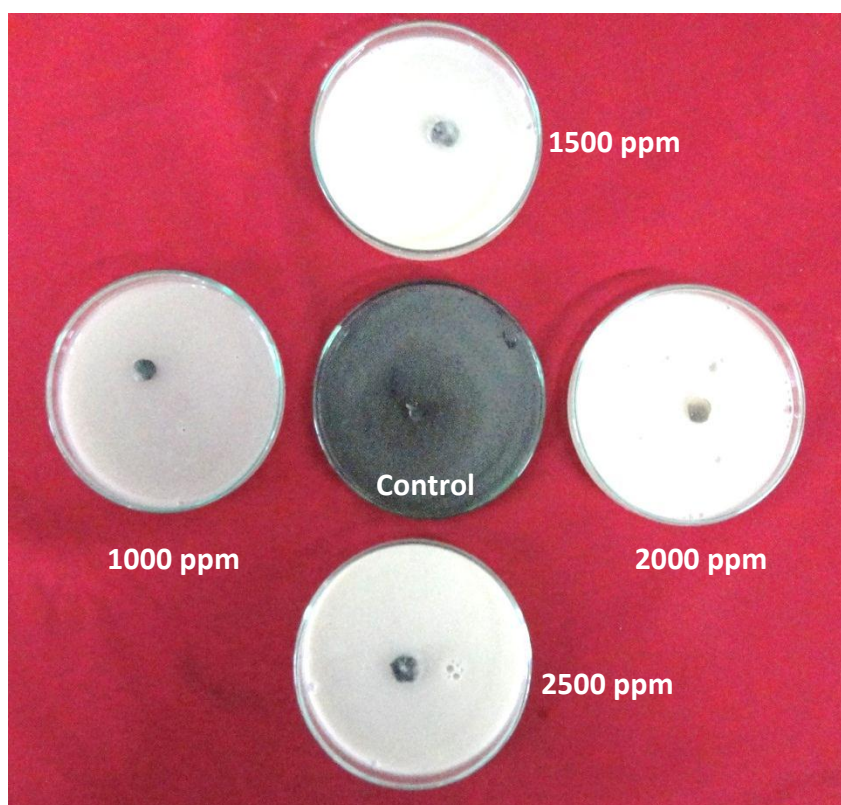


Plate 8: *In vitro* efficacy of captan on mycelial growth of *Rhizoctonia bataticola* by poisoned food technique

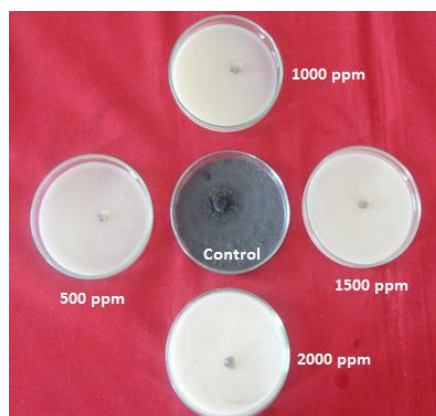


Plate 9: *In vitro* efficacy of hexaconazole on mycelial growth of *Rhizoctonia bataticola* by poisoned food technique

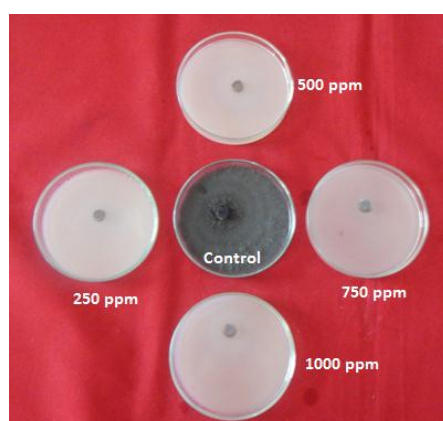


Plate 10: *In vitro* efficacy of tebuconazole on mycelial growth of *Rhizoctonia bataticola* by poisoned food technique

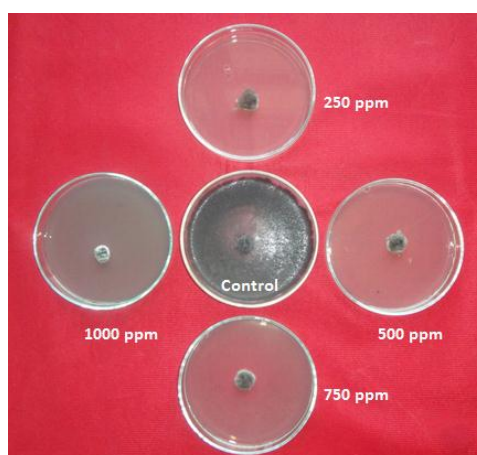


Plate 11: *In vitro* efficacy of validamycin on mycelial growth of *Rhizoctonia bataticola* by poisoned food technique

root rot disease in chickpea. Konde *et al.* (2008) evaluated the fungicides against *R. bataticola* causing root rot of soybean and revealed that combination of carbendazim + thiram (0.1 + 0.2 % concentration), penconazole (0.1%) and thiophanate-M (0.1%) were significantly effective in completely (100%) inhibiting the radial growth of *R. bataticola*.

4.3 POTENTIAL BIOCONTROL AGENTS

4.3.1 Isolation and identification of native antagonistic mycoflora from rhizosphere soil and root endophytic region of chickpea against *R. bataticola*

Antagonistic mycoflora from rhizosphere soil and roots of healthy chickpea plants were isolated as per procedure given in section 3.5.1 and 3.5.2. The mycoflora were isolated on Rose Bengal Agar (RBA) medium (Plate 12 and 13). The fungal antagonists were purified by single spore method and were maintained on PDA medium. A total of 10 fungi were obtained from rhizosphere soil and root endosphere.

Based on colony and morphological characters, mycoflora were identified. Ten *Trichoderma* isolates designated from CT1 to CT10 were purified (Plate 14).

During the present investigation, preference was given for the *Trichoderma* isolates compared to other mycoflora since *Trichoderma* is known to be a potential fungal antagonist.

Ramesh and Korikanthimath (2006) isolated biocontrol agents like *Trichoderma viride*, *T. harzianum* from rhizosphere of various crops and tested their efficacy against the *Macrophomina phaseolina* causing root rot of groundnut.

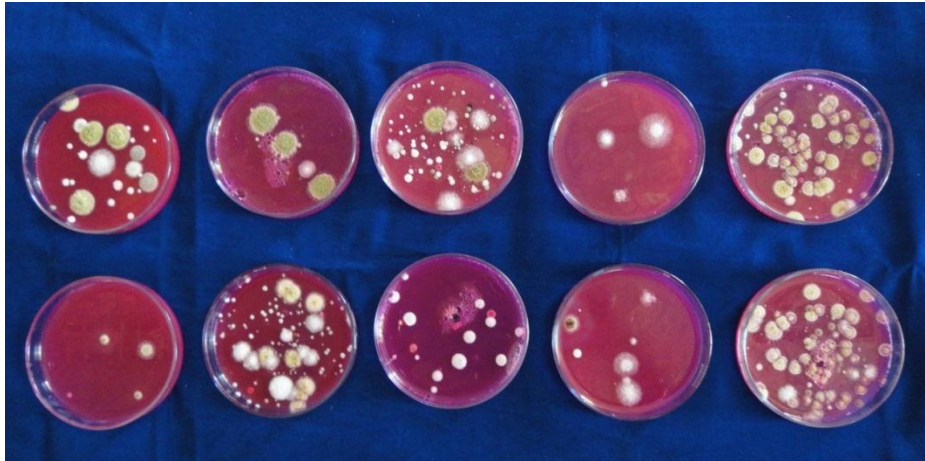


Plate 12: Colonies of antagonistic mycoflora isolated from rhizosphere zone of chickpea



Plate 13: Colonies of antagonistic mycoflora isolated from root endophytic zone of chickpea



Plate 14: Pure cultures of *Trichoderma* isolates CT1, CT2, CT3, CT4, CT5, CT6, CT7, CT8, CT9 and CT10
CT: Chickpea *Trichoderma*

4.3.2 Isolation and identification of native antagonistic bacteria from rhizosphere soil and root endophytic region of chickpea against *R. bataticola*

Antagonistic bacteria from rhizosphere soil and roots of healthy chickpea plants were isolated as per the procedure given in section 3.5.1 and 3.5.2. The bacteria were isolated on Nutrient Agar (NA) medium (Plate 15 and 16). The bacteria were purified by streak plate method and maintained on Nutrient agar medium. A total of 40 bacteria were isolated out of which 20 bacteria were obtained from rhizosphere soil and 20 bacteria from root as root endophytes.

The bacterial isolates from rhizosphere soil were designated as CRB-1 to CRB-20 (Plate 17) followed by root endophytic bacteria as CREB-1 to CREB-20 (Plate 18).

Parmer and Dadarwal (1997) isolated *Pseudomonas* and *Bacillus* sp. from rhizosphere and rhizoplane of healthy chickpea plants. Rangeswaran and Prasad (2000) isolated three hundred antagonistic bacteria against root rot and wilt from chickpea rhizosphere. Sharma and Sarita (2005) reported bacteria belonging to the *Rhizobium* and *Agrobacterium* group as wheat root endophytes. Rangeswaran *et al.* (2008) isolated five endophytic and two rhizospheric bacteria from healthy chickpea plants and identified them based on morphological, physiological and biochemical tests.

During the present investigation also several biocontrol agents were isolated from both rhizosphere and endophytic zone to evaluate their efficacy against *Rhizoctonia bataticola*.

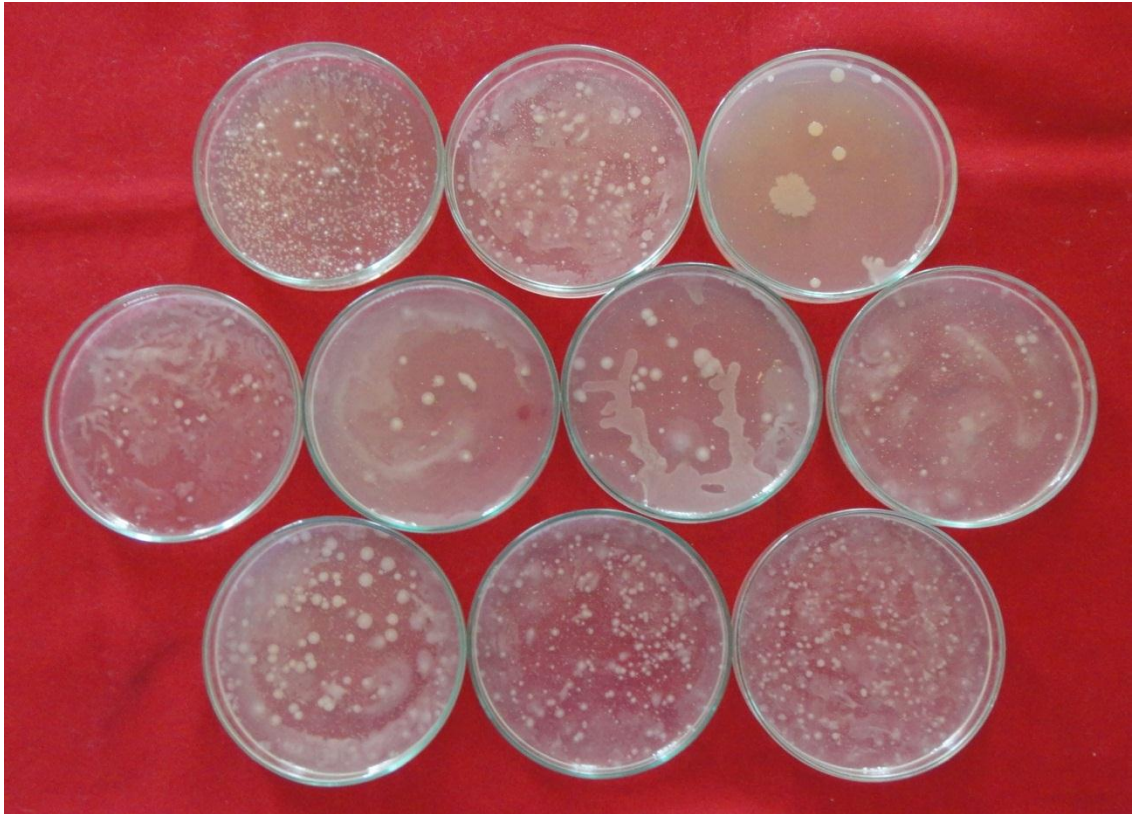


Plate 15: Colonies of antagonistic bacteria isolated from rhizosphere zone of chickpea

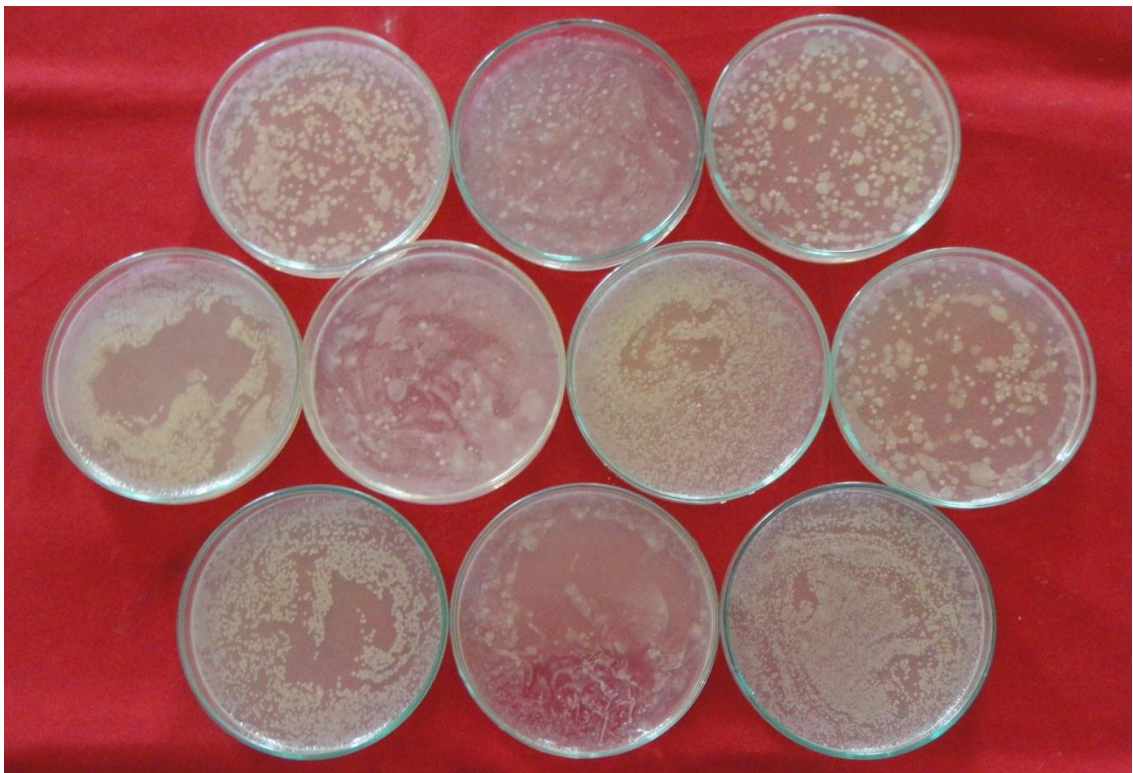


Plate 16: Colonies of antagonistic bacteria isolated from root endophytic zone of chickpea

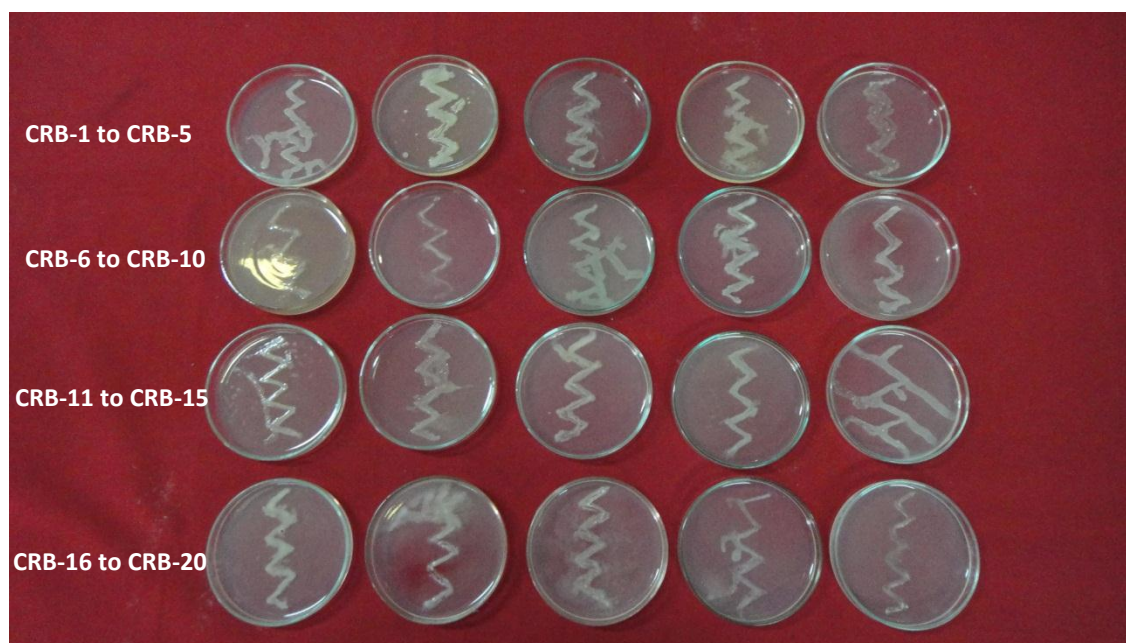


Plate 17: Pure cultures of Chickpea Rhizosphere Bacterial isolates (CRB 1 to 20)

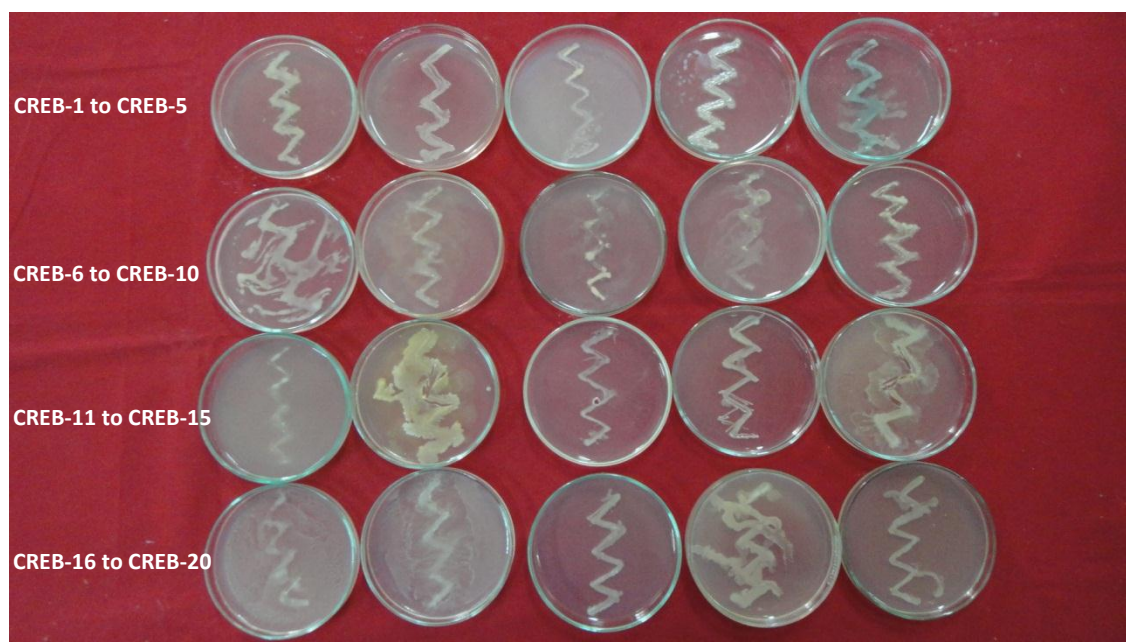


Plate 18: Pure cultures of Chickpea Root Endophytic Bacterial isolates (CREB 1 to 20)

4.4 IN VITRO EVALUATION OF THE ANTAGONISTIC ACTIVITY OF MICROFLORA AGAINST *Rhizoctonia bataticola* IN DUAL CULTURE.

The antagonistic effect of native microflora was assessed based on their ability to inhibit the pathogen growth in dual culture technique as per the procedure given in section 3.5.3.1. The effect of these native antagonists on the mycelial growth of the pathogen was calculated and expressed as per cent inhibition.

4.4.1 *In vitro* evaluation of efficacy of antagonistic mycoflora (*Trichoderma* spp.) against *R. bataticola* in dual culture technique.

All the native antagonists i.e *Trichoderma* isolates showed significant reduction in mycelial growth of *R. bataticola* when compared to control. The data pertaining to per cent inhibition of mycelial growth of *R. bataticola* due to *Trichoderma* spp. are presented in Table 4.2 and Fig 4.1.

Among 10 *Trichoderma* isolates, *Trichoderma* isolate-7 (CT7) showed maximum inhibition of growth of *Rhizoctonia bataticola* (83.33%) followed by *Trichoderma* isolate-4 (81.11%) and *Trichoderma* isolate-10 (74.44%) which were on par with each other (Plate 19). The least inhibition was shown by *Trichoderma* isolate-9 (45.56%).

These results were in agreement with Bandyopadhyay *et al.* (2003) who reported that strain of *Trichoderma* inhibited the growth of *Rhizoctonia bataticola* by 51.1 per cent under *in vitro* conditions. Paul *et al.* (2008) reported that among 11 *Trichoderma* isolates tested, *Trichoderma harzianum* showed higher inhibition of the growth of *Rhizoctonia solani* by 77 per cent under *in vitro* conditions. Kaushal (2008) reported that *T. harzianum* was effective in inhibiting the mycelial growth of *R. bataticola* the causal organism of chickpea dry root rot. Pan (2009) reported the antagonistic potential of *Trichoderma* isolates through production of volatile and non-volatile

Table 4.2. *In vitro* evaluation of efficacy of antagonistic *Trichoderma* isolates against *Rhizoctonia bataticola* in dual culture technique

S. No.	Antagonist	Linear growth of <i>R. bataticola</i> (Cm)*	Per cent inhibition of mycelial growth of <i>Rhizoctonia bataticola</i>
1.	<i>Trichoderma</i> isolate-1 (CT1)	2.6	71.11 (57.50)
2.	<i>Trichoderma</i> isolate-2 (CT2)	3.6	60.00 (50.77)
3.	<i>Trichoderma</i> isolate-3 (CT3)	3.8	57.77 (49.52)
4.	<i>Trichoderma</i> isolate-4 (CT4)	1.7	81.11 (64.28)
5.	<i>Trichoderma</i> isolate-5 (CT5)	4.5	50.00 (45.04)
6.	<i>Trichoderma</i> isolate-6 (CT6)	4.7	47.78 (43.72)
7.	<i>Trichoderma</i> isolate-7 (CT7)	1.5	83.33 (65.96)
8.	<i>Trichoderma</i> isolate-8 (CT8)	4.0	55.55 (48.24)
9.	<i>Trichoderma</i> isolate-9 (CT9)	4.9	45.56 (42.44)
10	<i>Trichoderma</i> isolate-10 (CT10)	2.3	74.44 (59.64)
11	Control	9.0	-
	S. Em $\pm$		2.42
	C.D (0.05)		7.21

* Mean of three replications

Figures in parenthesis are angular transformed values

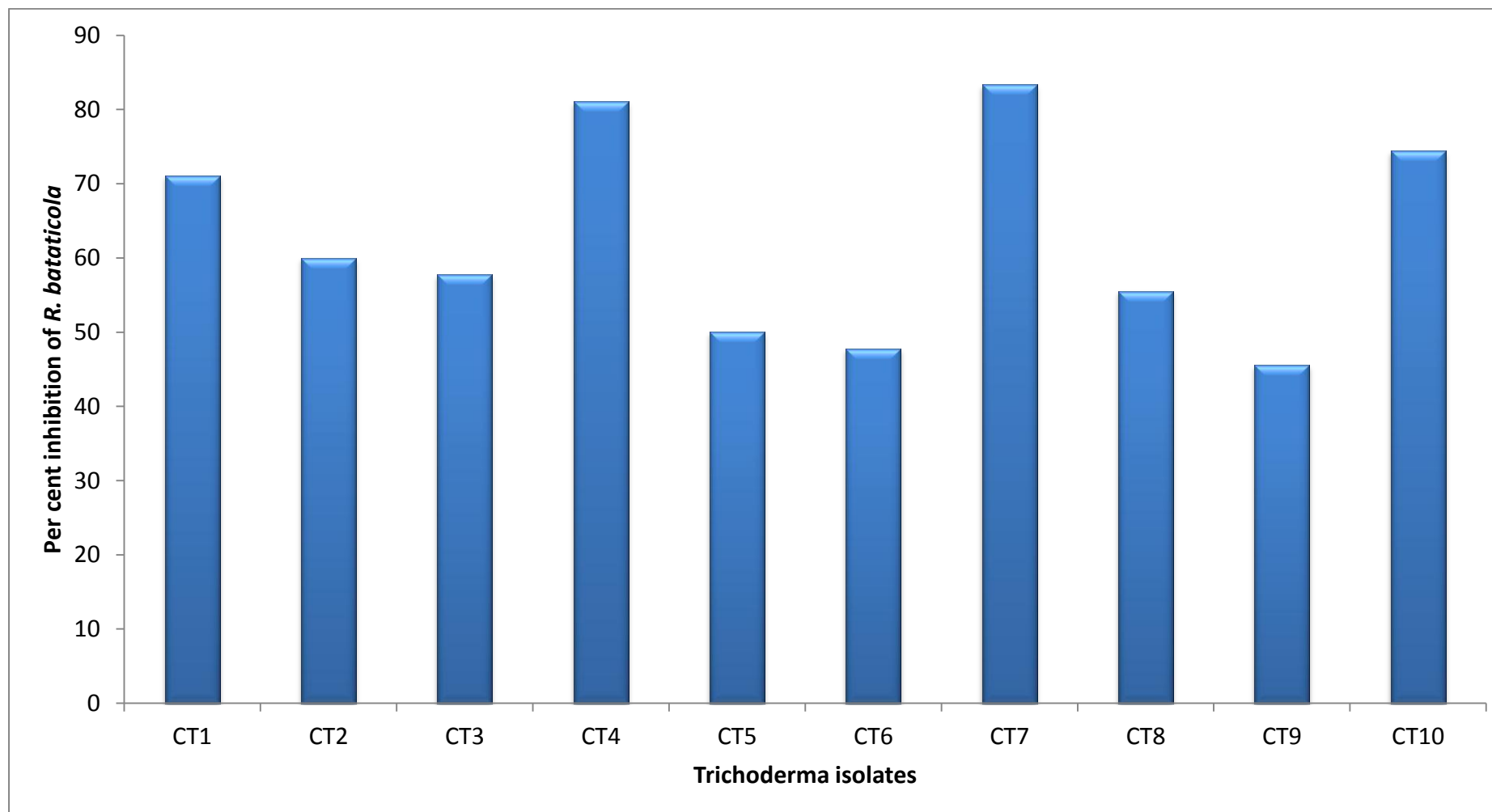


Fig. 4.1. *In vitro* evaluation of efficacy of *Trichoderma* isolates against *Rhizoctonia bataticola* in dual culture technique

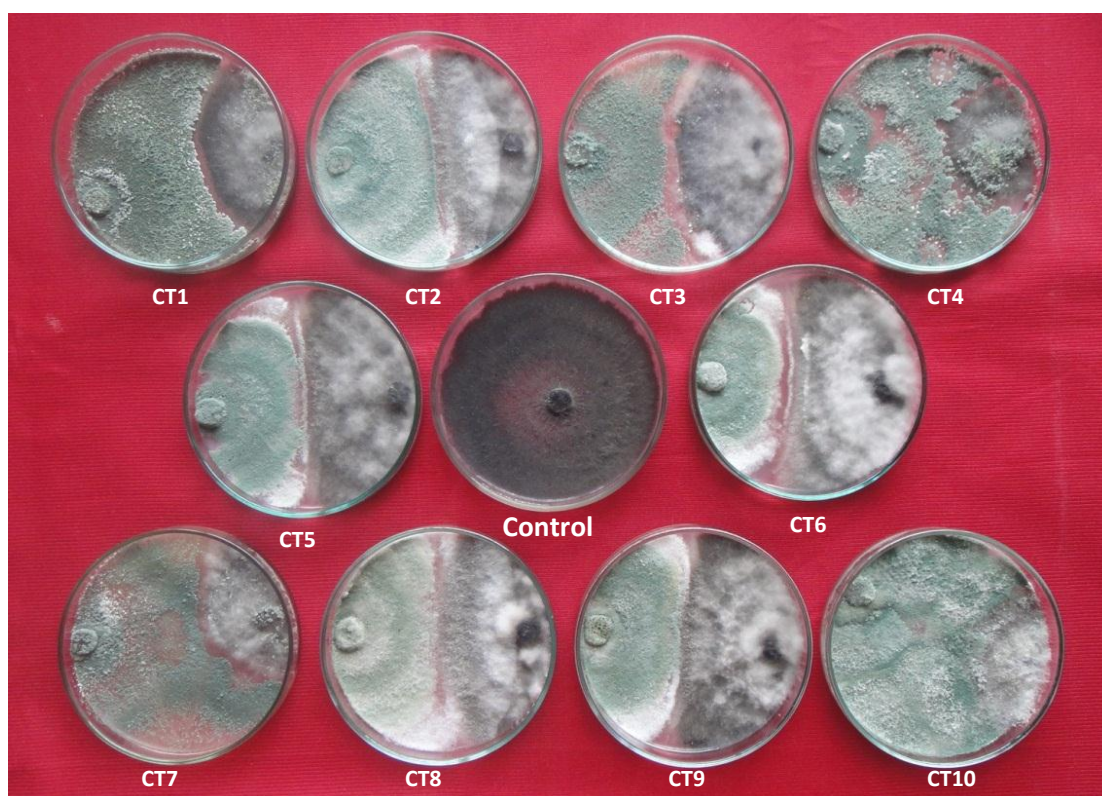


Plate 19: *In vitro* efficacy of *Trichoderma* isolates on mycelial growth of *Rhizoctonia bataticola* in dual culture technique

substances against *Macrophomina phaseolina*. Choudhary *et al.* (2010) tested four bioagents viz, *Trichoderma viride*, *Trichoderma harzianum*, *Aspergillus versicolor* and *Bacillus firmus* for the control of dry root rot in mungbean. In dual cultures, *Trichoderma viride*, *Trichoderma harzianum* and *Aspergillus versicolor* were effective in inhibiting the growth of *Macrophomina phaseolina* to an extent of 61 to 65%.

4.4.2 *In vitro* evaluation of efficacy of antagonistic bacteria against *R. bataticola* in dual culture technique.

Efficacy of antagonistic bacteria against *Rhizoctonia bataticola* was evaluated by using dual culture technique. The data pertaining to per cent inhibition of mycelial growth of *R. bataticola* due to antagonistic bacteria are presented in Table 4.3 and 4.4 and Fig 4.2 and 4.3.

Among the 20 rhizospheric bacterial antagonists tested (Plate 20) the isolate CRB-13 showed the maximum inhibition (86.66%) of growth of *R. bataticola* followed by CRB-18 (83.33%) and CRB-2 (81.48%). However, both were found to be on par with each other in inhibiting the pathogen. The isolate CRB-6 recorded least (27.77%) per cent inhibition (Fig 4.2).

Among the 20 root endophytic bacterial antagonists tested (Plate 21) the isolate CREB-13 showed the maximum inhibition (95.55%) of growth of *R. bataticola* that was on par with CREB-6 (94.44%) and CREB-16 (93.33%) The isolate CREB-12 recorded least (44.44%) per cent inhibition (Fig 4.3).

These results were in agreement with Khot *et al.* (1996) who reported that five rhizobacteria isolated from rhizosphere of chickpea plants inhibited the growth of *Fusarium oxysporum* f. sp. *ciceri* and *Rhizoctonia bataticola* under *in vitro* conditions. Khan and Gangopadhyay (2008) reported that the maximum reduction of mycelial growth of *R. bataticola*, incitant of dry root rot of chickpea in dual culture technique by *Pseudomonas fluorescens* strains PFBC-25 and 26. Vinod Kumar *et al.* (2007) tested the efficacy of

Table 4.3. *In vitro* evaluation of efficacy of chickpea rhizosphere bacterial isolates against *Rhizoctonia bataticola* in dual culture technique

S. No	Bacterial antagonists	Linear growth of <i>R. bataticola</i> (Cm)*	Per cent inhibition of mycelial growth of <i>Rhizoctonia bataticola</i>
1	CRB-1	2.3	74.44 (59.63)
2	CRB-2	1.6	81.48 (64.52)
3	CRB-3	3.9	56.66 (48.83)
4	CRB-4	4.5	50.00 (44.99)
5	CRB-5	3.0	66.66 (54.74)
6	CRB-6	6.5	27.77 (31.79)
7	CRB-7	5.2	42.22 (40.51)
8	CRB-8	4.0	55.55 (48.20)
9	CRB-9	5.5	38.88 (38.56)
10	CRB-10	4.5	50.00 (44.99)
11	CRB-11	2.8	68.88 (56.10)
12	CRB-12	3.5	61.11 (52.43)
13	CRB-13	1.2	86.66 (68.73)
14	CRB-14	2.2	75.55 (60.40)
15	CRB-15	4.2	53.33 (46.90)
16	CRB-16	4.4	51.11 (45.62)
17	CRB-17	4.0	55.55 (48.18)
18	CRB-18	1.5	83.33 (65.91)
19	CRB-19	4.8	46.66 (43.07)
20	CRB-20	5.7	36.66 (37.24)
21	Control	9.0	-
	S. Em $\pm$		1.05
	C D (0.05)		3.02

* Mean of three replications

Figures in parenthesis are angular transformed values

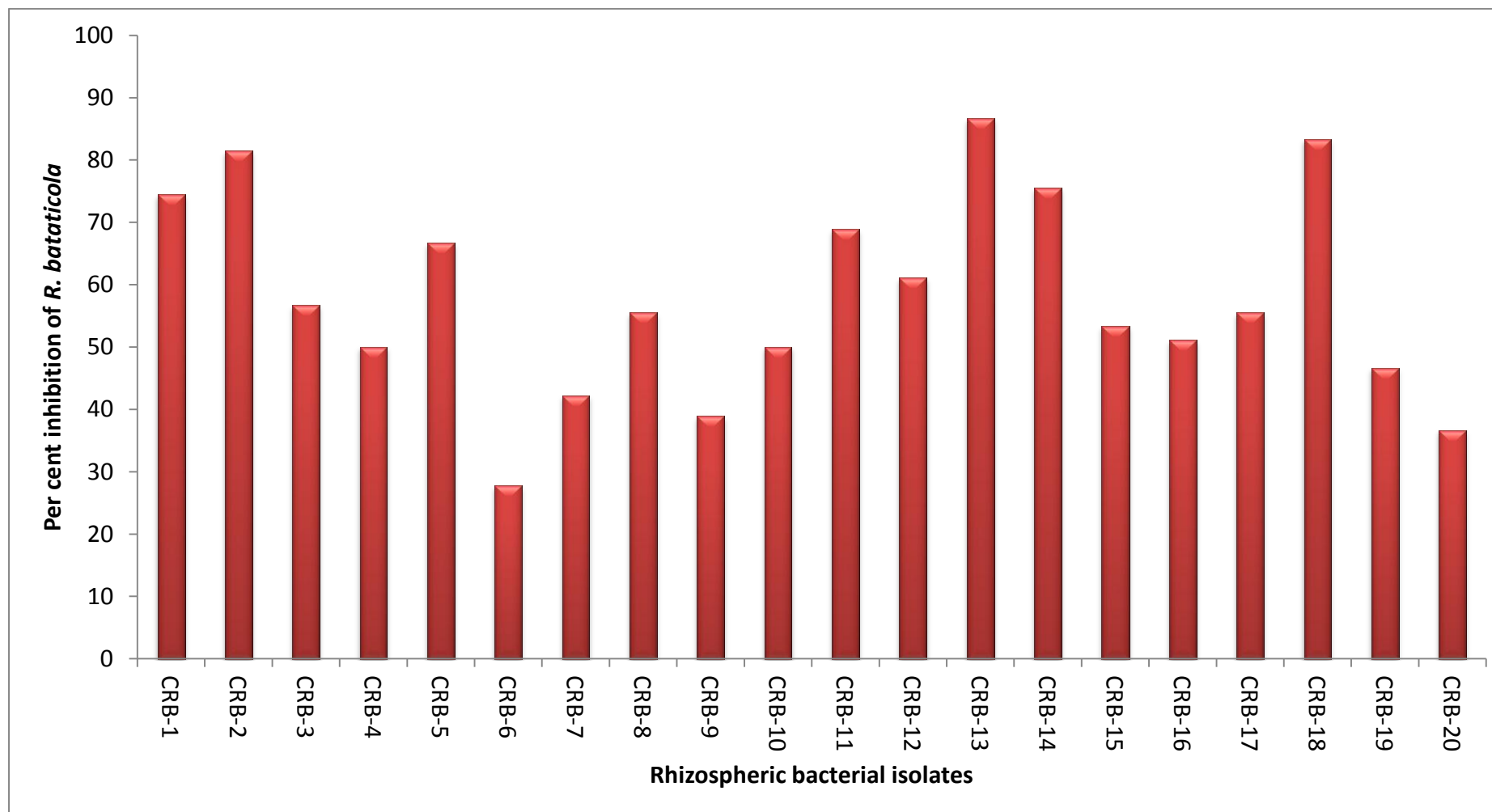


Fig. 4.2. *In vitro* evaluation of efficacy of chickpea rhizospheric bacteria against *Rhizoctonia bataticola* in dual culture technique

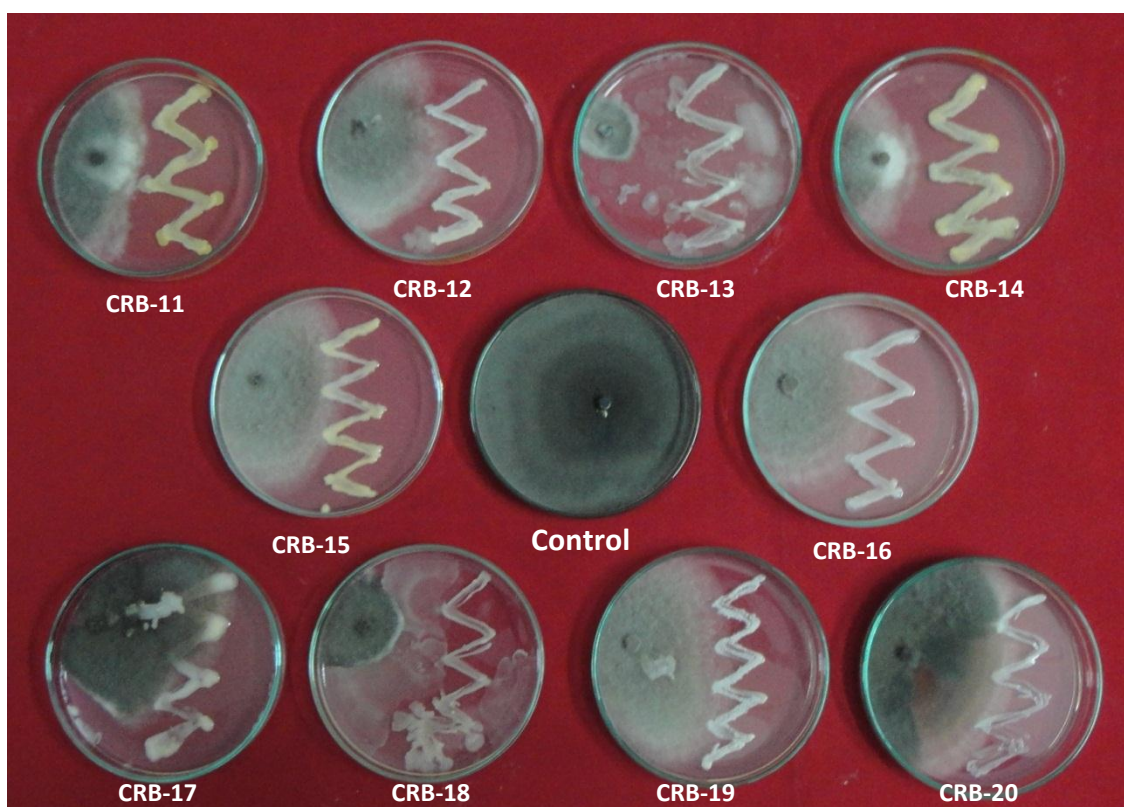
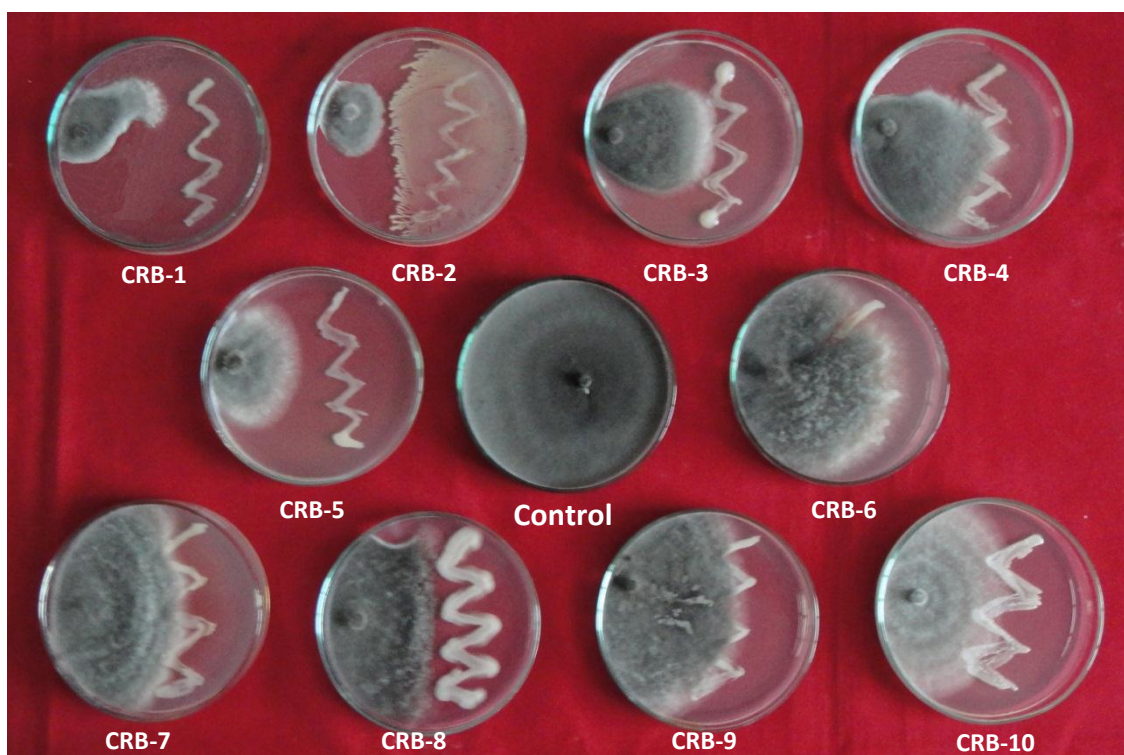


Plate 20: *In vitro* efficacy of rhizospheric bacterial isolates on mycelial growth of *Rhizoctonia bataticola* in dual culture technique

Table 4.4. *In vitro* evaluation of efficacy of chickpea root endophytic bacterial isolates against *Rhizoctonia bataticola* in dual culture technique

S. No	Bacterial antagonists	Linear growth of <i>R. bataticola</i> (Cm)*	Per cent inhibition of mycelial growth of <i>Rhizoctonia bataticola</i>
1	CREB-1	2.8	68.88 (56.13)
2	CREB-2	3.0	66.66 (54.74)
3	CREB-3	3.5	61.11 (51.41)
4	CREB-4	3.6	60.00 (50.77)
5	CREB-5	4.8	46.66 (43.07)
6	CREB-6	0.5	94.44 (76.50)
7	CREB-7	4.3	52.22 (46.27)
8	CREB-8	4.6	48.88 (44.35)
9	CREB-9	1.9	78.88 (62.67)
10	CREB-10	3.2	64.44 (53.44)
11	CREB-11	4.7	47.77 (43.71)
12	CREB-12	5.0	44.44 (41.75)
13	CREB-13	0.4	95.55 (77.88)
14	CREB-14	4.8	46.66 (43.08)
15	CREB-15	3.3	63.33 (52.76)
16	CREB-16	0.6	93.33 (75.06)
17	CREB-17	1.4	84.44 (66.82)
18	CREB-18	1.2	86.66 (68.66)
19	CREB-19	2.0	77.77 (61.90)
20	CREB-20	2.6	71.11 (57.49)
21	Control	9.0	-
	S. Em $\pm$		1.40
	CD (0.05)		3.94

* Mean of three replications

Figures in parenthesis are angular transformed values

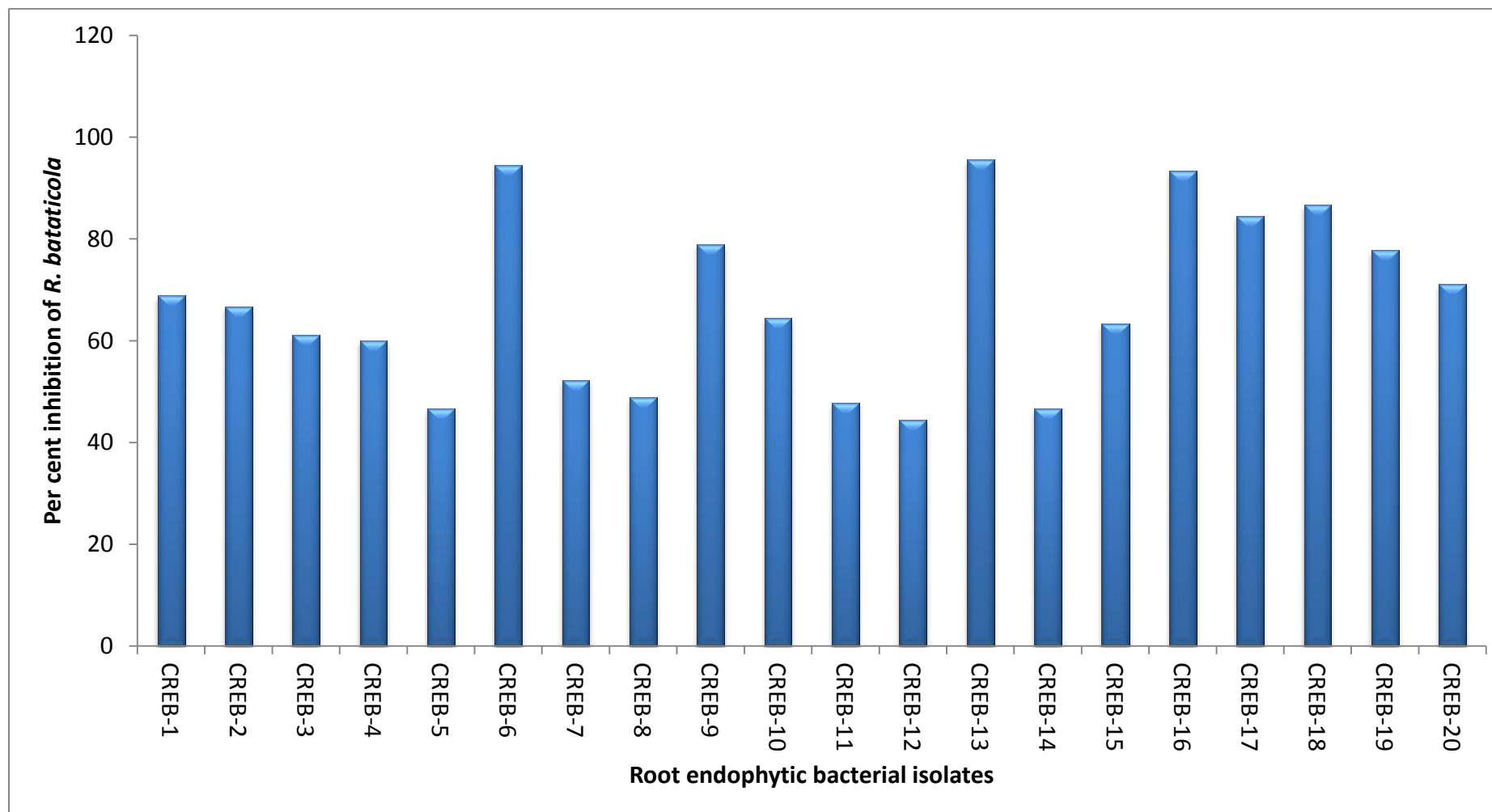


Fig. 4.3. *In vitro* evaluation of efficacy of chickpea root endophytic bacteria against *Rhizoctonia bataticola* in dual culture technique

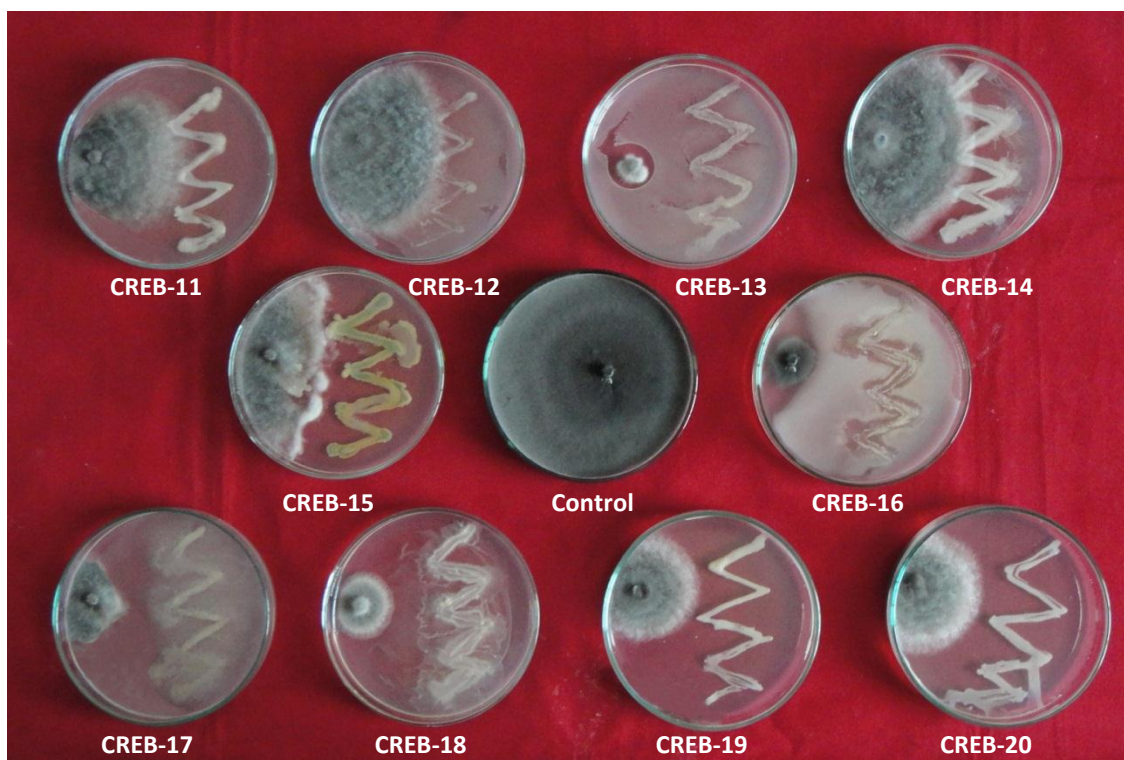
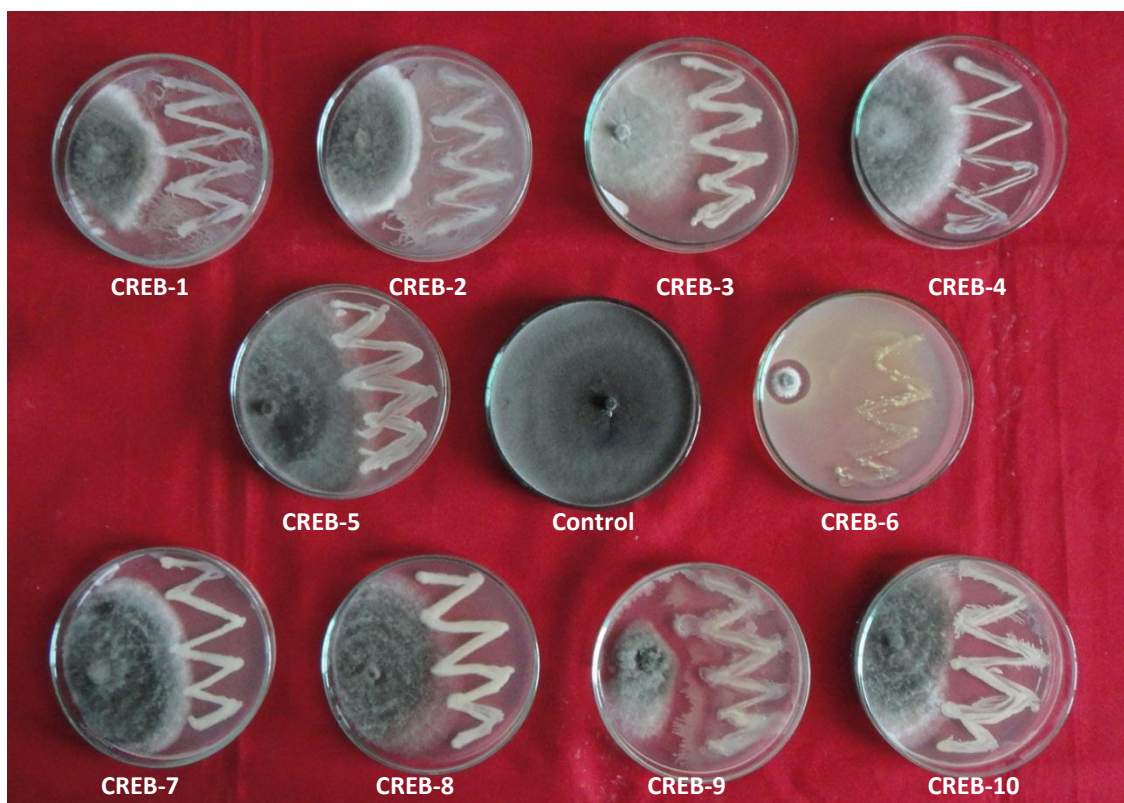


Plate 21: *In vitro* efficacy of root endophytic bacterial isolates on mycelial growth of *Rhizoctonia bataticola* in dual culture technique

Pseudomonas fluorescens isolates against *M. phaseolina* incitant of charcoal rot of chickpea under *in vitro* and reported that the isolate Pf 4-99 was found to be effective in inhibiting the mycelial growth of pathogen. Uppala *et al.* (2009) isolated 46 bacterial endophytes from amaranth and screened against *Rhizoctonia solani* under *in vitro* conditions using dual culture method. Among the endophytes screened, only six bacteria showed antagonistic activity against *R. solani*.

4.5 IN VITRO EVALUATION OF THE COMPATIBILITY OF POTENTIAL ANTAGONISTS WITH DIFFERENT FUNGICIDES.

A biocontrol agent must be effective and compatible with latest crop production practices so that its use can be integrated into the production system. In such an approach, biocontrol agents have been used without any toxic effect on antagonists (Papavizas and Lumsden, 1980).

4.5.1 *In vitro* evaluation of the compatibility of potential fungal antagonists with different fungicides.

The highly potential fungal antagonist *Trichoderma isolate-7* (CT7) was selected for fungicidal compatibility studies since it has shown maximum inhibition of *Rhizoctonia bataticola* growth in dual culture studies when compared to all other antagonists. Poisoned food technique was used to evaluate the compatibility of *Trichoderma isolate-7* (CT7) with different fungicides and the results are presented in Table 4.5 and Fig 4.4.

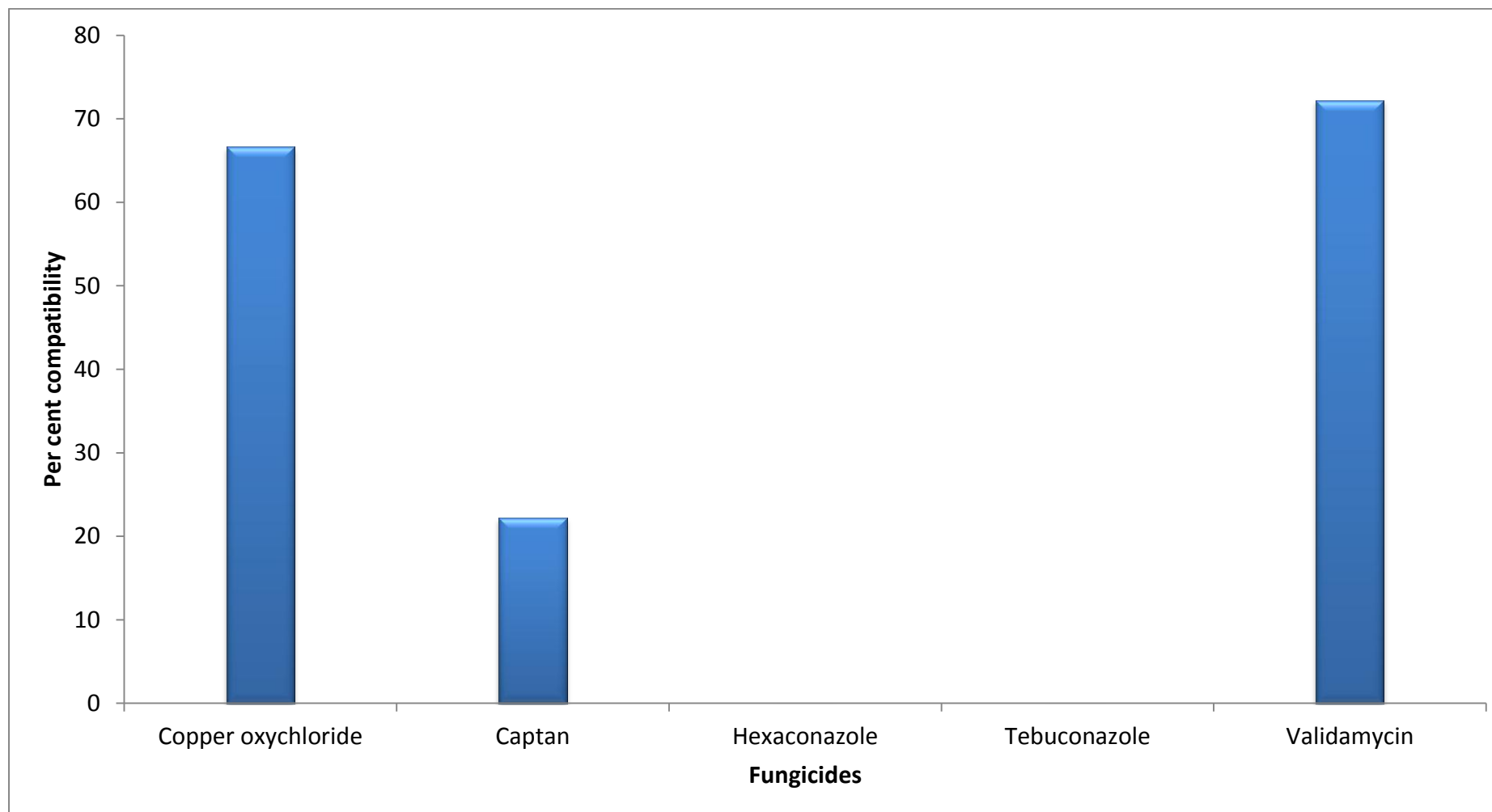
The results revealed that the *Trichoderma isolate-7* (CT7) was more compatible with validamycin (72.22%), followed by copper oxychloride (66.66%). The *Trichoderma isolate-7* (CT7) showed complete incompatibility with the fungicides hexaconazole and tebuconazole, whereas the fungicide captan has shown 22.22 per cent compatibility (Plate 22).

Table 4.5. *In vitro* evaluation of compatibility of the potential fungal antagonist *Trichoderma* isolate-7 (CT7) with different fungicides in poisoned food technique

S. No.	Fungicides	Linear growth of <i>Trichoderma</i> isolate-7 (CT7) (Cm)*	Per cent compatibility
1	Copper oxychloride (0.25%)	6.0	66.66 (54.77)
2	Captan (0.25%)	2.0	22.22 (28.04)
3	Hexaconazole (0.2%)	0	0.00 (0.00)
4	Tebuconazole (0.1%)	0	0.00 (0.00)
5	Validamycin (0.1%)	6.5	72.22 (58.24)
6	Control	9.0	-
	S.Em $\pm$		1.61
	CD (0.05)		5.26

* Mean of three replications

Figures in parenthesis are angular transformed values



85 **Fig. 4.4.** *In vitro* evaluation of compatibility of the potential fungal antagonist *Trichoderma* isolate-7 (CT7) with different fungicides in poisoned food technique



Plate 22: *In vitro* evaluation of compatibility of the potential fungal antagonist *Trichoderma* isolate-7 (CT7) with different fungicides in poisoned food technique

The present results were in agreement with Tiwari and Singh (2004) who reported that the mycelial growth of *T. harzianum* was completely inhibited by carbendizam and hexaconazole @ 1500 ppm, whereas copper oxychloride and mancozeb @ 1500 ppm has shown 90 and 41 per cent inhibition respectively. Naseema Beevi *et al.* (2005) tested the *in vitro* compatibility of *T. harzianum* with mancozeb, carbendazim and copper oxychloride and found that carbendazim at 0.1 per cent completely inhibited the mycelial growth while mancozeb and copper oxychloride showed compatibility with the antagonist at 0.2 and 0.1 per cent respectively. Parab *et al.* (2009) studied the sensitivity of *T. harzianum* to different fungicides and the results indicated that all systemic fungicides were found completely inhibiting growth of *T. harzianum* whereas zineb, copper hydroxide, mancozeb and copper oxychloride were found safe for the growth of *T. harzianum*. Gaur and Sharma (2010) tested the *in vitro* compatibility of *Trichoderma harzianum* (TG-1) against different fungicides. Metalaxyl, fosetyl-Al, mancozeb, cymoxanil 8% + Mancozeb 64% mixture and copper oxychloride fungicides were found compatible with tolerance limits (ED₅₀) of >1000 µg/ml.

4.5.2 *In vitro* evaluation of the compatibility of potential bacterial antagonists with different fungicides.

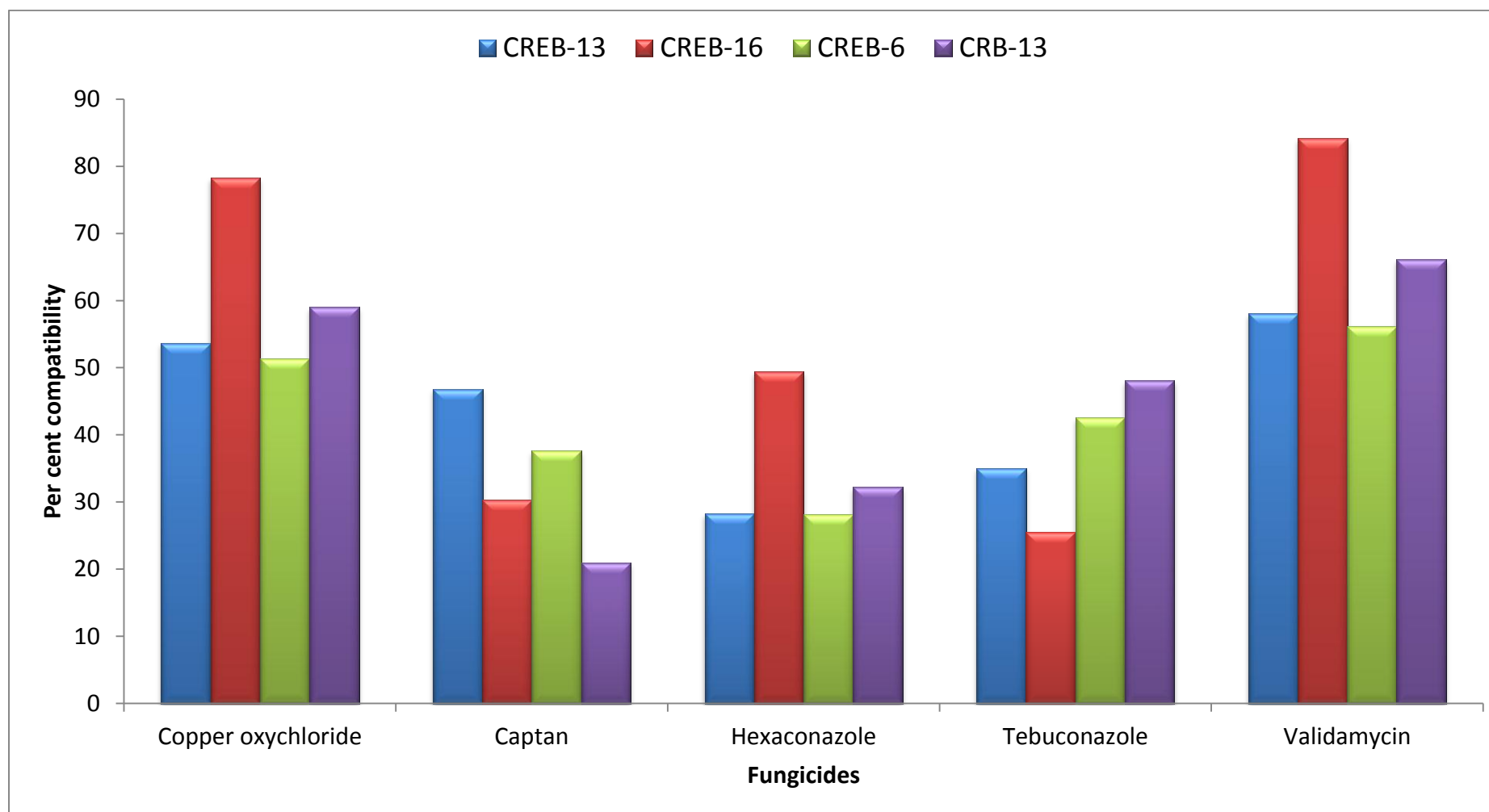
The highly potential antagonistic bacteria CREB-13, CREB-6, CREB-16 from root endophytic isolates and CRB-13 from rhizosphere isolates were selected for fungicidal compatibility studies since they have shown maximum inhibition of *Rhizoctonia bataticola* growth in dual culture studies when compared to all other antagonists. Spectrophotometric method was used to evaluate the compatibility of CREB-13, CREB-6, CREB-16 and CRB-13 bacterial isolates with different fungicides and the results are presented in Table 4.6 and Fig 4.5.

Table 4.6. *In vitro* evaluation of the compatibility of the potential bacterial isolates CRB-13, CREB-6, CREB-13 and CREB-16 with different fungicides by spectrophotometry

S. No.	Fungicides	CRB-13*	% Compatibility	CREB-6*	% Compatibility	CREB-13*	% Compatibility	CREB-16*	% Compatibility
1	Copper oxychloride (0.25%)	0.732	59.03 (50.24)	0.845	51.33 (45.76)	0.797	53.70 (47.18)	1.135	78.27 (63.25)
2	Captan (0.25%)	0.260	20.96 (27.19)	0.620	37.66 (37.84)	0.695	46.83 (43.17)	0.440	30.34 (33.37)
3	Hexaconazole (0.2%)	0.400	32.25 (34.49)	0.463	28.12 (32.00)	0.420	28.30 (32.09)	0.716	49.37 (44.64)
4	Tebuconazole (0.1%)	0.596	48.06 (43.81)	0.700	42.52 (40.68)	0.520	35.04 (36.28)	0.370	25.51 (30.15)
5	Validamycin (0.1%)	0.820	66.12 (54.74)	0.925	56.19 (48.59)	0.862	58.08 (49.65)	1.220	84.13 (68.25)
6	Control	1.240		1.646		1.484		1.450	
	S. Em $\pm$		3.04		1.48		1.44		3.98
	CD (0.05)		9.93		4.84		4.72		12.99

* Mean of three replications

Figures in parenthesis are angular transformed values



68 **Fig. 4.5. *In vitro* evaluation of the compatibility of potential bacterial isolates CREB-13, CREB-16, CREB-6 and CRB-13, with different fungicides by spectrophotometry**

Higher OD values at 600 nm indicate high compatibility of the antagonist with that specific fungicides.

It is evident from the data (Table 4.6) that the isolate CREB-13 was more compatible with validamycin (58.08%) followed by copper oxychloride (53.70%) and least compatible with hexaconazole (28.30%) and tebuconazole (35.04%).

The isolate CREB-6 was more compatible with validamycin (56.19%) followed by copper oxychloride (51.33%), while it has shown least compatibility with hexaconazole (28.12%).

The isolate CREB-16 was more compatible with validamycin (84.13%) followed by copper oxychloride (78.27%). However, both were found to be on par with each other. It has shown least compatibility with tebuconazole (25.51%).

In the case of CRB-13, more compatibility was recorded with validamycin (66.12%) that was on par with copper oxychloride (59.03%) and least compatible with captan (20.96%).

The present results indicated that the isolate CREB-16 was found to be most compatible antagonistic bacteria as it has shown highest per cent compatibility when compared to CRB-13, CREB-13 and CREB-6 isolates. Hence the isolate CREB-16 was selected for further studies.

The fungicide validamycin showed highest compatibility with all the four isolates followed by copper oxychloride and less compatibility was recorded in case of captan, tebuconazole and hexaconazole.

Similarly Vidhyasekaran and Muthamilan (1995) reported that thiram and carbendazim were not inhibitory to *P. fluorescens* under *in vitro* conditions. Dean *et al.* (2006) carried out *in vitro* compatibility of the antagonistic bacterium *Bacillus subtilis* BSF4 strain with some fungicides and

the results showed that it was compatible with azoxystrobin, penconazole, triadimenol and sulphur fungicides. Khan and Gangopadhyay (2008) who reported that carbendazim and carboxin were least toxic to *Pseudomonas fluorescens* with the fungicides and revealed that carboxin and carbendazim were least toxic to *Pseudomonas fluorescens* strain PFBC-25 whereas captan was most inhibitory.

The *in vitro* compatibility results of both fungal and bacterial isolates revealed that the fungicide validamycin and copper oxychloride are on par statistically with respect to compatibility with CREB-16 isolate and *Trichoderma* isolate-7. Moreover, validamycin is available in liquid formulation which is difficult for seed treatment, hence the next best compatible fungicide copper oxychloride which is available in wettable powder formulation was selected to integrate in integrated disease management.

4.6 IN VITRO MODE OF ACTION OF POTENTIAL ANTAGONISTS AGAINST *Rhizoctonia bataticola*

4.6.1 Mode of action of *Trichoderma* isolate-7 (CT7) against *Rhizoctonia bataticola* under *in vitro*

The photomicrograph (Plate 23) obtained from slide prepared from dual culture as described in section 3.7.1 showed hyphal interaction between the antagonist and the pathogen. The mycelium of *Trichoderma* may coil around the mycelium of *Rhizoctonia bataticola* and later penetrate into it leading to disruption of mycelial contents of the pathogen.

Similar observations were made by Manjunatha and Naik (2010) who selected two isolates of *Trichoderma* sp, viz., *Trichoderma viride* and *Trichoderma harzianum* for intensive studies on the basis of their *in vitro* antagonistic activity towards *Rhizoctonia bataticola*, the inducer of chickpea dry root rot. On microscopic examination, these fungal antagonists were found parasitic on *Rhizoctonia bataticola* as evidenced by coiling,

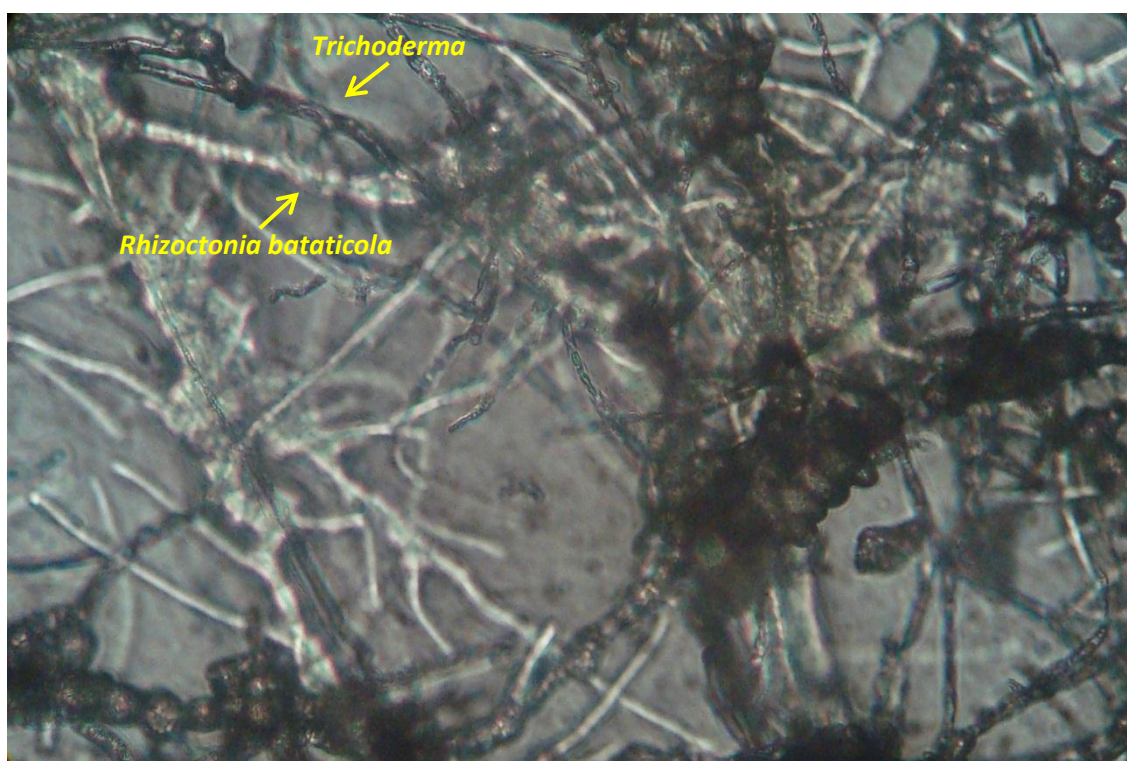


Plate 23: Photomicrograph showing mycoparasitism of *Trichoderma* isolate-7 (CT7) on *Rhizoctonia bataticola*

penetration, disintegration and overlapping on germinating sclerotia as well as hyphae of the pathogen. Manoranjitham *et al.* (2000) reported the antagonistic activity of *Trichoderma* spp. and *Gliocladium virens* against a number of soil borne pathogens. Similarly Pant and Mukhopadhyay (2001) reported that production of knob like, haustorial structure with infection peg followed by penetration of host hyphae and finally coagulation, vacuolation and lysis of host protoplasm as the principal mechanism in the mycoparasitism of *Rhizoctonia solani* by *Trichoderma virens*.

Bose *et al.* (2005) reported strong antagonism by *Trichoderma* spp. against a range of soilborne plant pathogens. Chakravarthy and Nagamani (2007) reported that the inhibition of *Rhizoctonia solani* by *Trichoderma* spp. was due to competition for food, space, production of antibiotics and mycoparasitism.

4.6.2 Antibiosis

Antibiosis is the antagonism mediated by specific or non-specific metabolites of microbial origin, viz., lytic agents, enzymes, volatile compounds or other toxic substances.

The effect of volatile and non-volatile metabolites produced by the potential antagonistic bacteria on *Rhizoctonia bataticola* was studied under *in vitro*.

4.6.2.1 Effect of volatile metabolites of potential antagonistic bacteria against *Rhizoctonia bataticola*

The *in vitro* experimental results (Table 4.7 and Fig 4.6) revealed that volatile metabolites produced by antagonistic bacteria CREB-16 has shown maximum inhibition of the mycelial growth of *Rhizoctonia bataticola* to an extent of 77.77 per cent, followed by CRB-6 (72.22 per cent) and CREB-13 (71.11 per cent). CRB-13 showed least inhibition (55.55 per cent) against

Rhizoctonia bataticola (Plate 24). The inhibitory effect of the bacterial antagonist on *Rhizoctonia bataticola* may be due to the production of some volatile metabolites like hydrogen cyanide.

Prasanna Reddy *et al.* (2010) isolated eight *Pseudomonas fluorescens* strains from rhizosphere of rice seedlings collected from Andhra Pradesh and TamilNadu. Crude metabolites from one particular isolate of *Pseudomonas fluorescens* (P.f 003) gave 78 per cent inhibition against *Rhizoctonia solani* compared to control.

4.6.2.2 Effect of non volatile compounds of potential antagonistic bacteria against *Rhizoctonia bataticola*

The results of the effect of non-volatile compounds of the bacterial antagonist against the test pathogen are presented in Plate 25 and Table 4.8. The effect of culture filtrates of antagonistic bacteria on radial growth of *Rhizoctonia bataticola* indicated that the antagonists did not have any inhibitory action on the growth of *Rhizoctonia bataticola* over control under *in vitro*.

The results revealed that there was no reciprocal relationship between the culture filtrates of antagonistic bacteria and the radial growth of *Rhizoctonia bataticola* indicating that they did not produced any inhibitory non-volatile metabolites against *Rhizoctonia bataticola*.

4.7 IDENTIFICATION OF COMPATIBILITY BETWEEN POTENTIAL FUNGAL AND BACTERIAL ANTAGONISTS UNDER *IN VITRO*

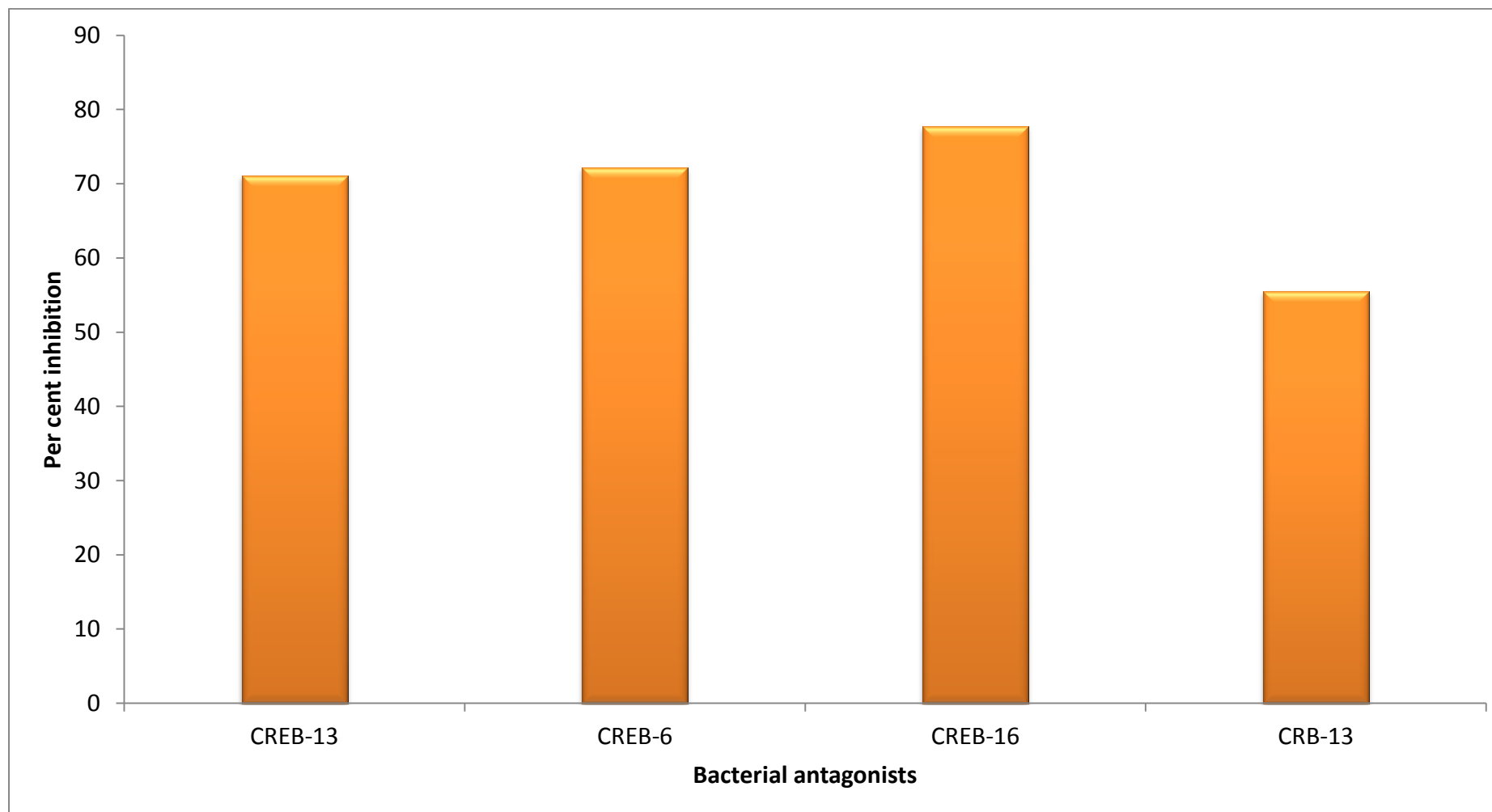
The compatibility between potential fungal and bacterial antagonists was determined by dual culture technique (Morton and Stroube, 1995) under *in vitro* conditions. The highly potential and fungicidal tolerant bacteria CREB-13, CREB-6 and CREB-16 were selected for compatibility studies. Among the fungal antagonists *Trichoderma* isolate -7 (CT7) was selected

Table 4.7. *In vitro* evaluation of efficacy of volatile metabolites produced by antagonistic bacteria on the growth of *Rhizoctonia bataticola*

S. No.	Bacterial Antagonists	Linear growth of <i>Rhizoctonia bataticola</i> (Cm)*	Per cent inhibition
1	CREB-13	2.6	71.11 (57.48)
2	CREB-6	2.5	72.22 (58.18)
3	CREB-16	2.0	77.77 (61.82)
4	CRB-13	4.0	55.55 (48.16)
5	Control	9.0	-
	S. Em $\pm$		0.68
	CD (0.05)		2.37

* Mean of three replications

Figures in parenthesis are angular transformed values



96 **Fig. 4.6.** *In vitro* evaluation of efficacy of volatile metabolites produced by antagonistic bacteria on the growth of *Rhizoctonia bataticola*

Table 4.8. *In vitro* evaluation of efficacy of non-volatile metabolites produced by antagonistic bacteria on the growth of *Rhizoctonia bataticola*

S. No.	Bacterial Antagonists	Linear growth of <i>Rhizoctonia bataticola</i> (Cm)	Per cent inhibition
1	CREB-13	9.0	0.00
2	CREB-6	9.0	0.00
3	CREB-16	9.0	0.00
4	CRB-13	9.0	0.00
5	Control	9.0	-

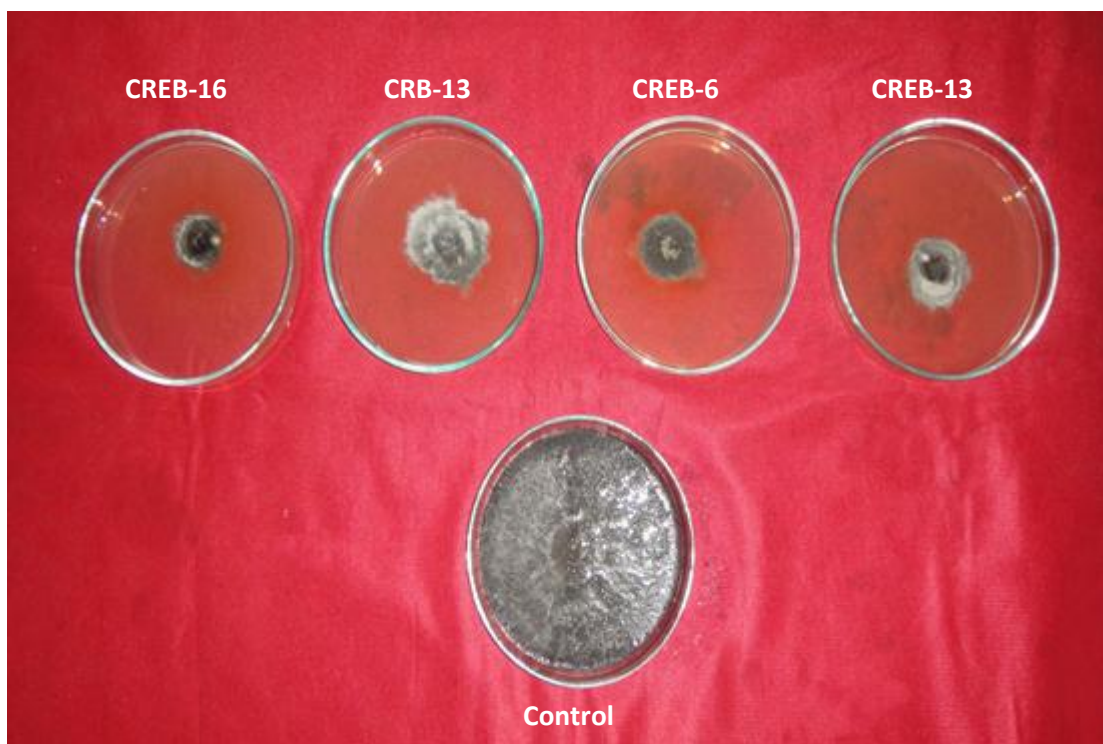


Plate 24: *In vitro* evaluation of efficacy of volatile metabolites produced by antagonistic bacteria on the growth of *Rhizoctonia bataticola*

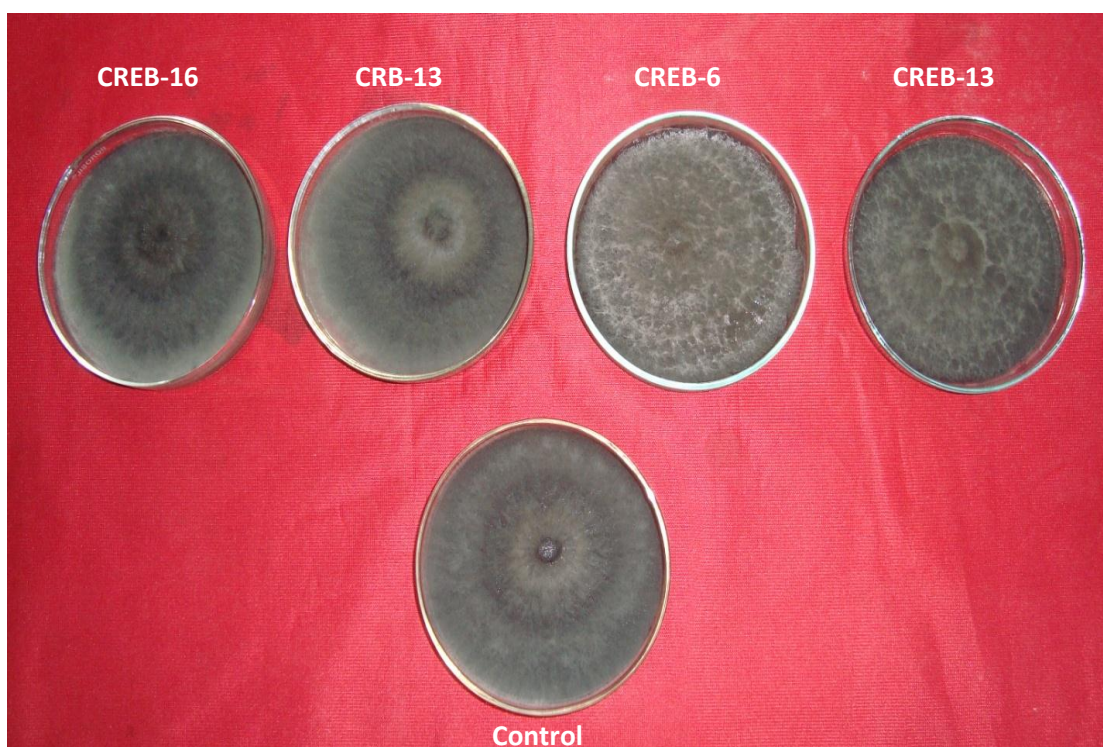


Plate 25: *In vitro* evaluation of efficacy of non volatile metabolites produced by antagonistic bacteria on the growth of *Rhizoctonia bataticola*

since it has shown maximum inhibition of *Rhizoctonia bataticola* growth in dual culture studies when compared to all other antagonists and the results are presented in the Table 4.9 and Fig 4.7.

The results revealed that the *Trichoderma* isolate-7 (CT7) has shown highest compatibility with the bacterial isolate CREB-16 (72.22%), followed by CREB-13 (66.66%) and 50% compatibility with the bacterial isolate CREB-6 (Plate 26).

The bacterial isolate CREB-16 was selected to integrate in IDM as it has shown highest compatibility (72.2%) with the *Trichoderma isolate-7* (CT7).

Similar observations were made by Pan and Das (2010) who reported that the highest growth inhibition (59%) of *Trichoderma* (Th3) was noted when *Pseudomonas fluorescens* was first inoculated in KMB medium 24 h before inoculation of *Trichoderma harzianum* in the same medium. In contrast, the least inhibition (9%) was recorded when *Trichoderma viride* (Tv1) was first inoculated in KMB medium before inoculation of *Pseudomonas fluorescens*.

4.8 INTEGRATED MANAGEMENT OF DRY ROOT ROT OF CHICKPEA

The Chickpea variety JG-11 popularly grown in Rayalaseema region of A.P was used for pot culture studies. The different treatments were imposed as indicated in the section 3.9.1.

The chemical management is costly and leads to residual effect so in order to minimize the residual effects due to continuous and indiscriminate use of toxic chemicals for the disease management, it is worthwhile to search for an alternative method for the management of this disease. Integrated disease management is gaining importance which involves blending of compatible

Table 4.9. *In vitro* evaluation of compatibility between highly potential bacterial isolates CREB-6, CREB-13, CREB-16 and fungal antagonist *Trichoderma* isolate-7 (CT7)

S. No	Bacterial antagonists	Linear growth of <i>Trichoderma</i> isolate-7 (CT7) (Cm)*	Per cent compatibility
1	CREB-6	4.5	50.00 (45.00)
2	CREB-13	3.0	33.33 (35.24)
3	CREB-16	6.5	72.22 (58.21)
4	Control	9.0	-
	S. Em $\pm$		0.99
	CD (0.05)		3.90

* Mean of three replications

Figures in parenthesis are angular transformed values

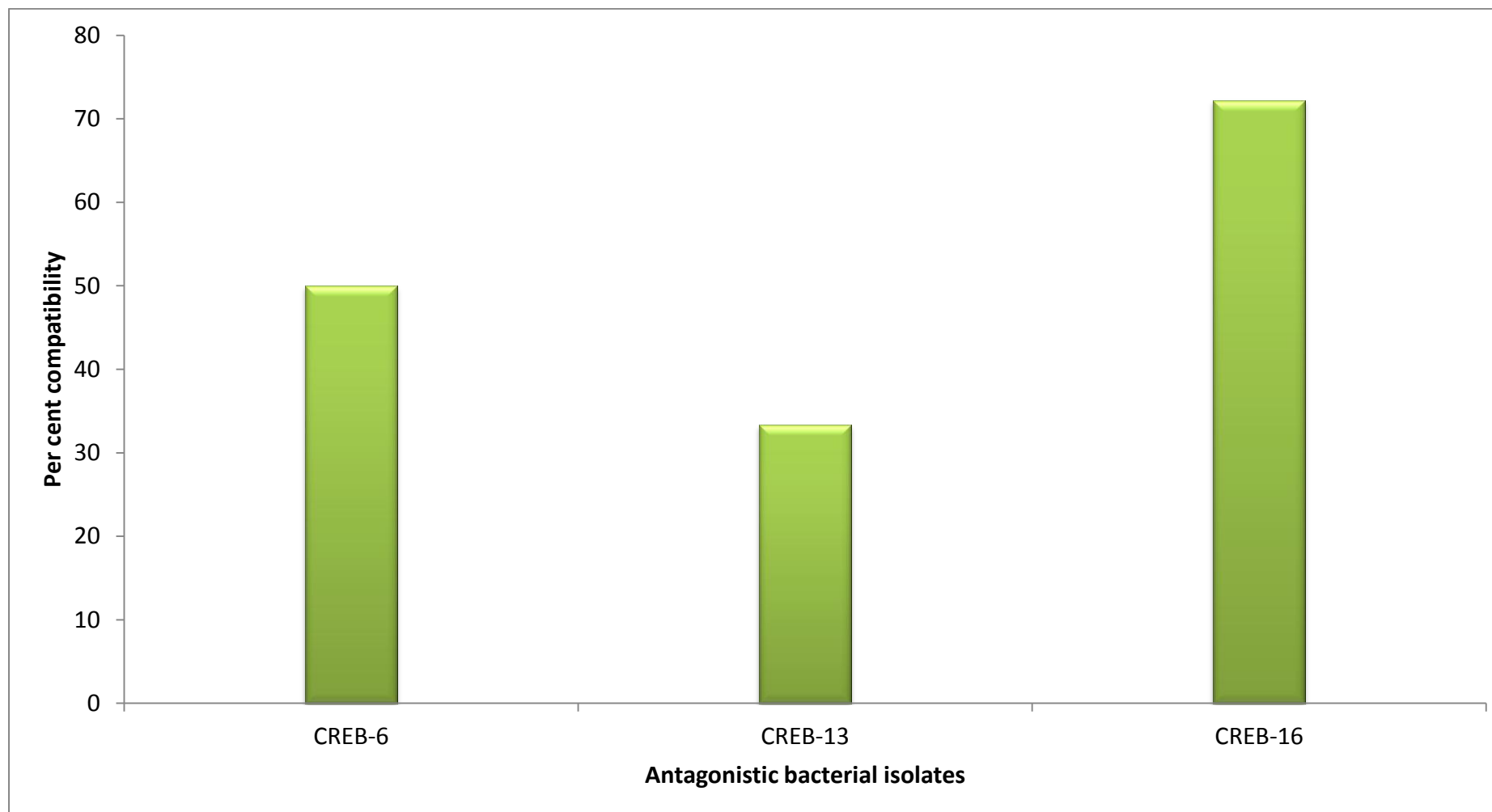


Fig. 4.7. *In vitro* evaluation of compatibility between highly potential bacterial isolates CREB-6, CREB-13, CREB-16 and fungal antagonist *Trichoderma* isolate-7 (CT7)



a) *Trichoderma* CT7 vs. CREB-6



b) *Trichoderma* CT7 vs. CREB-13



c) *Trichoderma* CT7 vs CREB-16

Plate 26: *In vitro* evaluation of compatibility between highly potential fungal antagonist *Trichoderma* isolate-7 (CT7) and bacterial isolates

systems of control measures for effective management of disease from profitability to food and environmental safety (Jacksen and Backmon, 1993).

In this contest the biological control, organic amendments, along with fungicidal treatment was found to be a more reliable approach to manage soil borne plant pathogens (Mukhopadhyay, 1987).

4.8.1 Mass multiplication of *Rhizoctonia bataticola*

The pathogen was mass multiplied on sorghum seeds (Plate 27) and added to soil @ 100 g kg⁻¹ at the time of sowing.

Rajeswari *et al.* (1999) and Durai, (2004) reported that sterilized sorghum seeds were found to be suitable for mass multiplication of *R. bataticola*. Ram and Pandey (2010) reported that sand maize medium was used for mass multiplication of *Fusarium solani* incitant of black root rot of chickpea.

4.8.2 Preparation of talc based formulation of potential fungicidal tolerant fungal antagonist *Trichoderma* isolate-7 (CT7)

Talc based formulation of potential fungal antagonist *Trichoderma* isolate-7 (CT7) was prepared by following the procedure given in section 3.9.3 (Plate 28 and 29). The population was estimated in talc based formulation after preparation and before use and it was 2.0×10^5 cfu/g (Plate 30). The talc based formulation of potential fungal antagonist was applied @ 100 g per pot before sowing.

Panicker and Jeyarajan (1993) reported that FYM to be the most suitable carrier for *Trichoderma* multiplication. Gaur *et al.* (2005) used the talc based formulation of *Trichoderma harzianum* multiplied on yeast molasses broth containing 2×10^6 cfu/g talc for root rot control in chickpea. Chauhan *et al.* (2008) used 10 media for growth of fungal bioagent and reported that Potato Dextrose broth had maximum mycelial growth and



Plate 27: Mass multiplication of *Rhizoctonia bataticola* on sorghum grains



Plate 28: Mass multiplication of *Trichoderma* isolate-7 (CT7) on Potato dextrose broth



Plate 29: Talc based formulation of *Trichoderma* isolate-7 (CT7)



Plate 30: Population of potential fungal antagonist *Trichoderma* isolate-7 (CT7) at the time of application in talc based formulation (cfu/g)

sporulation. In general Potato Dextrose agar and Cellulose Agar Medium were found to be best media for maximum growth of *Trichoderma* spp. for mass multiplication.

4.8.3 Preparation of talc based formulation of potential fungicide tolerant bacterial antagonist (CREB-16)

Talc based formulation of potential bacterial biocontrol agent (CREB-16) was prepared by following the procedure given in section 3.9.4 (Plate 31 and 32). The population was estimated in talc based formulation and it was 1.0×10^8 cfu/g (Plate 33). The talc based formulation of potential biocontrol agent (CREB-16) was applied @ 100 g per pot before sowing.

Similarly Rangeshwaran *et al.* (2001) also prepared talc based formulations of *Pseudomonas fluorescens* (PDBCA 2) and *Pseudomonas putida* (PDBCAB 19) which were effective in inhibiting the wilt and root rot of chickpea *in vitro*.

4.8.4 In vitro evaluation of efficacy of different seed treatments on seed germination by blotter technique

The effect of different seed treatments as indicated in the section 3.9.5 on per cent seed germination was calculated by blotter technique under *in vitro* conditions and the results are presented in Table 4.10 and Plate 34.

The results revealed that seed treatment with fungicide, biocontrol agents alone and in combination, has given 100 per cent seed germination under *in vitro* conditions. There was no significant difference between the treatments, all the treatments has given good germination percentage.

The seeds started germinating at 24 hours after incubation and the time taken for 100 per cent seed germination in all the treatments was 78 hours.



Plate 31: Mass multiplication of potential bacterial isolate CREB-16 on Nutrient broth



Plate 32: Talc based formulation of potential bacterial isolate CREB-16



Plate 33: Population of potential bacterial isolate (CREB-16) at the time of application in talc based formulation (cfu/g)

Table 4.10. *In vitro* evaluation of efficacy of different seed treatments on seed germination by blotter technique

S. No.	Treatment	Number of seeds germinated*	Per cent germination
1	Copper oxychloride @ 2.5 g/kg	5.00	100.00 (90.00)
2	<i>Trichoderma</i> isolate-7 (CT7) @ 10 g/kg	5.00	100.00 (90.00)
3	Bacterial isolate CREB-16 @ 10 g/kg	5.00	100.00 (90.00)
4	<i>Trichoderma</i> isolate-7 (CT7) @ 10 g/kg + Bacterial isolate CREB-16 @ 10 g/kg	5.00	100.00 (90.00)
5	Control	5.00	100.00 (90.00)
	S.Em $\pm$		-
	CD (0.05)		NS

* Mean of three replications

Figures in parenthesis are angular transformed values

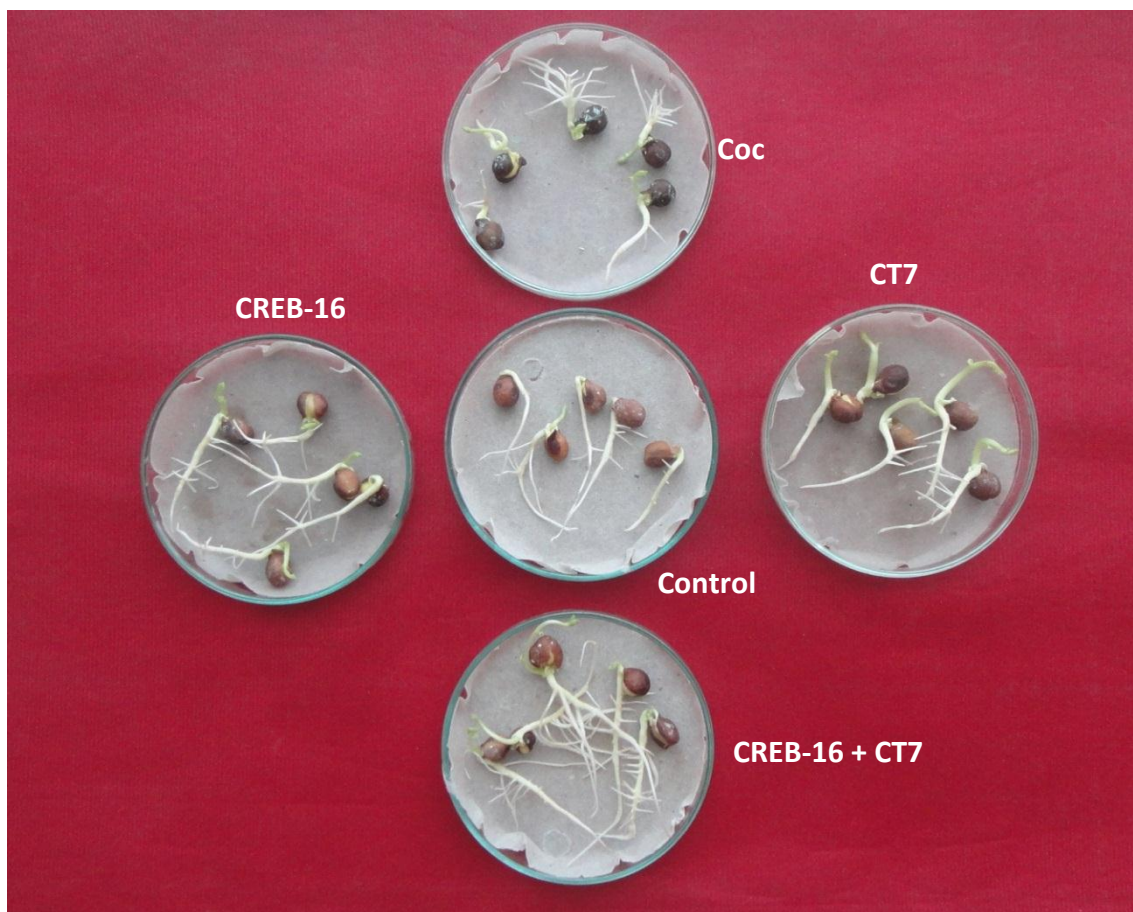


Plate 34: *In vitro* evaluation of efficacy of different seed treatments on seed germination by blotter technique

As antagonistic bacteria CREB-16 has shown more inhibition (93.33%) compared to *Trichoderma* isolate-7 (83.33%) under *in vitro* conditions in dual culture against the pathogen, hence it was selected as potential antagonist seed treatment in combination with organic amendments.

Similarly Rangeshwaran *et al.* (2001) reported that seed treatment with bioagents and fungicide gave 100 per cent germination of chickpea seedlings under *in vitro* conditions in the absence of pathogen. Nautiyal (1997) reported that seed bacterization with *Pseudomonas fluorescens* NBRI1303 increased the germination of chickpea by 25%.

4.8.5 Observations

The data on per cent germination, initial and final population of chickpea, per cent incidence of dry root rot and plant growth parameters *viz.*, plant height, root length, shoot dry weight and root dry weight of chickpea in each of the treatment were recorded and presented in Table 4.11 and Plate 35.

4.8.5.1 Per cent germination

Maximum per cent germination (100 per cent) was recorded in treatment T₈ (seed treatment with fungicide and soil application of potential fungal and bacterial biocontrol agent) and T₇ (soil application of potential fungal and bacterial biocontrol agent). There was no significant difference between the treatments T₆, T₅, T₁₄ and T₁₃. It is evident from Fig 4.8 that least germination percentage (60.00) was recorded in inoculated control (T₁₅).

Seed treatment with talc based formulations of biocontrol agents proved better and thus may serve as ecofriendly alternative to copper oxychloride.

Our results showed that the antagonists have some stimulatory effect on seed germination. This might be due to the production of some hormones.

Table 4.11. Effect of antagonists, fungicide and organic amendments on per cent germination, initial population, final population and per cent disease incidence of chickpea cv. JG-11 in *Rhizoctonia bataticola* infested soil in pot culture

Treatment	*Per cent germination	*Initial population	*Final population	*Per cent disease incidence
T1 : Seed treatment with potential fungal bio-control agent @ 10 g/kg	83.33 (66.14)	8.33 (16.77)	7.33 (15.65)	37.80 (37.93)
T2 : Seed treatment with potential bacterial bio-control agent @10g/kg	83.33 (70.07)	8.33 (16.73)	7.33 (15.70)	36.40 (37.10)
T3 : Seed treatment with potential fungal and bacterial bio-control agents @10g/kg	86.66 (68.85)	8.66 (17.11)	7.66 (16.06)	32.00 (34.44)
T4 : Seed treatment with fungicide	86.66 (72.28)	8.66 (17.09)	7.66 (16.02)	34.50 (35.96)
T5 : Soil application of potential fungal bio-control agent	93.33 (77.71)	9.33 (17.78)	8.66 (17.09)	21.40 (27.55)
T6 : Soil application of potential bacterial bio-control agent	96.66 (83.85)	9.66 (18.10)	9.00 (17.44)	20.50 (26.91)
T7 : Soil application of potential fungal and bacterial bio-control agents @10g/kg	100.00 (90.00)	10.00 (18.43)	9.66 (18.10)	18.20 (25.25)
T8 : T ₄ +T ₅ +T ₆	100.00 (90.00)	10.00 (18.43)	9.66 (18.10)	16.00 (23.55)
T9 : Application of Neem cake @ 12.5 q/ha	76.66 (61.21)	7.66 (16.06)	6.00 (14.09)	52.00 (46.14)
T10 : Application of FYM @ 5 t/ha	80.00 (63.92)	8.00 (16.40)	6.66 (14.95)	47.50 (43.56)
T11 : Application of Vermicompost @ 5 q/ha	80.00 (63.43)	8.00 (16.42)	6.66 (14.92)	49.80 (44.88)
T12 : Seed treatment with potential bacterial bio-control agent and soil application of Neem cake	90.00 (71.56)	9.00 (17.45)	8.33 (16.77)	27.80 (31.81)
T13 : Seed treatment with potential bacterial bio-control agent and soil application of FYM	90.00 (75.00)	9.00 (17.44)	8.33 (16.73)	22.50 (28.31)
T14 : Seed treatment with potential bacterial bio-control agent and soil application of Vermicompost	90.00 (78.90)	9.00 (17.40)	8.33 (16.69)	25.60 (30.38)
T15 : Control	60.00 (51.14)	6.00 (14.09)	3.00 (9.88)	83.00 (65.66)
S Em ±	5.79	0.56	0.78	0.52
CD (0.05)	16.80	1.63	2.27	1.52

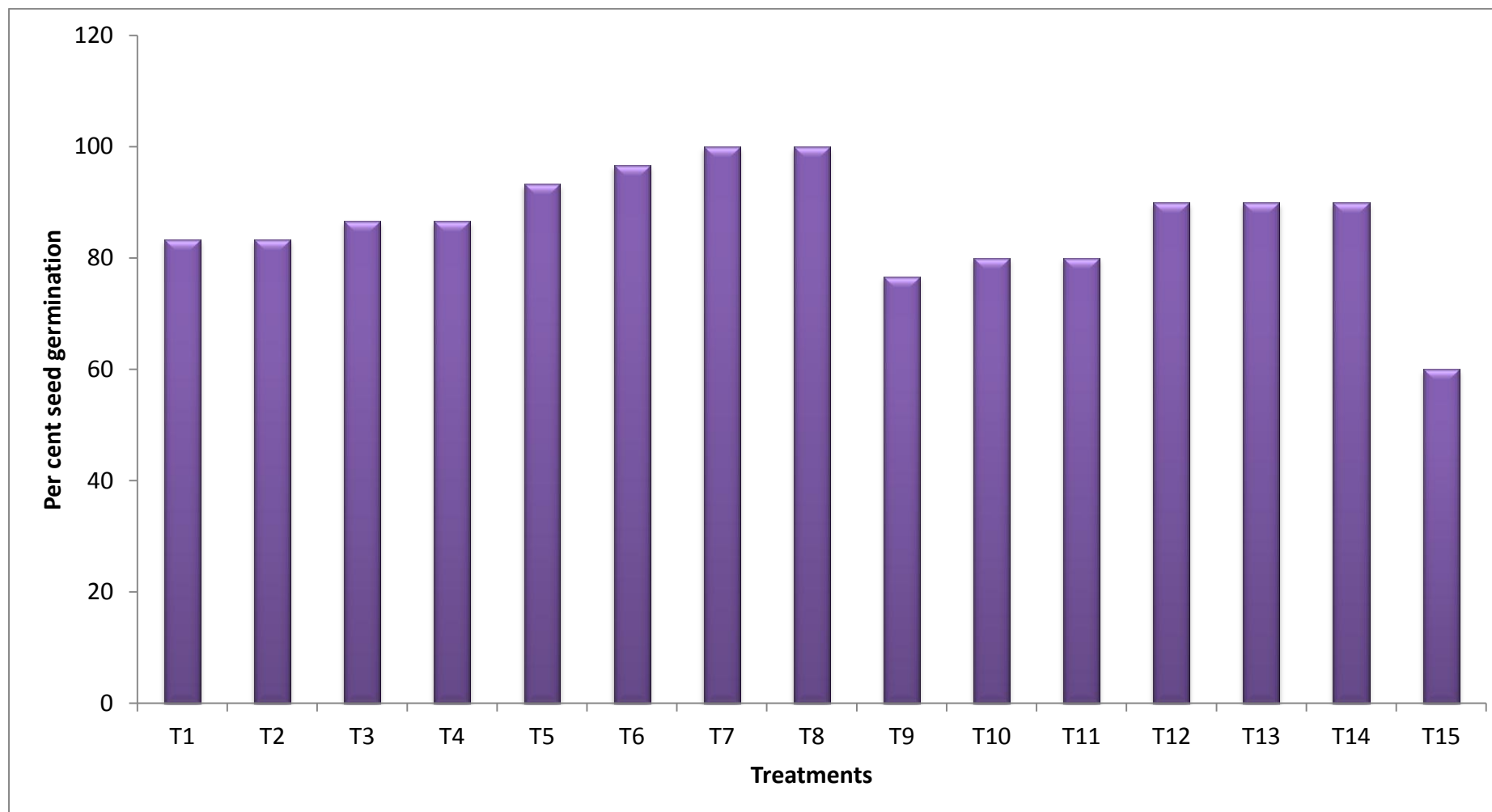


Fig. 4.8. Effect of antagonists, fungicide and organic amendments on per cent germination of chickpea in pot culture

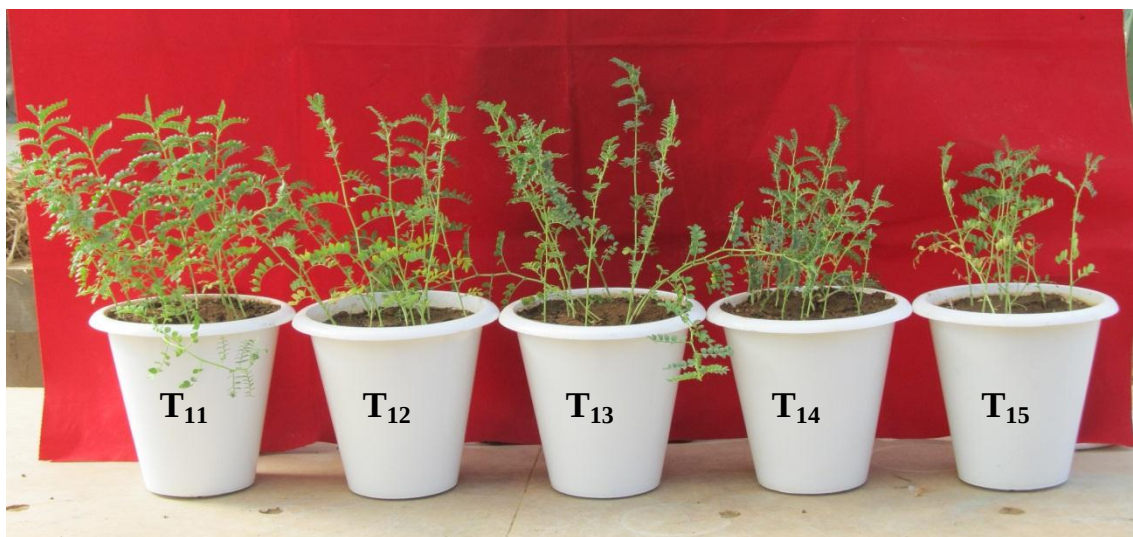


Plate 35: Pot culture experiment showing the integrated disease management of *Rhizoctonia bataticola* in chickpea (variety JG-11)

It is interesting to identify those compounds involved in stimulation of seed germination.

Bagri *et al.* (2004) reported that seed treatment with captan gave 86 per cent germination and minimum pre and post emergence mortality (11 and 5%) respectively against *Macrophomina phaseolina* causing root rot disease in chickpea. Gaur *et al.* (2005) reported that dry seed treatment with talc based formulation of *Trichoderma harzianum* @ 10 g/kg seed has given maximum germination percentage (84.8%).

4.8.5.2 Initial and final population

From the data (Table 4.11 and Fig 4.9) it is evident that initial population was highest in the treatments T₈ (10.00) and T₇ (10.00) and the lowest population was recorded in the treatment T₁₅ (6.00). Similarly regarding final population also, highest number was recorded in the treatments T₈ (9.66) and T₇ (9.66) and the lowest population was recorded in the treatment T₁₅ (3.00). The final population were in accordance with the per cent disease incidence in each treatment.

4.8.5.3 Per cent disease incidence

From the data (Table 4.11 and Fig 4.10) it is evident that all the treatments were significantly superior over control in reducing the per cent disease incidence. Maximum reduction was observed in treatment T₈ (seed treatment with fungicide and soil application of potential fungal and bacterial biocontrol agent) in which PDI of 16.00 per cent was recorded followed by T₇ (18.20%), T₆ (20.50%) and T₅ (21.40%) in comparison to 83.00 % in inoculated control.

The reasons for a good control of root rot by fungal and bacterial antagonists in combination with copper oxychloride in our study may be due to the different or complementary modes of action of antagonists or their

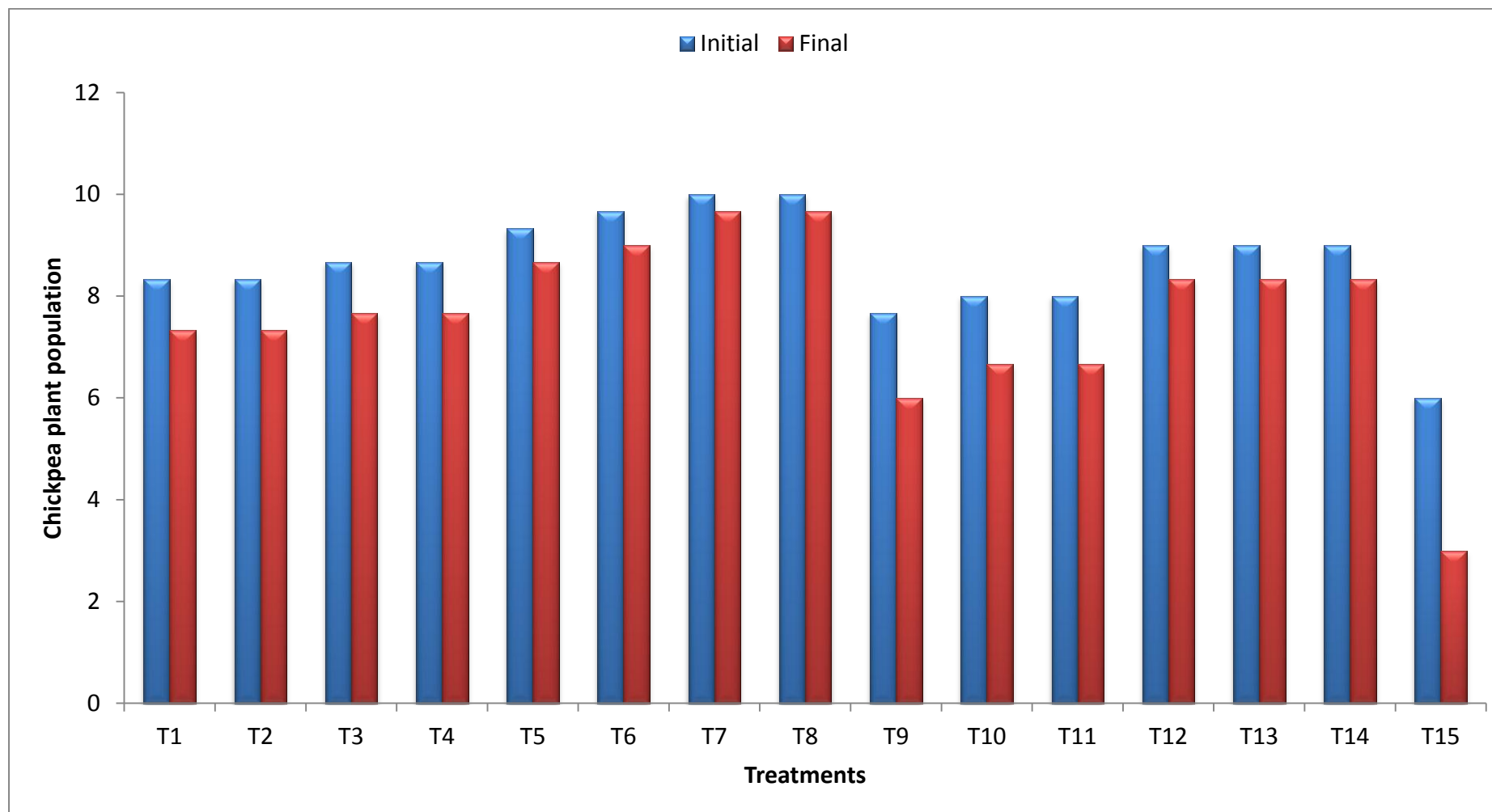


Fig. 4.9. Effect of fungicide, biocontrol agents and organic amendments on initial and final population of chickpea in pot culture

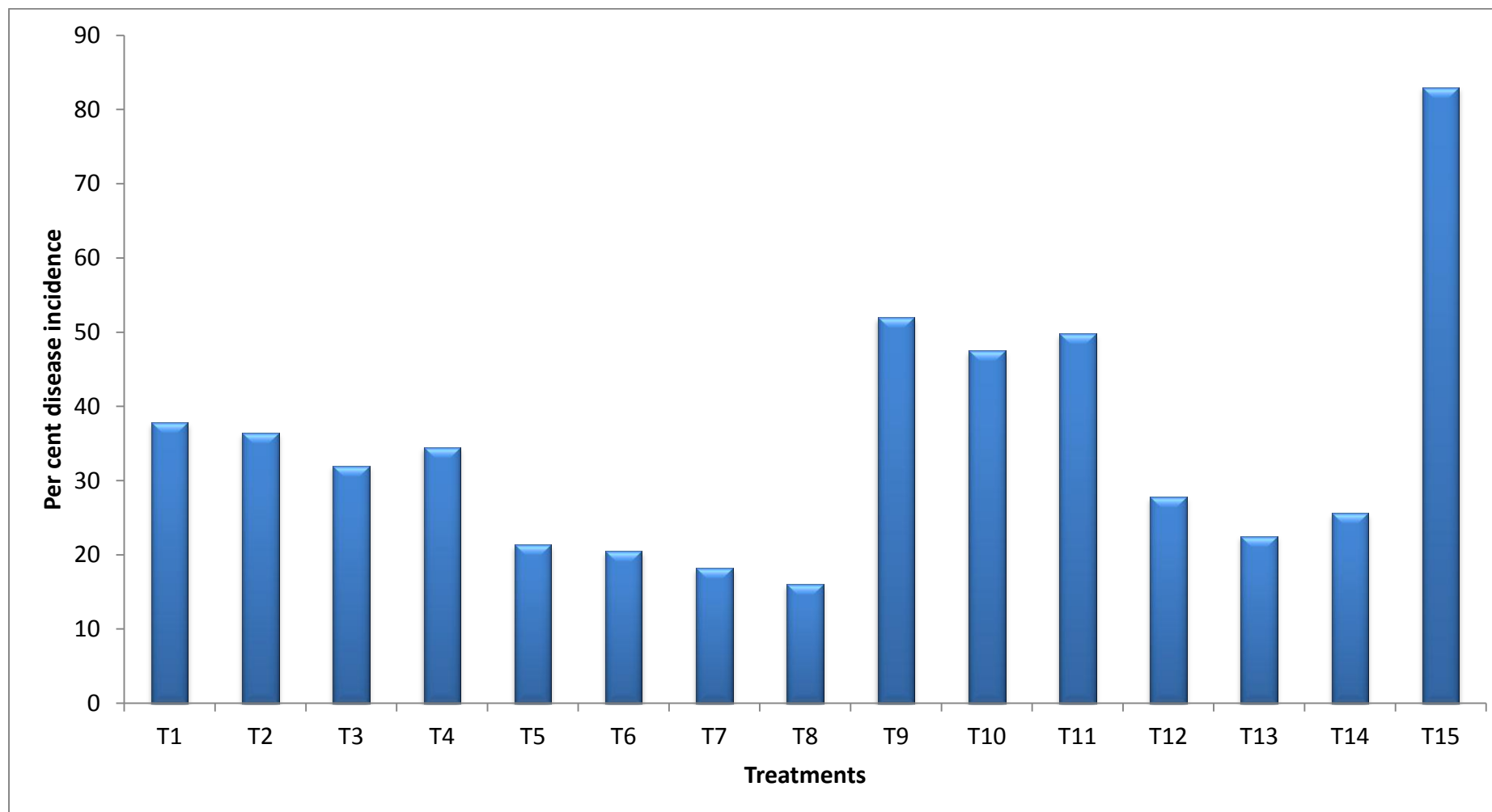


Fig. 4.10. Effect of antagonists, fungicide and organic amendments on per cent disease incidence of chickpea in pot culture

abilities to colonize root microsites or lesser sensitivity of antagonists to copper oxychloride or due to the sensitivity of *R. bataticola* as evident from poisoned food technique.

The results of per cent disease incidence in treatments including seed treatment with bacterial biocontrol agent and soil application of organic amendments revealed that lowest PDI was observed in T₁₃ (22.50%) i.e seed treatment with bacterial biocontrol agent and soil amendment of FYM, followed by T₁₄ (25.60%) i.e seed treatment with bacterial biocontrol agent and soil amendment of vermicompost, T₁₂ (27.80%) i.e seed treatment with bacterial biocontrol agent and soil amendment of neem cake.

Increase in disease control in combined treatments i.e seed treatment with bacterial biocontrol agent and soil amendment of FYM or neem cake or vermicompost observed may be due to enhanced population of antagonists or due to suppression of *Rhizoctonia bataticola* or due to inhibition of growth of *Rhizoctonia bataticola* by toxic and volatile products released during breakdown of organic amendment in soil.

Seed treatment with both fungal and bacterial biocontrol agents was found to be significantly superior in reducing the disease incidence when compared to individual treatments.

The per cent disease incidence was low when antagonists were applied in soil compared to seed treatment alone or in combination of seed treatment and soil application of organic amendments.

This might be due to the application of more number of antagonistic propagules which have multiplied at faster rate and inhibited the growth of pathogen.

Similar results have been reported by several workers in the management of several soil borne diseases by integration of biocontrol agents with chemicals.

Biocontrol potential of *Trichoderma* spp. against soil borne diseases of chickpea has been reported (Prasad and Rangeshwaran, 1999; Sindhan *et al.*, 2002). Sindhan *et al.* (2002) reported satisfactory disease control when *Pseudomonas fluorescens* was used as seed treatment along with carbendazim against *Macrophomina phaseolina* in chickpea. Gaur *et al.* (2005) reported that dry root rot of chickpea (16%) can be effectively managed by the soil application of 10-15 days pre-incubated *Trichoderma harzianum* @10 kg ha⁻¹ in 200 kg of FYM when compared to inoculated control (79.5%).

Seed treatment with *Pseudomonas fluorescens* plus soil amendment with mustard cake provided best control of dry root rot caused by *Macrophomina phaseolina* in chickpea (Khan and Gangopadhyay, 2008). The results of Harlapur and Wali (2009) against charcoal stalk rot of maize caused by *Macrophomina phaseolina* revealed that combined seed treatment with *Trichoderma harzianum* Sk₁, phosphate solubilizing bacteria (PSB) Acd-15 is significantly superior over individual treatments, which recorded lower disease incidence (9.64%) and highest grain yield (56.97 q/ha). Usha Rani *et al.* (2009) reported that seed treatment along with soil application of *Trichoderma viride* was significantly superior in controlling root rot disease incidence by 14.1 per cent in groundnut caused by *Macrophomina phaseolina*.

4.8.5.4 Effect of different treatments on plant growth parameters

In the present investigation, an attempt was made to observe whether the treatments imposed have any stimulatory (or) inhibitory effect on mean plant height, root length and shoot and root dry weight of chickpea plants

a) Plant height

Maximum plant height (25.61cm) was recorded in treatment T₈ (seed treatment with fungicide and soil application of potential fungal and bacterial biocontrol agent) followed by treatment T₇ (24.88 cm), T₆ (24.64 cm) and T₅ (24.13 cm). It is evident from Table 4.12 that least plant height (14.43cm) was recorded in inoculated control (T₁₅).

It is evident from Fig 4.11 that treatment T₈ stimulated the plant growth and development (25.61cm) when compared to inoculated control (14.43cm). It is attributed that copper oxychloride could arrest the pathogen and antagonists could parasitize the pathogen and promote growth by secreting growth promoting metabolites.

The results were in agreement with Vinod Kumar *et al.* (2007) who reported that *Pseudomonas fluorescens* isolate pf 4-99 reduced the incidence of charcoal rot of chickpea and enhanced the plant height by 29.40 per cent against the control. Lynch and Whipps (1991) proved that plant growth promotion by rhizobacteria is due to chemical and physical stimulation of plant roots resulting in more rapid emergence, higher chlorophyll level and increased stature. Rajan *et al.* (2000) demonstrated the involvement of endophytic bacteria in plant growth promotion and disease suppression. Rangeshwaran *et al.* (2001) reported that combined seed treatment of *Pseudomonas fluorescens* + *Pseudomonas putida* at 10g/kg gave maximum shoot length of 31.1 cm when compared to individual treatments.

b) Root length

Maximum root length was recorded in treatment T₈ (13.40 cm) that was on par with treatment T₇ (12.90 cm) and T₆ (12.78 cm). It is evident from Table 4.12 and Fig 4.12 that least root length (7.4 cm) was recorded in inoculated control (T₁₅).

Table 4.12. Effect of fungicide, biocontrol agents and organic amendments on growth parameters of chickpea cv. JG-11 in *Rhizoctonia bataticola* infested soil in pot culture

Treatment	*Plant height (cm)	*Root length (cm)	*Dry weight of shoot (g)	*Dry weight of root (g)
T1 : Seed treatment with potential fungal bio-control agent @ 10 g/kg	19.84 (26.43)	10.20 (18.61)	0.24 (2.80)	0.06 (1.42)
T2 : Seed treatment with potential bacterial bio-control agent @10g/kg	20.57 (26.96)	10.63 (19.02)	0.25 (2.86)	0.06 (1.44)
T3 : Seed treatment with potential fungal and bacterial bio-control agents @10g/kg	21.40 (27.55)	11.25 (19.59)	0.29 (3.08)	0.07 (1.51)
T4 : Seed treatment with fungicide	21.09 (27.33)	11.00 (19.34)	0.26 (2.91)	0.06 (1.48)
T5 : Soil application of potential fungal bio-control agent	24.13 (29.42)	12.20 (20.43)	0.40 (3.61)	0.09 (1.71)
T6 : Soil application of potential bacterial bio-control agent	24.64 (29.76)	12.78 (20.94)	0.42 (3.71)	0.09 (1.77)
T7 : Soil application of potential fungal and bacterial bio-control agents @10g/kg	24.88 (29.92)	12.90 (21.04)	0.46 (3.88)	0.10 (1.81)
T8 : T ₄ +T ₅ +T ₆	25.61 (30.40)	13.40 (21.46)	0.49 (4.01)	0.11 (1.89)
T9 : Application of Neem cake @ 12.5 q/ha	16.86 (24.24)	8.80 (17.25)	0.17 (2.35)	0.05 (1.33)
T10 : Application of FYM @ 5 t/ha	19.43 (26.15)	9.82 (18.25)	0.22 (2.68)	0.05 (1.36)
T11 : Application of Vermicompost @ 5 q/ha	19.14 (25.94)	9.30 (17.75)	0.20 (2.54)	0.05 (1.31)
T12 : Seed treatment with potential bacterial bio-control agent and soil application of Neem cake	21.65 (27.72)	11.50 (19.82)	0.32 (3.24)	0.07 (1.57)
T13 : Seed treatment with potential bacterial bio-control agent and soil application of FYM	23.49 (28.99)	12.13 (20.38)	0.37 (3.47)	0.08 (1.68)
T14 : Seed treatment with potential bacterial bio-control agent and soil application of Vermicompost	22.61 (28.39)	11.90 (20.17)	0.34 (3.33)	0.08 (1.62)
T15 : Control	14.43 (22.32)	7.40 (15.76)	0.14 (2.12)	0.05 (1.28)
S Em ±	0.342	0.36	0.14	0.03
CD (0.05)	0.99	1.04	0.42	0.11

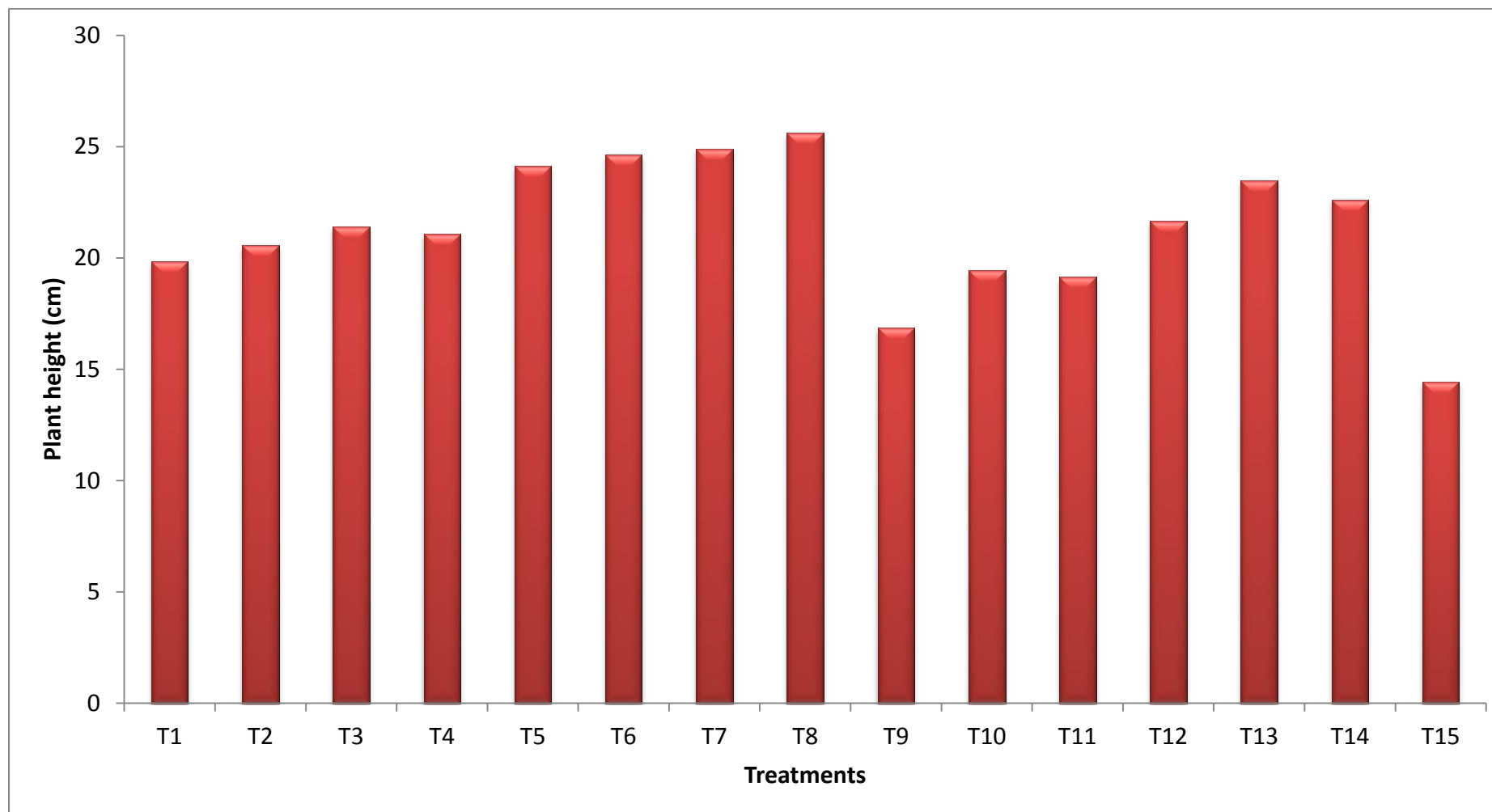


Fig. 4.11. Effect of antagonists, fungicide and organic amendments on plant height of chickpea in pot culture

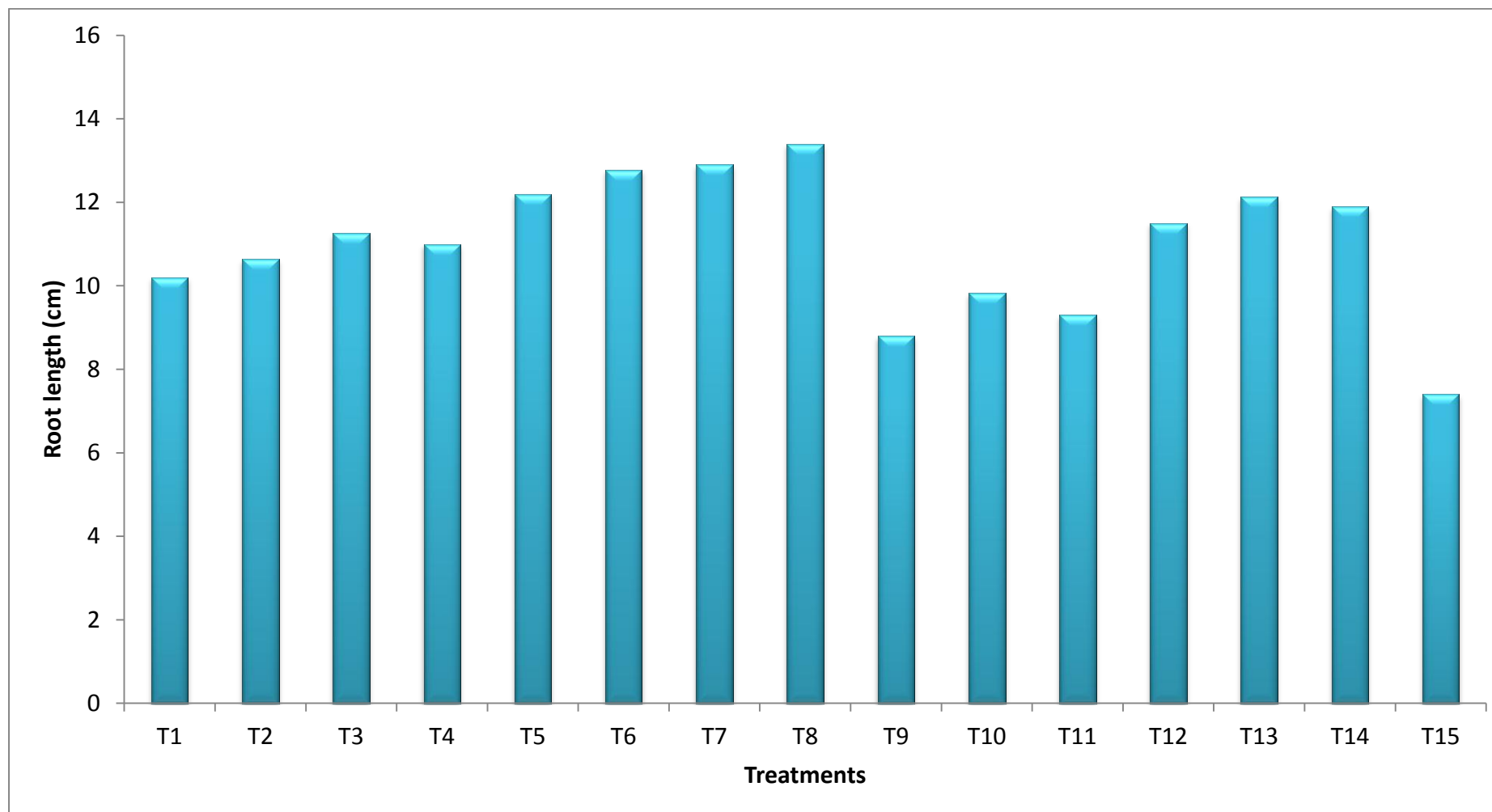


Fig. 4.12. Effect of antagonists, fungicide and organic amendments on root length of chickpea in pot culture

Similarly Bharati *et al.* (2004) reported that soil application of *Trichoderma harzianum* and *B. subtilis* reduced the incidence of damping-off in tomato caused by *Pythium aphanidermatum* and also increased shoot and root length and biomass production. Sendhilvel *et al.* (2005) used *Pseudomonas fluorescens* isolate SVPF₂ in combination of seed treatment and soil application against cowpea root rot and recorded less disease incidence and maximum root length (22.38 cm) under green house conditions. Jayasree *et al.* (2000) reported that maximum root length and shoot length of sesame and black gram were recorded in combined seed treatment and soil application of *T. harzianum* and *P. fluorescens*. Rangeshwaran *et al.* (2008) reported that maximum root length (6.7 cm) and shoot length (5.9 cm) was recorded in seed treatment with *Bacillus* sp. (UASEBCH2) after 14 days of germination against *Rhizoctonia solani* in chickpea.

c) Dry weight of shoot and root

The maximum shoot weight was recorded in treatment T₈ (0.49 g) that was on par with T₇ (0.46 g), T₆ (0.42 g) and T₅ (0.40 g). It is evident from Table 4.12 and Fig 4.13 that least shoot weight (0.14 g) was recorded in inoculated control (T₁₅) .

Maximum root weight (0.11g) was recorded in treatment T₈ that was on par with T₇ (0.10 g), T₆ (0.09 g) and T₅ (0.09 g) with the least (0.05 g) recorded in inoculated control (T₁₅) (Table 4.12 and Fig 4.14).

Rajan *et al.* (2000) reported that the endophytic bacteria enhanced tillering, overall growth of the plants and suppressed the pathogens and disease incidence in ginger.

The results of the present investigation were in agreement with earlier reports showing that the integrated treatment with biocontrol agents increased weight of shoot and root of chickpea. Maximum shoot and root dry weights

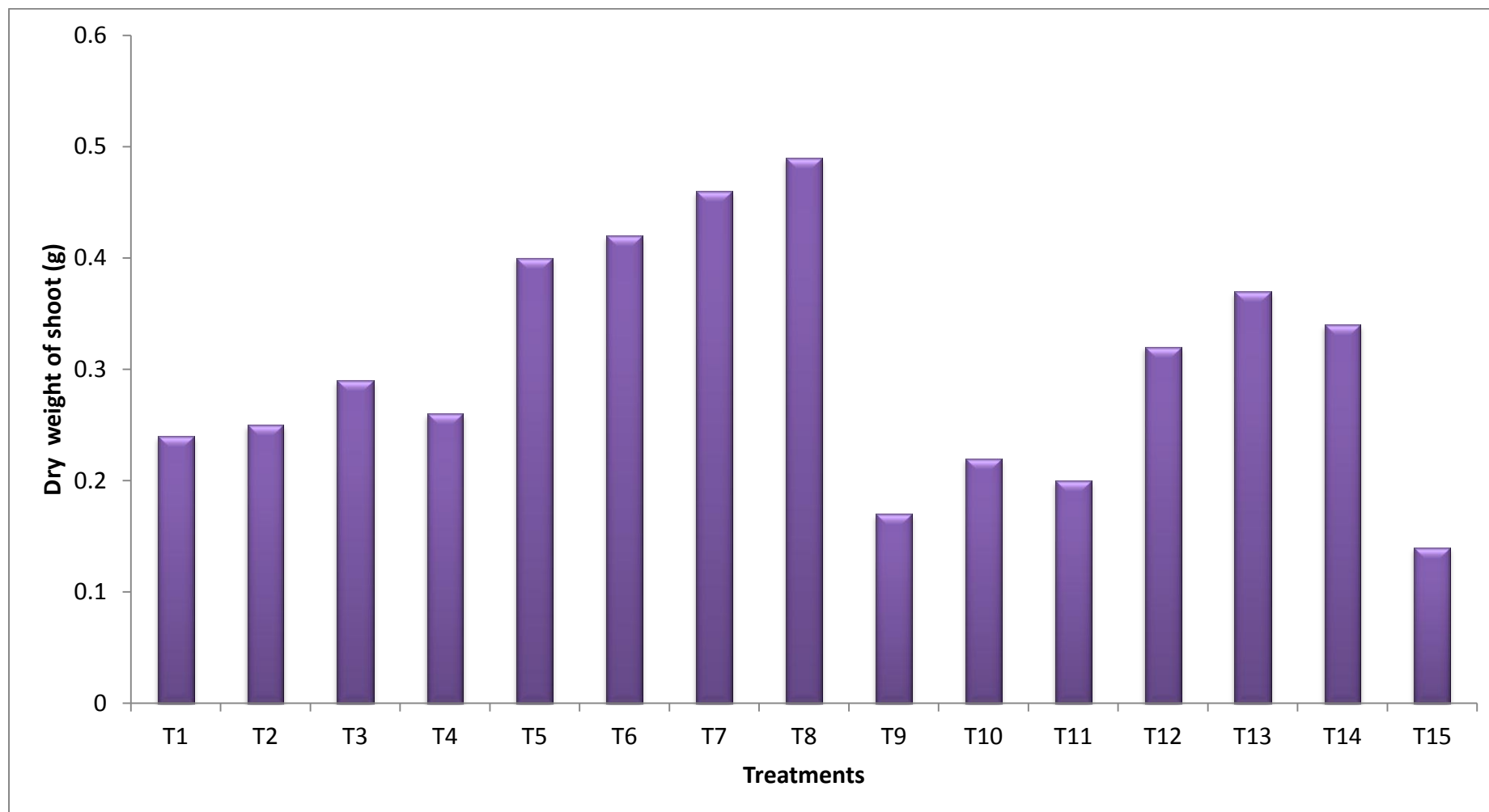


Fig. 4.13. Effect of antagonists, fungicide and organic amendments on shoot weight of chickpea in pot culture

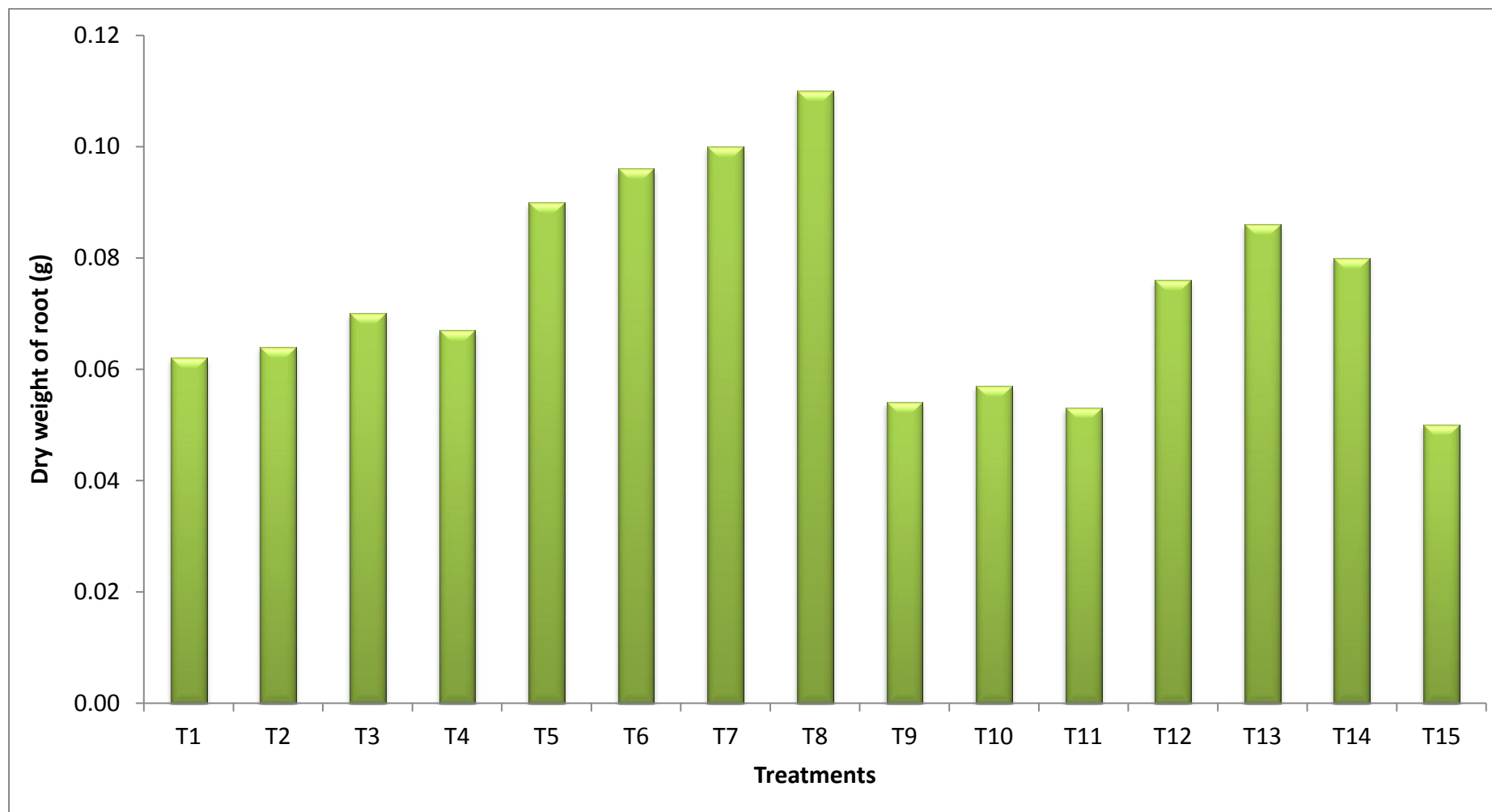


Fig. 4.14. Effect of antagonists, fungicide and organic amendments on root weight of chickpea in pot culture

were recorded in treatment T₈ (seed treatment with fungicide and soil application of potential fungal and bacterial biocontrol agent).

Thus overall, the efficacy of treatment T₈ (seed treatment with fungicide and soil application of potential fungal and bacterial biocontrol agent) was found to be superior which recorded highest germination per cent, least PDI, highest initial and final population of chickpea, maximum plant height, root length, shoot and root dry weight when compared to other treatments. The present findings are supported by other workers according to whom the integration of biocontrol agent with compatible fungicide gave significantly higher disease control in several crops than obtained by either biocontrol agent (or) fungicide. (Henis *et al.* 1978, Sawant and Mukhopadhyay, 1990).

4.9 ESTIMATION OF POPULATION LEVELS OF PATHOGEN, FUNGAL AND BACTERIAL ANTAGONISTS

4.9.1 Population dynamics of potential fungal isolate *Trichoderma* isolate-7 (CT7)

Population dynamics of potential fungal isolate *Trichoderma* isolate-7 (CT7) was estimated at two different intervals, initially 7 DAS and then on 45 DAS and the results are presented in the Table 4.13.

It is evident from the Fig 4.15 that maximum population was obtained in the treatment T₅ (7.80×10^5), followed by T₇ (7.00×10^5) and T₈ (6.80×10^5) which were on par with each other. The least population was obtained from the treatment T₃ (4.00×10^5) at 7 DAS. Similarly at 45 DAS, maximum population was obtained in the treatment T₅ (15.20×10^5), followed by T₇ (13.80×10^5) and T₈ (13.60×10^5) which were on par with each other. The least population was obtained from the treatment T₃ (9.60×10^5).

Table 4.13. Population dynamics of Potential fungal isolate *Trichoderma* isolate-7 (CT7)

Treatment	*Population of <i>Trichoderma</i> isolate-7 cfu/g soil ($\times 10^5$)	
	7 DAS	45 DAS
T1 : Seed treatment with potential fungal bio-control agent @ 10 g/kg	4.60 (12.38)	10.50 (18.91)
T2 : Seed treatment with potential bacterial bio-control agent @10g/kg	-	-
T3 : Seed treatment with potential fungal and bacterial bio-control agents @10g/kg	4.00 (11.54)	9.60 (18.05)
T4 : Seed treatment with fungicide	-	-
T5 : Soil application of potential fungal bio-control agent	7.80 (16.22)	15.20 (22.95)
T6 : Soil application of potential bacterial bio-control agent	-	-
T7 : Soil application of potential fungal and bacterial bio-control agents	7.00 (15.34)	13.80 (21.81)
T8 : T ₄ +T ₅ +T ₆	6.80 (15.11)	13.60 (21.64)
T9 : Application of Neem cake @ 12.5 q/ha	-	-
T10: Application of FYM @ 5 t/ha	-	-
T11: Application of Vermicompost @ 5 q/ha	-	-
T12: Seed treatment with potential bacterial bio-control agent and soil application of Neem cake	-	-
T13: Seed treatment with potential bacterial bio-control agent and soil application of FYM	-	-
T14: Seed treatment with potential bacterial bio-control agent and soil application of Vermicompost	-	-
T15: Control	-	-
S Em $\pm$	0.21	0.15
CD (0.05)	0.69	0.51

* Mean of three replications

Figures in parenthesis are angular transformed values

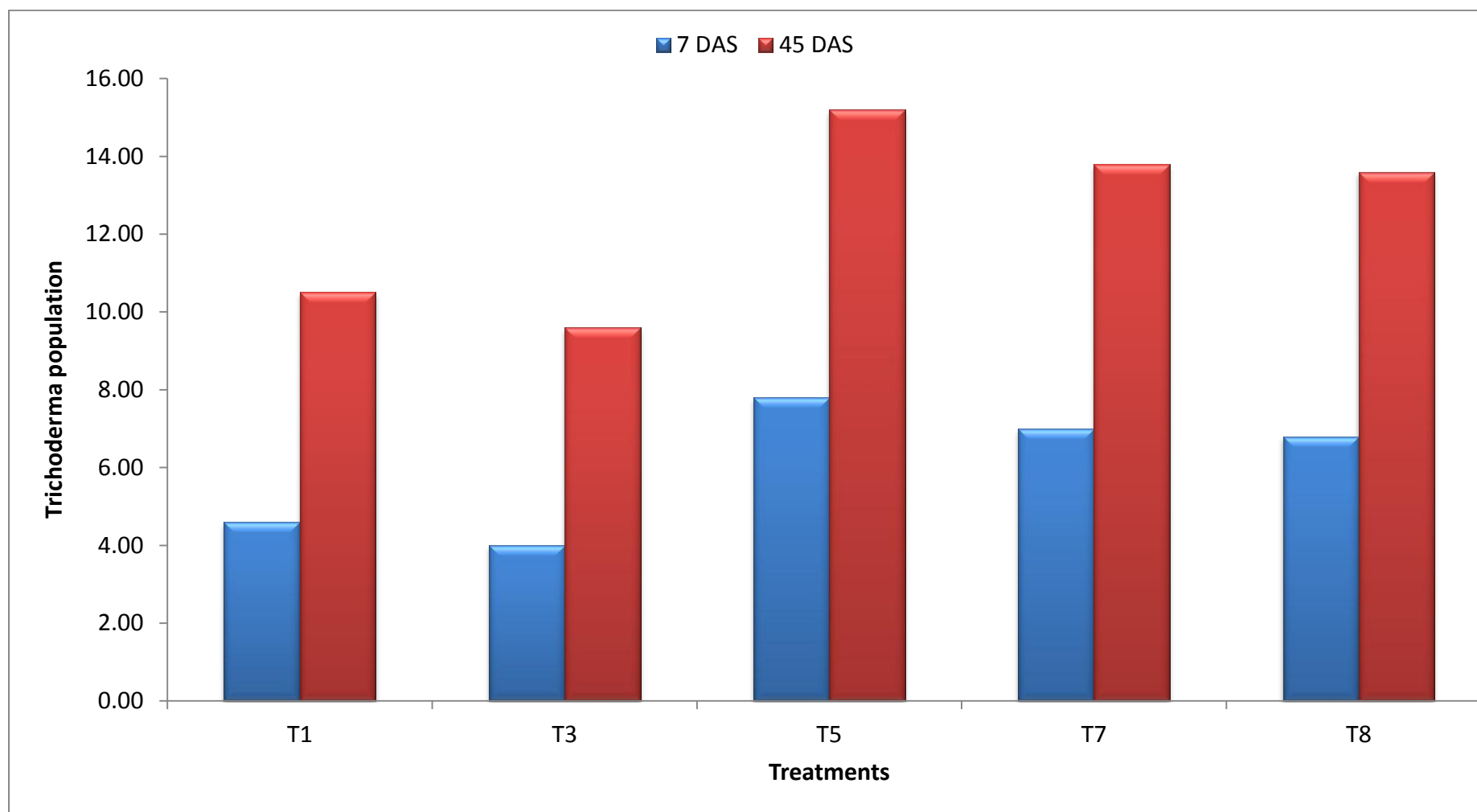


Fig. 4.15. Population dynamics of *Trichoderma* isolate-7 cfu/g of soil ($\times 10^5$) at two different intervals associated with chickpea cv. JG-11 in pot culture

In the present investigation, a rapid increase in the antagonist population of the rhizosphere soil was observed from 7 DAS to 45 DAS. The initial increase in population might be due to germination of different spore forms and their subsequent proliferation with or without food base. Odunfa and Oso (1979) also found that initial increase of antagonist population in rhizosphere could be due to the abundance of sugar and amino acids exuded from roots while the subsequent decrease could be ascribed to a reduction or change in composition of root exudates as the plants age.

The above results showed that the populations levels of potential fungal isolate *Trichoderma* isolate-7 (CT7) was more when applied as soil treatment compared to seed treatment. Further, fungal population was more when applied singly compared to combination treatment with bacteria and fungicide. This might be due to lack of complete compatibility of bacteria and fungicide with *Trichoderma* isolate-7 (CT7) as evident from *in vitro* results or due to competition in the zone of rhizosphere.

Application of suitable biocontrol agents capable of competing with other microbes in the soil can result in great success of biological control of crop diseases.

Gaur *et al.* (2005) reported that the initial mean population of *Trichoderma* in talc based formulation stored at 25°C on 0 days of storage was 270×10^6 cfu/g and it declined to 20×10^6 cfu/g after 105 days of storage indicating a reduction of 93 per cent. Paul *et al.* (2008) studied the effect of organic amendments on population dynamics of *Trichoderma harzianum* (JMA-4) and revealed that high population 103×10^6 cfu/g soil was obtained in soil amended with *Eupatorium odoratum*. Choudhary *et al.* (2010) reported that the mean population densities of *Trichoderma harzianum* (10.20×10^6) recorded minimum dry root rot incidence (16.67%) on mungbean when compared to control (66.67%).

4.9.2 Population dynamics of potential bacterial isolate CREB-16

Population dynamics of potential bacterial isolate CREB-16 was estimated at two different intervals, initially 7 DAS and then at 45 DAS and the results are presented in the Table 4.14.

It is evident from the Fig 4.16 that maximum population was obtained in the treatment T₆ (3.90×10^8), followed by T₇ (3.37×10^8) and T₈ (3.20×10^8) which were on par with each other. The least population was obtained from the treatment T₃ (1.30×10^8) at 7 DAS. Similarly at 45 DAS, maximum population was obtained in the treatment T₆ (4.97×10^8), followed by T₇ (4.60×10^8) and T₈ (4.50×10^8) which were on par with each other. The least population was obtained from the treatment T₃ (2.40×10^8).

The results of the treatments including organic amendments along with seed treatment with bacterial isolate CREB-16, showed that the population levels of bacteria were more with FYM both at 7 DAS (2.80×10^8) and at 45 DAS (3.67×10^8) compared to vermicompost and neem cake.

The above results showed that the population levels of potential bacterial isolate CREB-16 was more when applied as soil treatment compared to seed treatment. FYM was found to be the best suitable substrate for multiplication of the bacteria.

Khan and Gangopadhyay (2008) reported that the population of *Pseudomonas fluorescens* in FYM amended soil were increased from 17.3 to 28.7×10^{10} cfu/g soil with the increase in FYM dose from 5 g to 20 g/kg soil. Choudhary *et al.* (2010) reported that the mean population densities of *Bacillus firmus* (22.04×10^8) recorded minimum dry root rot incidence (33.33%) on mungbean when compared to control (58.33%).

Table 4.14. Population dynamics of Potential bacterial isolate CREB-16

Treatment	*Population of CREB-16 cfu/g soil ($\times 10^8$)	
	7 DAS	45 DAS
T1 : Seed treatment with potential fungal bio-control agent @ 10 g/kg	-	-
T2 : Seed treatment with potential bacterial bio-control agent @10g/kg	1.67 (7.41)	2.80 (9.63)
T3 : Seed treatment with potential fungal and bacterial bio-control agents @10g/kg	1.30 (6.54)	2.40 (8.90)
T4 : Seed treatment with fungicide	-	-
T5 : Soil application of potential fungal bio-control agent	-	-
T6 : Soil application of potential bacterial bio-control agent	3.90 (11.39)	4.97 (12.88)
T7 : Soil application of potential fungal and bacterial bio-control agents	3.37 (10.57)	4.60 (12.38)
T8 : T ₄ +T ₅ +T ₆	3.20 (10.30)	4.50 (12.24)
T9 : Application of Neem cake @ 12.5 q/ha	-	-
T10 : Application of FYM @ 5 t/ha	-	-
T11 : Application of Vermicompost @ 5 q/ha	-	-
T12 : Seed treatment with potential bacterial bio-control agent and soil application of Neem cake	2.23 (8.59)	3.10 (10.14)
T13 : Seed treatment with potential bacterial bio-control agent and soil application of FYM	2.80 (9.63)	3.67 (11.04)
T14 : Seed treatment with potential bacterial bio-control agent and soil application of Vermicompost	2.43 (8.95)	3.33 (10.52)
T15 : Control	-	-
S Em $\pm$	0.22	0.14
CD (0.05)	0.66	0.43

* Mean of three replications

Figures in parenthesis are angular transformed values

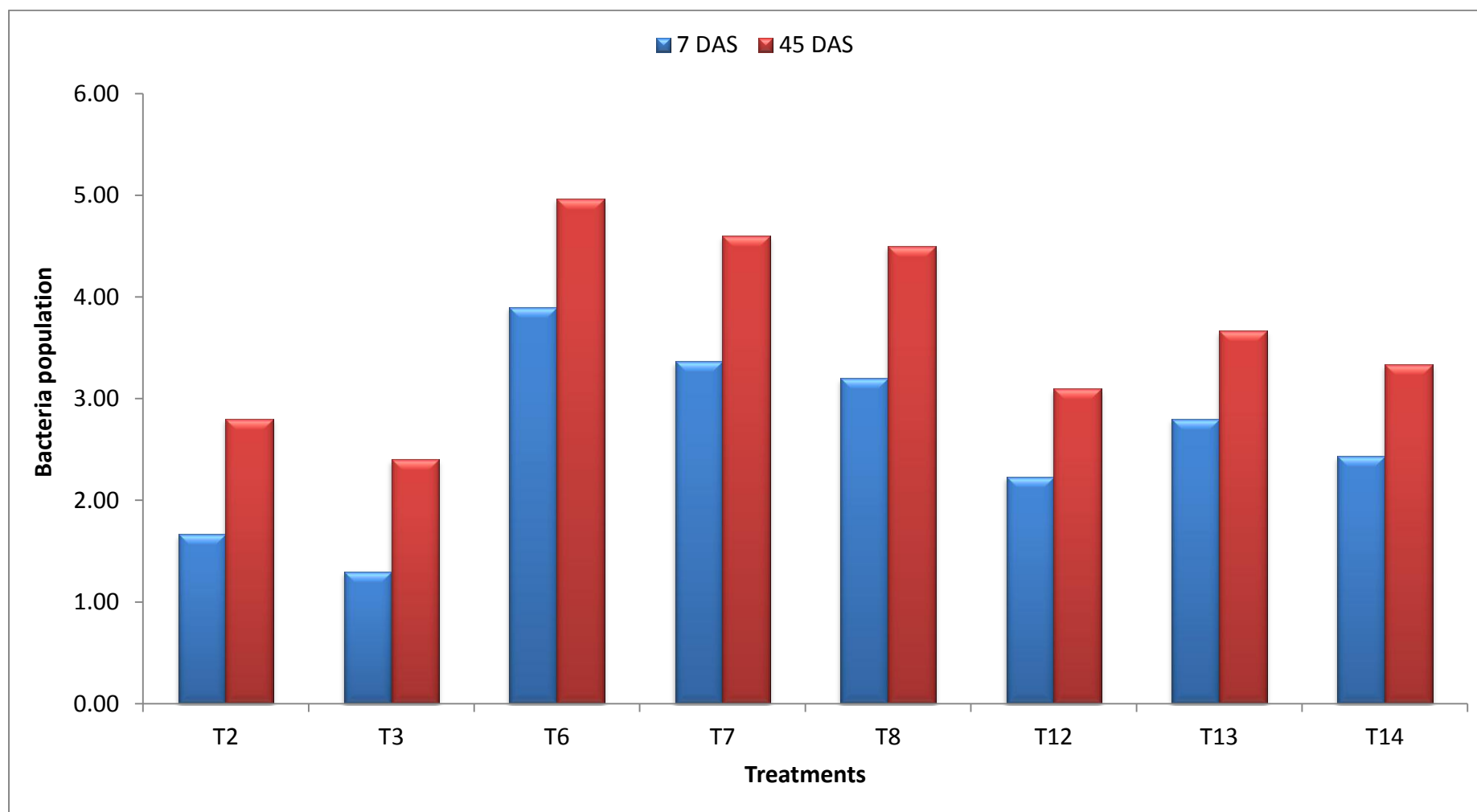


Fig. 4.16. Population dynamics of CREB-16 cfu/g of soil ($\times 10^8$) at two different intervals associated with chickpea cv. JG-11 in pot culture

4.9.3 Effect of fungicide, biocontrol agents and organic amendments on population levels of *Rhizoctonia bataticola* at different intervals associated with chickpea cv. JG-11

Population dynamics of *Rhizoctonia bataticola* was estimated at two different intervals, initially 7 DAS and then at 45 DAS and the results are presented in the Table 4.15.

It is evident from the Fig 4.17 that there was no significant difference between the treatments regarding population levels of the pathogen at 7 DAS. Maximum reduction in population levels of *Rhizoctonia bataticola* was observed in the treatment T₈ (3.00×10^4) followed by T₇ (4.33×10^4) and the least reduction was obtained in the treatment T₉ (15.33×10^4) at 45 DAS.

The above results indicated that the pathogen population were found to be decreased from 7 DAS to 45 DAS in all the treatments imposed. Maximum reduction in pathogen population was observed in treatment T₈ (seed treatment with fungicide and soil application of potential fungal and bacterial biocontrol agent). The decrease in pathogen population may be due to increased levels of antagonists and fungitoxicity of copper oxychloride. Whereas the pathogen population was very high in control, which may be due to lack of competition with any other microbes and fungitoxicants in soil.

Choudhary *et al.* (2010) reported that high levels of suppression of *Macrophomina phaseolina* (1.10×10^3) were observed in combination treatment of *Trichoderma harzianum* and *Bacillus firmus* at the mean population densities of 10.14×10^6 and 23.10×10^8 respectively, when compared to control (1.60×10^3).

Table 4.15. Effect of antagonists, fungicide and organic amendments on population levels of *Rhizoctonia bataticola*

Treatment	*Population of <i>Rhizoctonia bataticola</i> cfu/g soil ($\times 10^4$)	
	7 DAS	45 DAS
T1 : Seed treatment with potential fungal bio-control agent @ 10 g/kg	34.00 (35.66)	11.33 (19.61)
T2 : Seed treatment with potential bacterial bio-control agent @10g/kg	33.33 (35.26)	11.00 (19.35)
T3 : Seed treatment with potential fungal and bacterial bio-control agents @10g/kg	33.00 (35.05)	9.66 (18.09)
T4 : Seed treatment with fungicide	33.00 (35.06)	10.00 (18.42)
T5 : Soil application of potential fungal bio-control agent	31.00 (33.82)	6.00 (14.14)
T6 : Soil application of potential bacterial bio-control agent	31.00 (33.80)	5.33 (13.34)
T7 : Soil application of potential fungal and bacterial bio-control agents	30.33 (33.41)	4.33 (11.99)
T8 : T ₄ +T ₅ +T ₆	30.00 (33.20)	3.00 (9.88)
T9 : Application of Neem cake @ 12.5 q/ha	35.00 (36.26)	15.33 (23.03)
T10 : Application of FYM @ 5 t/ha	34.33 (35.86)	13.00 (21.09)
T11 : Application of Vermicompost @ 5 q/ha	35.00 (36.26)	14.00 (21.94)
T12 : Seed treatment with potential bacterial bio-control agent and soil application of Neem cake	31.66 (34.23)	8.66 (17.11)
T13 : Seed treatment with potential bacterial bio-control agent and soil application of FYM	32.00 (34.42)	7.33 (15.70)
T14 : Seed treatment with potential bacterial bio-control agent and soil application of Vermicompost	31.66 (34.23)	8.00 (16.40)
T15 : Control	36.00 (36.86)	65.00 (53.73)
S Em $\pm$	0.812	0.693
CD (0.05)	NS	2.01

* Mean of three replications

Figures in parenthesis are angular transformed values

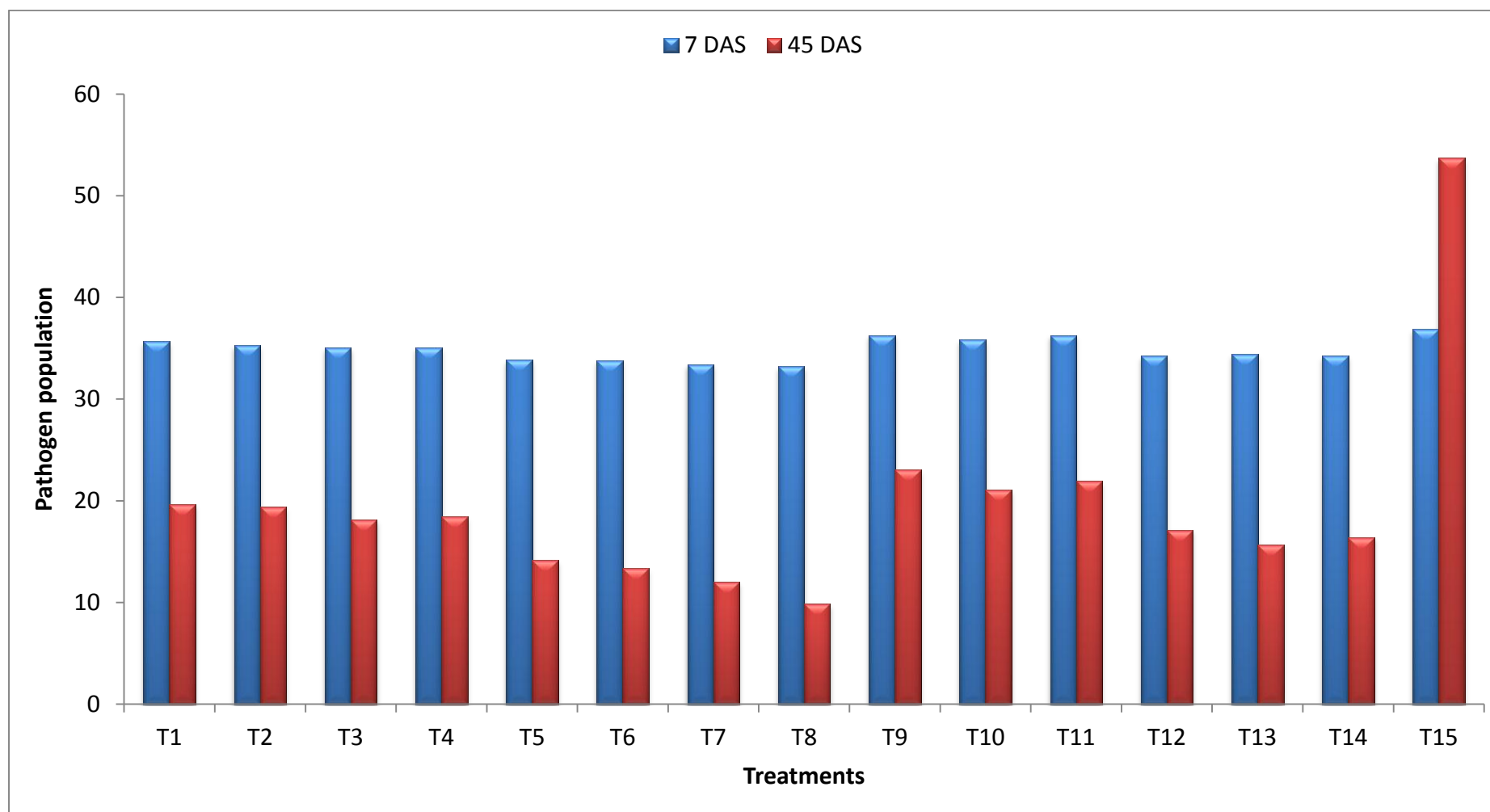


Fig. 4.17. Effect of antagonists, fungicide and organic amendments on population levels of *Rhizoctonia bataticola* cfu/g of soil (x10⁴) at two different intervals associated with chickpea cv. JG-11 in pot culture

4.10 MOLECULAR CHARACTERIZATION OF POTENTIAL BIOCONTROL AGENTS

The genetic variability among the isolates of biocontrol agents was studied by using molecular techniques like RAPD and 16S rDNA analysis.

Recently, molecular techniques have gained importance in characterization and diagnosis of microbial population. Moreover, these techniques are not influenced by environment, growth independent and are reproducible when compared to conventional methods. Hence, molecular characterization of potential biocontrol agents using Random Amplified Polymorphic DNA (RAPD) and analysis of 16S rDNA has been carried, which helps in identification of antagonists that are of immense use. These molecular techniques can also be used to develop markers in the diagnosis of potential biocontrol agents in future.

4.10.1 Qualitative and Quantitative verification of DNA from different isolates of biocontrol agents

Eight bacterial isolates having different degrees of antagonistic activity i.e. CREB-13, CREB-6, CREB-16, CREB-10, CRB-12, CRB-9, CRB-20 and CRB-6 were selected for molecular characterization.

The genomic DNA from the isolates under the investigation was extracted as per the procedure given in section 3.10.3. The quantity and quality of DNA was analyzed by running 2 µl of each sample in 1 per cent agarose gel (Figure not shown). Alternatively the quantity and quality of DNA was analyzed by spectrophotometer. These results indicated that the ratio between 260/280 was more than 1.6. The agarose gel analysis and spectrophotometer results indicated the good quality of DNA. The DNA concentration was adjusted to 100 ng/µl for the RAPD and 16S rDNA analysis.

4.10.2 Characterization of potential biocontrol agents by RAPD

RAPD is one of the methods of molecular analysis of natural microbial communities that lack resolving power for the identification at the species level and they also fail to give sufficient information to infer the functions of newly isolated organisms. However, these methods have a role in surveying the composition of microbial communities and in the characterization of bacterial isolates.

In the present investigation, the genetic variability among the eight isolates of bacterial biocontrol agents was analyzed by RAPD. Primer survey was carried out using seven primers from OPA series and five primers from OPD series of Operon technologies, Bangalore. Out of twelve primers used for amplification of DNA for the eight bacterial isolates, four primers gave reproducible and scorable bands with high percentage of polymorphism.

Four random primers *viz.*, OPA-12, OPA-18, OPD-04 and OPD-05 generated reproducible polymorphism among the isolates of biocontrol agents (Plate 36). Amplified products with all the primers have shown polymorphic and distinguishable banding pattern indicating genetic diversity among all the isolates. A total of 102 reproducible and scorable polymorphic bands ranging approximately as low as 50 bp to as high as 3500 bp were generated with four primers among the eight isolates.

The primer OPA-12 yielded a specific band approximately 850 bp in case of potential isolate CREB-16 and absent in all other isolates. The 420 bp bands were specifically amplified in case of isolate CREB-13 and CREB-6.

The primer OPA-18 amplified unique bands of approximately 450 bp and 280 bp in case of potential isolate CREB-16 which is absent in remaining isolates.

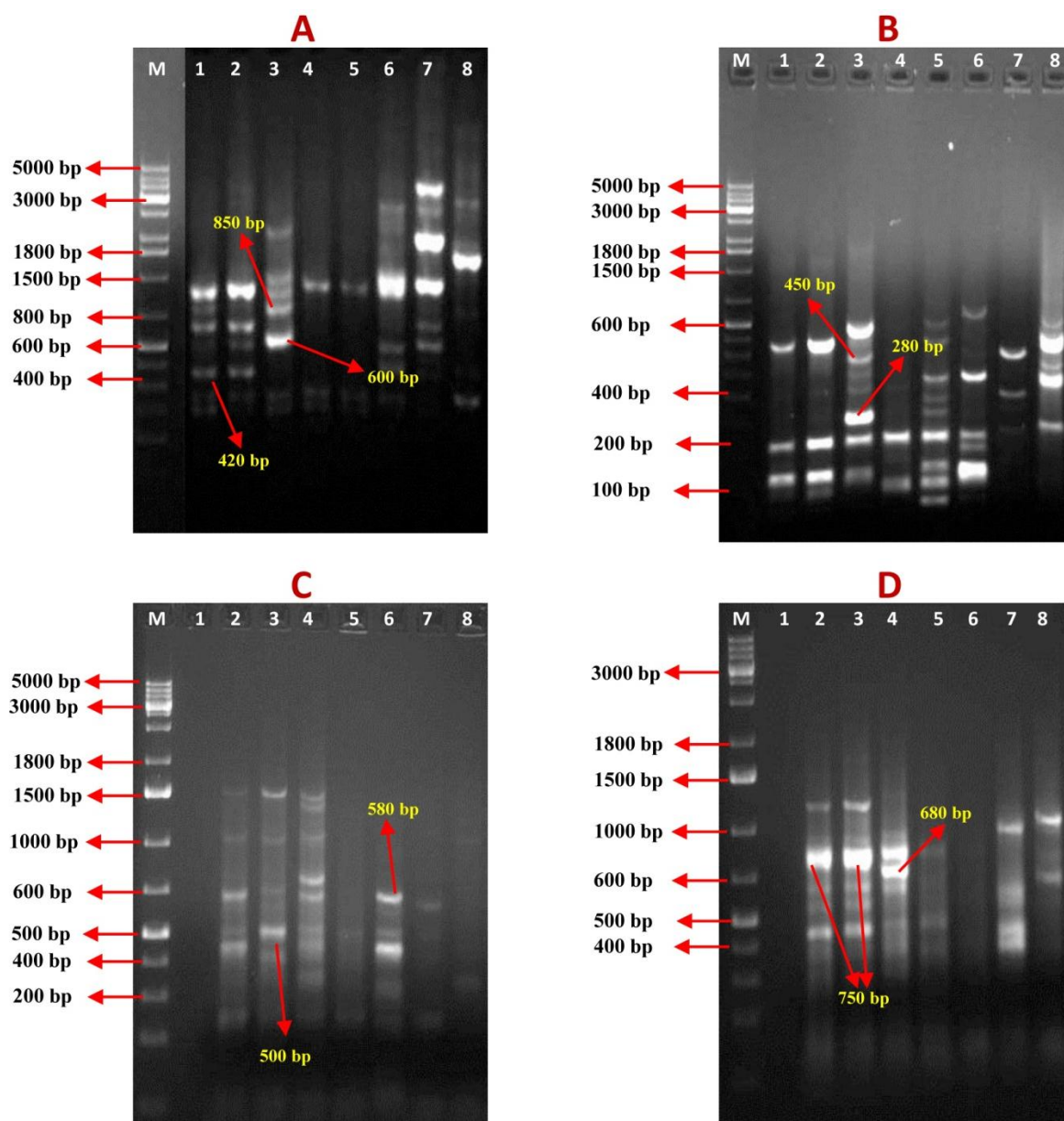


Plate 36. Random Amplified Polymorphic DNA (RAPD) profiles of antagonistic bacterial isolates with random primers

M: 1Kb DNA ladder. A: OPA-12, B: OPA-18, C: OPD-04, D: OPD-5.

Lane 1-8 are bacterial isolates CREB-13, CREB-6, CREB-16, CREB-10, CRB-12, CRB-9, CRB-20 and CRB-6.

The primer OPD-04 amplified unique band of approximately 500 bp in case of isolate CREB-16 and 580 bp in case of isolate CRB-20.

The primer OPD-05 amplified unique band of approximately 750 bp in case of isolate CREB-16 and CREB-6; whereas specific band of approximately 680 bp has been amplified in case of CREB-10 and absent in all other isolates.

PCR amplification with four primers was done twice before scoring for presence and absence of bands. Number of amplification products obtained was specific to each primer and it ranged from 2 to 10. All the primers used in the present analysis showed 25 per cent polymorphism as all the bands obtained were polymorphic with size ranging from 50 bp to 3.5 kb.

Megha *et al.* (2007) studied the diversity of 15 isolates of fluorescent pseudomonads using RAPD - PCR with eight random primers viz., OPC-9, OPD-2, OPD-3, OPO-6, OPO-09, OPO-13, 15 and 16. The PCR amplicons of fluorescent pseudomonads obtained from eight random primers produced 127 polymorphic bands. The minimum bands (9) were produced by the primer OPD-02 and maximum number of bands (25) was produced by OPO-16.

Relationship among the isolates was evaluated by cluster analysis of data based on similarity matrix. The dendrogram (Fig. 4.18) was generated using UPGMA package based on ward's squared Equalidean distance method. Based on the results obtained all the eight isolates were grouped into three main clusters.

Cluster I contains three isolates viz., CREB-13, CREB-6 and CRB-20, of which the first two and last one form two separate clusters. Cluster Ia contains CREB-13 and CREB-6; Ib containing CRB-20. Cluster II contains 2 isolates CREB-16 and CRB-6. Cluster III contains 3 isolates of which first two and last one form two separate clusters. Cluster Ic contains CRB-12, CRB-9 and cluster Id contains CREB-10.

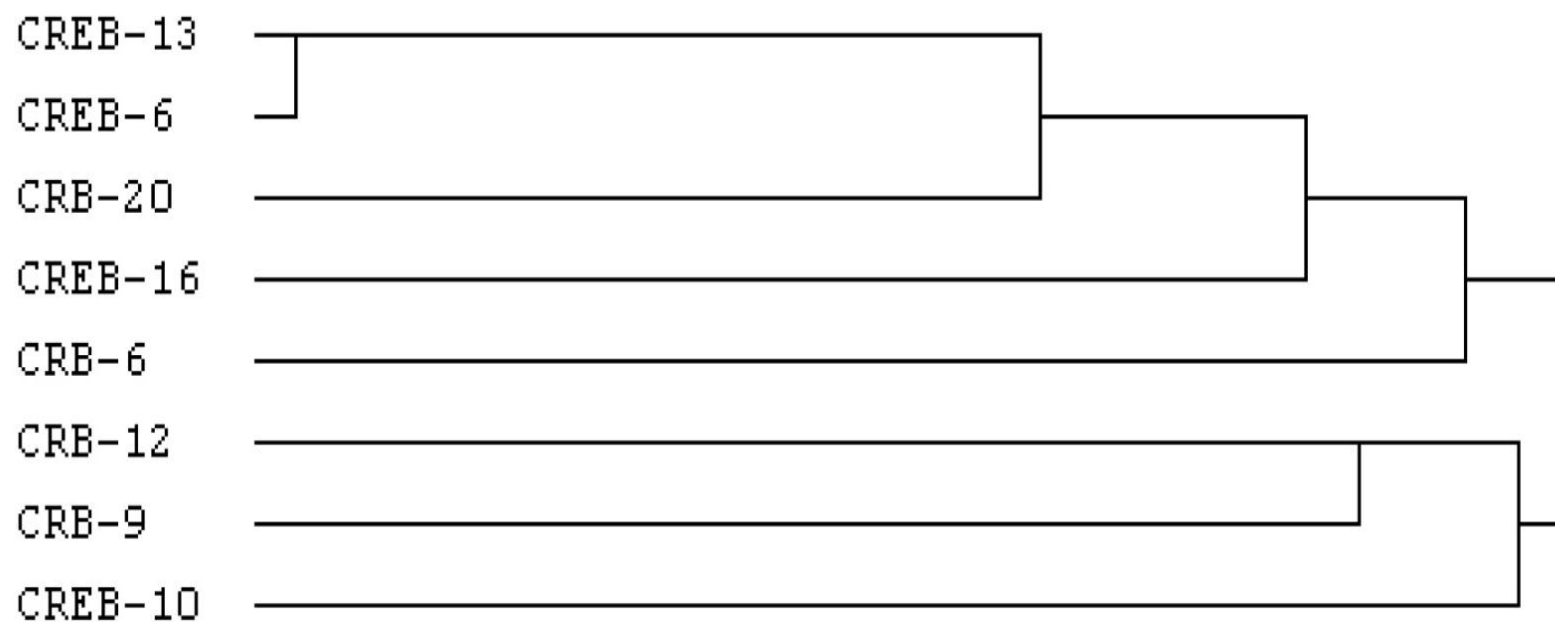


Fig. 4.18. Dendrogram generated using UPGMA analysis showing polymorphism between antagonistic bacterial isolates using RAPD markers

These results were in agreement with Ramesh Kumar *et al.* (2002) who studied the genetic variability among the isolates of *Pseudomonas* by RAPD with random primers, and the primer pgs3 produced several bands, including a unique band with size of 800 bp.

Gayathri *et al.* (2010) studied the polymorphism among the antagonistic seed and root endophytes against *Sclerotium rolfsii*. These results revealed that the isolates were grouped into two clusters with 133 reproducible and scorable polymorphic bands. Moreover, a primer OPD-2 yielded a unique band of 2600 bp in potential antagonists GSE-3 and GSE-4 which can be used to develop SCAR marker.

Jaccard's Similarity co-efficient among 8 isolates were calculated to establish the genetic relationships. The similarity index values ranged from 0.00 to 100 per cent indicating the presence of a high range of variability at nucleic acid level among the eight antagonistic bacterial isolates under the study.

RAPD markers could be used for developing Sequence Characterized Amplified Region (SCAR) markers linked to potential antagonists. SCAR marker is a specific and sensitive tool for monitoring the biocontrol in environmental samples. A specific band approximately 850 bp with the primer OPA-12, 450 bp with the primer OPA-18, 500 bp with the primer OPD-04 was amplified in CREB-16 can be used for the development of SCAR marker after cloning and sequencing. This SCAR marker may be validated further for confirmation.

These results were in agreement with Marta *et al.* (2005) who differentiated *Pseudomonas fluorescens* EPS62e from other *Pseudomonas fluorescens* strain by designing sequence characterized SCAR markers and two primer pairs from their differential amplified fragment for their specificity against EPS62e. A SCAR primer pair was evaluated and validated

for the assessment of population dynamics of EPS62e on pear plant under greenhouse condition.

It is worthwhile to analyse more RAPD markers including more number of isolates in order to observe genetic diversity among the isolates under the study which can help in identification of more specific bands that can be further developed into SCAR markers.

4.10.3 Characterization of bacterial isolates by 16S rDNA

The structure of rDNA cluster and the expected amplified products with 63F and 1387R primers are shown in Fig. 4.19.

16S rDNA specific target primers viz., 63F and 1387R were used for PCR amplification of 16S region rDNA cluster of potential isolate CREB-16. Both primers produced amplified product size of approximately 1300 bp in the isolate CREB-16 under the study as expected (Plate 37), indicating that the identified potential antagonistic isolate was bacteria and belongs to kingdom of prokaryotes.

4.10.4 Identification of potential bacterial antagonists based on sequence analysis of 16S rDNA.

The 16S rDNA sequence has been selected for the identification of the potential biocontrol agents.

4.10.5 Gel extraction and purification of 16S rDNA amplicon from potential bacterial antagonist CREB-16

The potential bacterial antagonist, CREB-16 which performed better in dual culture and pot culture studies was selected for further characterization. The 16S rDNA region of CREB-16 was amplified with the primer 63F and 1387R. The 1300bp amplified product of 16S rDNA was eluted from the gel using the kit M/s. Qualigens, Bangalore and then purified using the kit

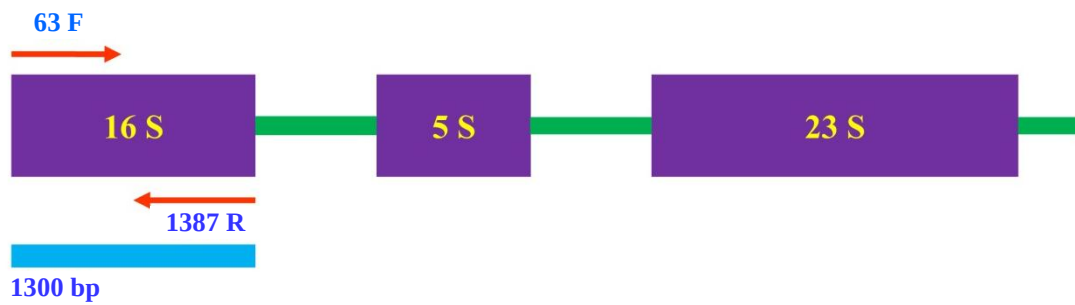


Fig. 4.21. Structure of rDNA cluster and the position of primers used in the PCR Amplification of 16S rDNA

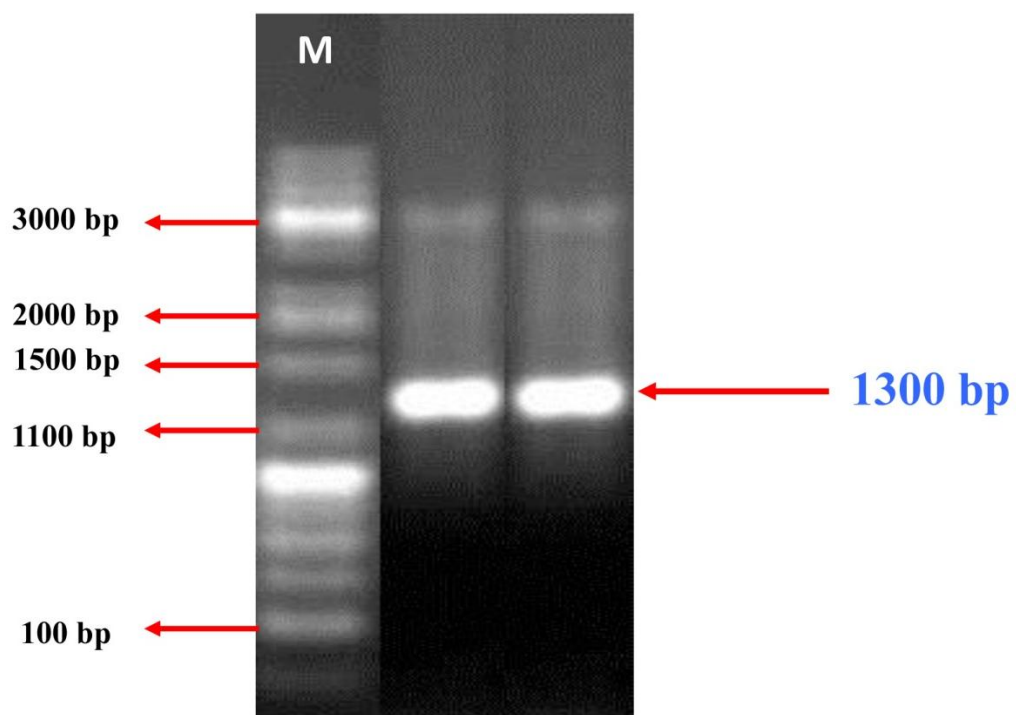


Plate 37. Amplification product of 16S rDNA with 63F and 1387R ribosomal DNA primers from CREB-16 isolate
Marker = 1 kb DNA ladder

M/s. Qiagen, Bangalore and sent for sequencing to Eurofins Genomics Private Limited, Bangalore and awaiting for sequence data for the identification of CREB-16 based on 16S rDNA sequence.

Misbah *et al.* (2005) identified the *Acinetobacter* of clinical isolates by amplification of 16S r DNA genome region consisting of approximately 1500 nucleotides using three primer pairs viz 27F, 780R; 529F, 1099R; 925F and 1491R.

Chapter ~ V

Summary & Conclusions

Chapter V

SUMMARY AND CONCLUSIONS

Chickpea (*Cicer arietinum* L.) is the third important food legume crop, which has world wide acceptance as a major source of protein and essential amino acids. India is the leading producer of chickpea contributing to about 70 per cent of the world's chickpea production. In India, it is grown over an area of 8.74 m.ha with an annual production of 7.35 million tonnes and productivity of 841 kg ha⁻¹ (Anonymous, 2009-10).

Chickpea suffers from several seed borne fungal diseases viz., black root rot, dry root rot, stem rot, crown rot, foot rot, *Sclerotinia* wilt and gray mould (Nene *et al.*, 1996). Among these diseases dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler, has been reported to cause severe losses from seedling to maturity stages of the crop.

The disease causes considerable yield losses in chickpea which may be as high as 50 to 71 per cent (Ahmed and Mohammad, 1986). The pathogen is soil- borne having a wide host range and survives well in light textured soil and under moisture stress conditions (Ghaffar and Erwin, 1969).

Keeping the importance of chickpea crop and the losses encountered by *Rhizoctonia bataticola*, the present work has been initiated with the following objectives.

- (i) Isolation and identification of the dry root rot pathogen from infected chickpea plants.
- (ii) To evaluate the efficacy of fungicides against causal agent of dry root rot *in vitro*.

- (iii) To isolate potential fungicide compatible biocontrol agents from rhizosphere and root endophytes under *in vitro*.
- (iv) Integrated disease management of dry root rot of chickpea under greenhouse conditions including organic amendments.
- (v) Molecular characterization of different isolates of potential antagonists was also studied by RAPD to detect polymorphism and analysis of 16S rDNA to identify the potential bacterial isolate.

The results obtained in the present investigation are summarized here.

The pathogen associated with dry root rot of chickpea was isolated from root rot affected plants and identified as *Rhizoctonia bataticola* based on standard mycological keys (Barnett and Barry, 1972).

In vitro efficacy of two non systemic fungicides viz., copper oxychloride and captan, two systemic fungicides viz., hexaconazole and tebuconazole and one antifungal antibiotic validamycin was evaluated against *Rhizoctonia bataticola* using poisoned food technique at different concentrations. Among all, the fungicides copper oxychloride, captan, hexaconazole and tebuconazole showed 100 per cent inhibition of mycelial growth, whereas the antibiotic validamycin has shown a mean inhibition of 91.66 per cent.

The potential antagonistic isolates were identified based on their ability to inhibit the growth of *Rhizoctonia bataticola* in dual culture technique.

A total of 10 *Trichoderma* isolates were obtained from rhizosphere soil and root endophytic zone of chickpea and were designated as CT1 to CT10. Among the ten fungal antagonistic isolates, *Trichoderma* isolate-7 (CT7) was superior with highest (83.33%) inhibition of the growth of

R. bataticola followed by *Trichoderma* isolate-4 (81.11%) and *Trichoderma* isolate-10 (74.44%) which were on par with each other.

A total of 20 rhizospheric bacterial isolates were obtained and designated as CRB-1 to CRB-20. Among all, the isolates, CRB-13 has shown highest inhibition (86.66%) followed by CRB-18 (83.33%) that were on par. Similarly a total of 20 root endophytic bacterial isolates were obtained and designated as CREB-1 to CREB-20. Regarding root endophytes, the isolate CREB-13 showed maximum inhibition (95.55%) that was on par with CREB-6 (94.44%) and CREB-16 (93.33%).

In vitro compatibility of potential *Trichoderma* isolate-7 was tested against four fungicides and one antifungal antibiotic at different concentrations using poisoned food technique. The compatibility was highest with validamycin (72.22%) followed by copper oxychloride (66.66%) and least with captan (22.22%). Hexaconazole and tebuconazole recorded 100% inhibition indicating complete incompatibility.

In vitro compatibility of four fungicides and one antifungal antibiotic used against *R. bataticola* were tested on potential antagonistic bacteria viz., CREB-6, CREB-13, CREB-16 and CRB-13 at different concentrations by using spectrophotometric method. The antibiotic validamycin showed highest compatibility followed by copper oxychloride, and least compatibility was recorded in case of captan and hexaconazole.

The *in vitro* compatibility results of both fungal and bacterial isolates revealed that the antibiotic validamycin and fungicide copper oxychloride are statistically on par with respect to compatibility with CREB-16 isolate. Moreover validamycin is available in liquid formulation which is difficult for seed treatment, hence the next best compatible chemical copper oxychloride which is available in wettable powder formulation and effective on pathogen was selected to integrate in IDM.

The *in vitro* mode of action of *Trichoderma* isolate-7 (CT7) against *Rhizoctonia bataticola* showed hyphal interaction between the antagonist and the pathogen. The mycelium of *Trichoderma* may coil around the mycelium of *Rhizoctonia bataticola* and later penetrate into it leading to disruption of mycelial contents of the pathogen.

The *in vitro* experimental results revealed that volatile metabolites produced by antagonistic bacteria inhibited the mycelial growth of *Rhizoctonia bataticola* to an extent of 77.77 per cent (CREB-16), 72.22 per cent (CREB-6), 71.11 per cent (CREB-13) and 55.55 per cent (CRB-13) over control after seven days of inoculation. In the case of non-volatile test, culture filtrate of antagonistic bacteria did not show any inhibition on the growth of *Rhizoctonia bataticola* over control *in vitro*.

The compatibility between highly potential and fungicide tolerant fungal isolate *Trichoderma* isolate-7 (CT7) and bacterial isolates CREB-6, CREB-13 and CREB-16 was tested using dual culture technique maintaining separate controls. The results revealed that the *Trichoderma* isolate-7 (CT7) has shown highest compatibility with the bacterial isolate CREB-16 (72.22%), followed by bacterial isolate CREB-6 (50.00%) and least compatibility was recorded with the bacterial isolate CREB-13 (33.33%).

So the bacterial isolate CREB-16 was selected to integrate in IDM as it has shown highest compatibility (72.22%) with the *Trichoderma* isolate-7 (CT7).

The effect of seed treatment with biocontrol agents and fungicide on seed germination was evaluated by blotter seed technique and the results revealed that all the treatments have given 100 per cent seed germination under *in vitro* conditions.

The efficacy of potential fungal antagonist *Trichoderma* isolate-7 (CT7), bacterial antagonist CREB-16, fungicide copper oxychloride and

organic amendments like FYM, vermicompost and neem cake were tested in pot culture against dry root rot of chickpea. Among the fifteen treatments imposed, treatment T₈ i.e. seed treatment with fungicide (Coc) + soil application of potential fungal (*Trichoderma* isolate-7) and bacterial biocontrol agent (CREB-16) was found to be superior as it recorded the highest germination percentage (100.00), highest initial population (10.00), final population (9.66) and least PDI of 16.00 per cent. This treatment also recorded maximum plant height (25.61 cm), root length (13.40 cm) and maximum shoot and root dry weights i.e. 0.49 g and 0.11 g respectively when compared to other treatments.

The population dynamics of both antagonists and pathogen were estimated at two different intervals initially at 7 DAS and then at 45 DAS. The population levels of *Trichoderma* were highest in the treatment T₅ (7.80×10^5), at 7 DAS and then increased to 15.2×10^5 (T₅) at 45 DAS.

The population dynamics of bacterial antagonist ranged between $1.30-3.90 \times 10^8$ at 7 DAS and then increased to a range of $2.40-4.97 \times 10^8$ at 45 DAS with the highest number in T₆ treatment and least number in T₃ treatment.

The population dynamics of pathogen at 7 DAS has no significant difference among the treatments with the highest number in inoculated control (36×10^4). However, there is a significant difference among the treatments at 45 DAS maximum being 65.00×10^4 in inoculated control and 3.00×10^4 in the T₈th treatment.

RAPD banding profile with four different random primers viz., OPA-12, OPA-18, OPD-4 and OPD-5 revealed the existence of genetic variability among the isolates that were classified into 3 main clusters. Cluster I is subdivided into cluster Ia (CREB-13 and CREB-6) and Ib (CRB-20). cluster-II

contains CREB-16 and CRB-6. Cluster III is sub-divided into cluster Ic (CRB-12 and CRB-9) and Id (CREB-10).

Amplification of 16S rDNA with 63F and 1387R primers which are specific to bacterial 16S rDNA produced approximately 1300 bp fragment. These results show that the potential antagonistic isolate CREB-16 was bacteria that belong to prokaryotes.

CONCLUSIONS

The following conclusions are drawn from the present investigation:

1. The pathogen associated with dry root rot of chickpea was isolated from root rot affected plants and identified as *Rhizoctonia bataticola*.
2. Among the four fungicides and one antibiotic tested against the pathogen copper oxychloride, captan, hexaconazole and tebuconazole showed 100 per cent inhibition of mycelial growth, whereas the antibiotic validamycin showed 91.66 per cent inhibition.
3. Among ten fungal antagonistic isolates, *Trichoderma* isolate-7 (CT7) was superior with highest per cent (83.33%) inhibition of the growth of *R. bataticola*.
4. Out of the 20 rhizosphere bacterial isolates CRB-13 showed highest inhibition (86.66%).
5. Of the 20 root endophytes, the isolate CREB-13 showed maximum inhibition (95.55%) followed by CREB-6 (94.44%) and CREB-16 (93.33%). Hence the isolates CREB-6, CREB-13 and CREB-16 were found to be more effective bacterial antagonists

against *Rhizoctonia bataticola* due to their high percentage of inhibition in dual culture technique.

6. *Trichoderma isolate-7* (CT7) was more compatible with validamycin (72.22%), followed by copper oxychloride (66.66%). The highest inhibition (100%) was observed with the fungicides hexaconazole and tebuconazole.
7. Isolate CREB-16 was found to be most compatible antagonistic bacteria with validamycin (84.13%) which was on par with copper oxychloride (78.27%) and least compatibility was recorded in case of tebuconazole (25.51%) in spectrophotometry.
8. *Trichoderma isolate-7* (CT7) showed mycoparasitism around the mycelium of *Rhizoctonia bataticola*.
9. CREB-16 antagonistic bacteria inhibited the mycelial growth of *Rhizoctonia bataticola* to an extent of 77.77 per cent followed by CREB-6 (72.22 %) and least with the isolate CRB-13 (55.55%) in the case of volatile test.
10. In the case of non-volatile test, culture filtrate of antagonistic bacteria did not show any inhibition on the growth of *Rhizoctonia bataticola* over control *in vitro*.
11. *Trichoderma isolate-7* (CT7) showed highest compatibility with the bacterial isolate CREB-16 (72.22%), followed by bacterial isolate CREB-6 (50%) and least compatibility with the bacterial isolate CREB-13 (33.33%) in compatibility tests between antagonistic fungus and bacteria.
12. In blotter seed technique, 100 per cent seed germination was recorded with all the treatments.

13. Treatment T₈ i.e. seed treatment with fungicide (Coc) + soil application of potential fungal (*Trichoderma isolate-7*) and bacterial biocontrol agent (CREB-16) was found to be superior as it recorded the highest germination percentage, highest initial and final population of chickpea, least PDI, maximum plant height, root length and maximum shoot and root dry weights.
14. The population dynamics of antagonists were increased from 7 DAS to 45 DAS while that of the pathogen were significantly reduced over control in all the treatments with the maximum reduction in T₈ treatment.
15. The RAPD banding profiles with random primers viz., OPA-12, OPA-18, OPD-4 and OPD-5 reflected the genetic diversity among the antagonistic bacterial isolates with formation of three main clusters. Amplified 16S rDNA with universal primers 63F and 1387R produced approximately 1300 bp fragments as expected.

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